



Leading in water supply

PP-R/PP-RCT Piping Systems

Pipes, fittings, valves, unions,
cutting tools, welding equipment,
welding devices and more



Made in Germany

Contact information

Germany

Address

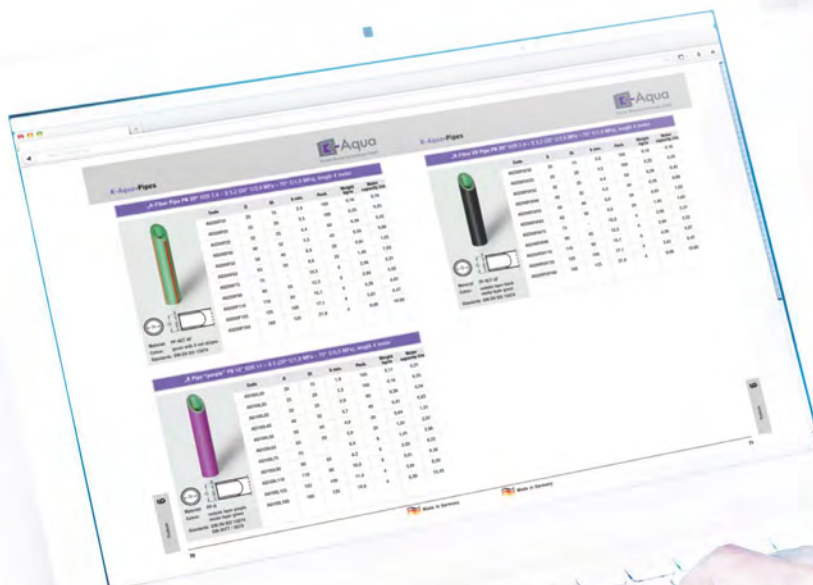
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Applying new strategies - our

Leading in water supply

The business field of KWT GmbH is manufacturing high-quality PP-R and PP-RCT piping systems designed for potable hot and cold water applications as well as industrial applications under the brand name **K-Aqua**.

KWT GmbH was founded in 2014 as independent unit on behalf of the premium manufacturer KESSEL AG, the market leader in the field of drainage.



KWT GmbH, Waldsolms-Brandobendorf, Germany

experience is your advantage

The name of KESSEL has stood for innovative products since more than 60 years. This knowledge, collected over decades, is not just flowing continuously into product development, but also into our customer service. This is how **K-Aqua** can help you to open up new markets and to extend existing business sectors.

K-Aqua provides innovative piping solutions tailored to the needs of industries worldwide.

With a commitment to German engineering excellence, we provide innovative piping solutions tailored to the needs of industries worldwide, from plumbing and heating systems to complex infrastructure projects.



K-Aqua- Impressions

KWT GmbH is a company with people working very efficient and autonomous for the goal to find the best market-oriented solutions for their customers.

We achieve at all times a balance of customer, business and employee benefits by all of our products, systems and processes. This is our very own key to long-term market success.

Our urge to improve processes and products constantly ensures the economic quality of our work and our existence.

Taking responsibility and market-oriented attitude of each of us is reflected in the quality of our work, our products and our togetherness. Openness to other cultures and languages, other products, markets and businesses, new ideas, technologies and organizations, partners and new collaborations is the reality of our being.

Comprehensive, cross-border thinking is common to all of us and is systematically encouraged!

Our organizational structure is consistently adapted to the needs of the market. Process chains are formed by the value chain of customers' requirements



K-Aqua-References

K-Aqua products have proven themselves countless times in destinations all over the world. The following references quickly mirror the extensive installation possibilities which K-Aqua products offer.

Diyar Al Muharaq, Bahrain
Al Kout Mall, Kuwait
Jahra Hospital, Kuwait
Blue Lagoon Resort, Iceland



PP-R/PP-RCT Pipes & Fittings

Company policy KWT GmbH

The goal is to create mutual benefits through the interaction of people and institutions in contact with us. There is no customer benefit without employee benefits, no employee benefits without company benefits and no company benefits without any benefit for our environment and the world around us.

Thus quality must always be compatible with the aspects of costs and deadlines. This will result in the involvement of all our employees in the process of continuous improvement.

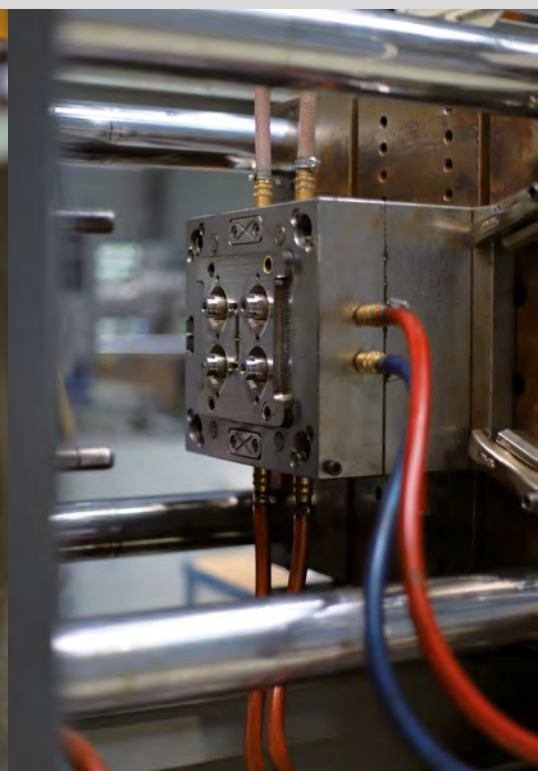
In compliance with legal and social standards, we will recognize, take seriously and implement the wishes of our customer and all other interested parties at all times.

Ecological and social responsibility is part of our understanding of quality, to which each employee of KWT GmbH assumes independently, promotes and consistently integrates into their daily actions and thinking.

For this reason, all company activities occur under consideration of resource conservation and environmental compatibility as well as a strict adherence to all commitments. The GENAU-management system is integrated into the House of KWT and for this reason an important element of our philosophy.

The Goal of the GENAU-management system is the continuous improvement of the company's work-, environment- and health protection. This also contains the optimization of energy efficiency. The prevention, continuous threat identification and risk evaluation as well as the implementation of remedial actions in the aspects of environment, energy and work-security are therefore an important components of the GENAU-policy.

The GENAU-policy is directed to all employees and departments of KWT. All employees daily work is equally responsible for the realization of the GENAU-actions. Every employee is obligated to report threats and actively participate in the improvement of the GENAU-management. The practiced GENAU-management meets the requirements of DIN EN ISO 14001 and 50001.



for drinkable hot & cold water

The K-Aqua PP-R/PP-RCT piping system includes different types of pipes (mono layer and multilayer reinforced pipes with glass fiber) with several wall thicknesses in SDR 6, SDR 7,4 , SDR 9, SDR 11 and SDR 17. Depending of the pipe type the available dimensions starts from size 20 mm up to 630 mm. The fittings are available in sizes 20 mm up to 315 mm as well as complementary products like valves, unions, cutting tools, welding equipment, welding devices, and more...



Pipes

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Fittings

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Transition Fittings

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Valves

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Tools

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K-Aqua-Leading in water supply

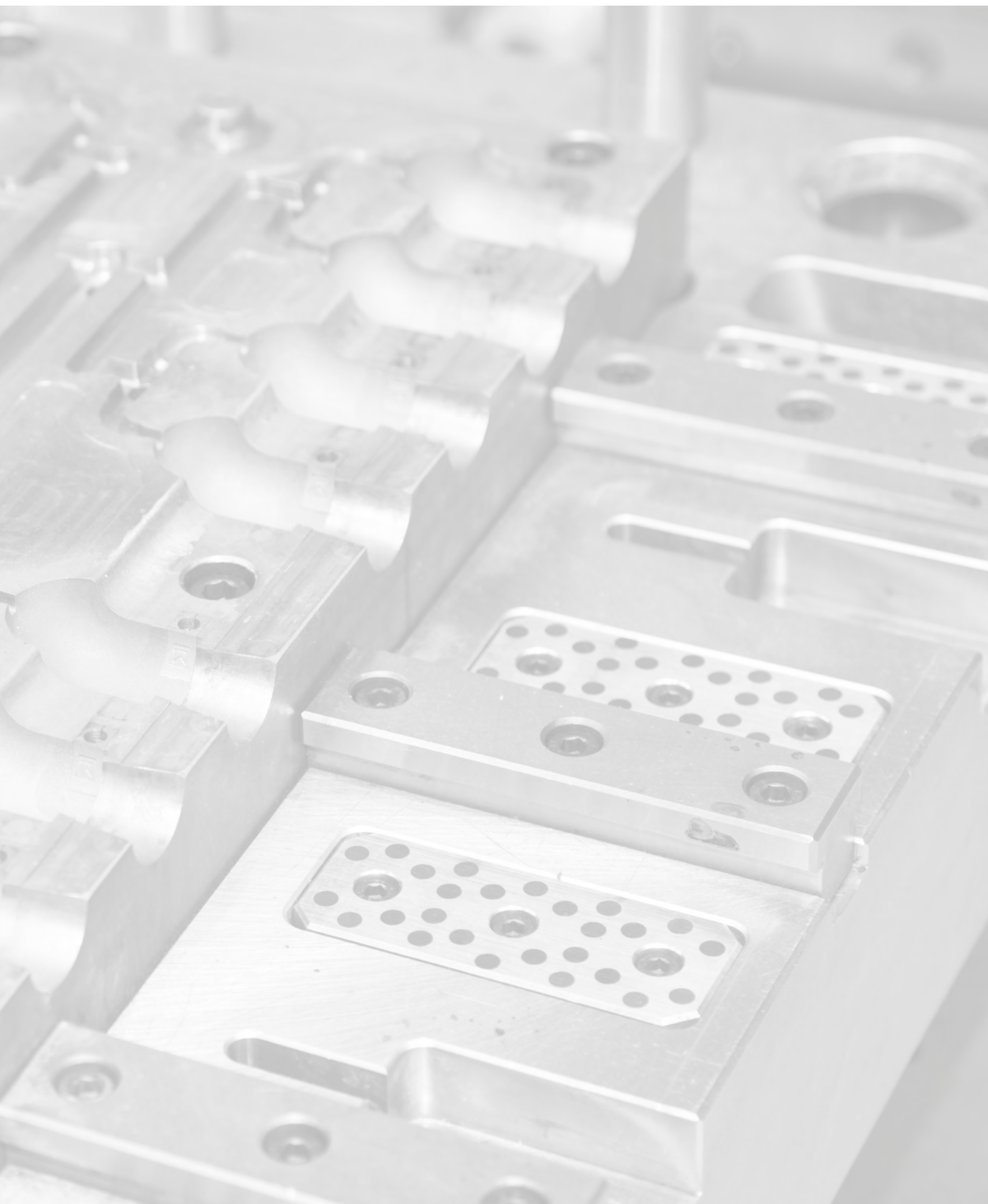




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Features

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Environmentally-friendly

Before polypropylene was invented, the gaseous waste from oil such as propylene and ethylene were simply burned, because they were useless. Today, these gases are used to produce polypropylene, thus drastically reducing atmospheric pollution. The production process also eliminates the potential pollution of rivers, streams and lakes, due to the use of water in a closed cooling cycle.



Recyclable

Polypropylene is commonly recycled, and has the number „5“ as its resin identification code, allowing efficient separation of different polymer types for recycling the raw material. The recycled raw materials have to conform with the manufacturing requirements of the environment commission, which stipulate that there is a minimum use of natural resources, reduced emissions, a long working life-span and optimum re-use. We recycle all production waste and cut-offs, noting that there is a high demand for polypropylene for recycling purposes as it can be recycled.



Superior

Polypropylene (PP), also known as polypropene, is a thermoplastic polymer used in a wide variety of applications, it is unusually resistant to many chemical solvents, bases and acids. This allows polypropylene to be used as an engineering plastic. Polypropylene is most commonly used for plastic moldings: it is injected into a mold while molten, forming complex shapes at relatively low cost and high volume, such as pipes and fittings.



Durable

Polypropylene products last much longer than those made from most similar materials. They are easy to wipe clean, hard wearing and withstand aging and extreme temperatures. Being more durable than alternative materials, products made from polypropylene don't need to be replaced as often, which means saving cost, resources, our environment and our future.

K-Aqua is designed for hot and cold water application and it is the latest and most suitable system for all plumbing applications. Besides plumbing, also it can be used for a variety of applications like air distribution, radiator heating, etc. The specific chemical structure of **K-Aqua** provides well balanced mechanical properties and superior long term heat resistance, ensuring the water flowing through which does not have any negative biological effect.

1.1 The Scope

K-Aqua products offer a superior German quality piping system, that could be equally installed in the residential and the industrial field:

- Sanitary applications.
- Heating & air-conditioning systems.
- Compressed air installations.
- Watering systems for greenhouses and gardens.
- Transporting liquid material.

For calculating the resistance, lifespan and safety of the **K-Aqua** system, depending on the average working temperature:

PN 20	10 bar	70° C
-------	--------	-------

The **K-Aqua** system should last for a minimum of 50 years. **K-Aqua** is an excellent choice for piping of clean hot & cold water.

- Potable water application
- Heating system construction
- Climate technology
- Chilled water technology
- Swimming-pool technology
- Chemical transport due to high chemical resistance
- Rainwater application
- Irrigation
- Compressed air systems
- Under- floor- heating- systems
- Application in the field of ship building
- Agriculture

1.2 Chemical Resistance

K-Aqua is a polyolefin polymer that features a high molecular weight. Therefore, it is more resistant to chemicals such as: acid, lime or cement. See table of chemical resistance. The resistance of **K-Aqua** products which are not submitted to the following factors: mechanical stress, various fluids, 20, 60 and 100° C temperatures according to ISO TR10358:1993.

1.3 Resistance to Current Strays

Like most thermoplastic products, **K-Aqua** is a poor electrical conductor. Therefore, there is no risk of stray currents occurring.

1.4 Soundproofness

The elasticity of **K-Aqua** pipes makes it viable to absorb and eliminate almost all vibrations, that would normally occur in the traditional cast iron pipes. Therefore, **K-Aqua** is highly soundproof at no extra cost.

1.5 Low Thermal Conductivity

K-Aqua has a low thermal conductivity (0.24 W/ m.K) that reduces the heat dispersion of the fluid that it conveys. Also, it reduces the condensation, which normally forms on the outside of the generic metal pipes, under specific hygrometric conditions.

1.6 Low Pressure Loss

The inside surface of **K-Aqua** is sleek, smooth with very few irregularities (0.0070 µ), which convey a significant reduction in pressure loss. As result, limestone cannot build up inside the pipe.

1.7 No-Toxicity

The raw material used for the production of **K-Aqua** is absolutely non-toxic and complies with the most up-to-date national and international regulations.

1.8 Easy Workability

One of the major attractions of the **K-Aqua** system is that it is extremely light and easy to weld and install. Our pipes with diameters ranging from 20 mm to 250 mm are extremely simple to assemble, providing the suitable polyfusion device. (See chapter 3)

1.9 UV Resistance

K-Fiber UV Pipes black cooled pipes are UV-resistant but must not be installed without protection in an exposed area. **K-Aqua** pipes and fittings are equipped with a stabilizer which allows for safe transport and installation. However, they should not be stored for more than six months in the open air.

1.10 Cracking Resistance Under Stress

The values determining the time resistance capacity of the **K-Aqua** system are the following:

Mechanical stress = Pressure



Thermal strain = Temperature



Stress duration = Time



The relationship between the above parameters can be controlled through regression curves.

K-Aqua minimum resistance values have been determined through internal pressure tests, at various temperature intervals: 20, 40, 60, 80, 95, 110° C. A logarithmic graphic representation shows the comparative tensions, the lifetime (in years), and the regression curves at various temperatures according to the DIN 8078 standard. See figure page 17 and 23

1.11 Advantages of the K-Aqua Piping System

K-Aqua provides all the necessary parts, for a complete and easy installation, from the beginning to the end. Saying goodbye to the conventional problems of the past. It is guaranteed to feel and see the difference with the **K-Aqua** piping system.

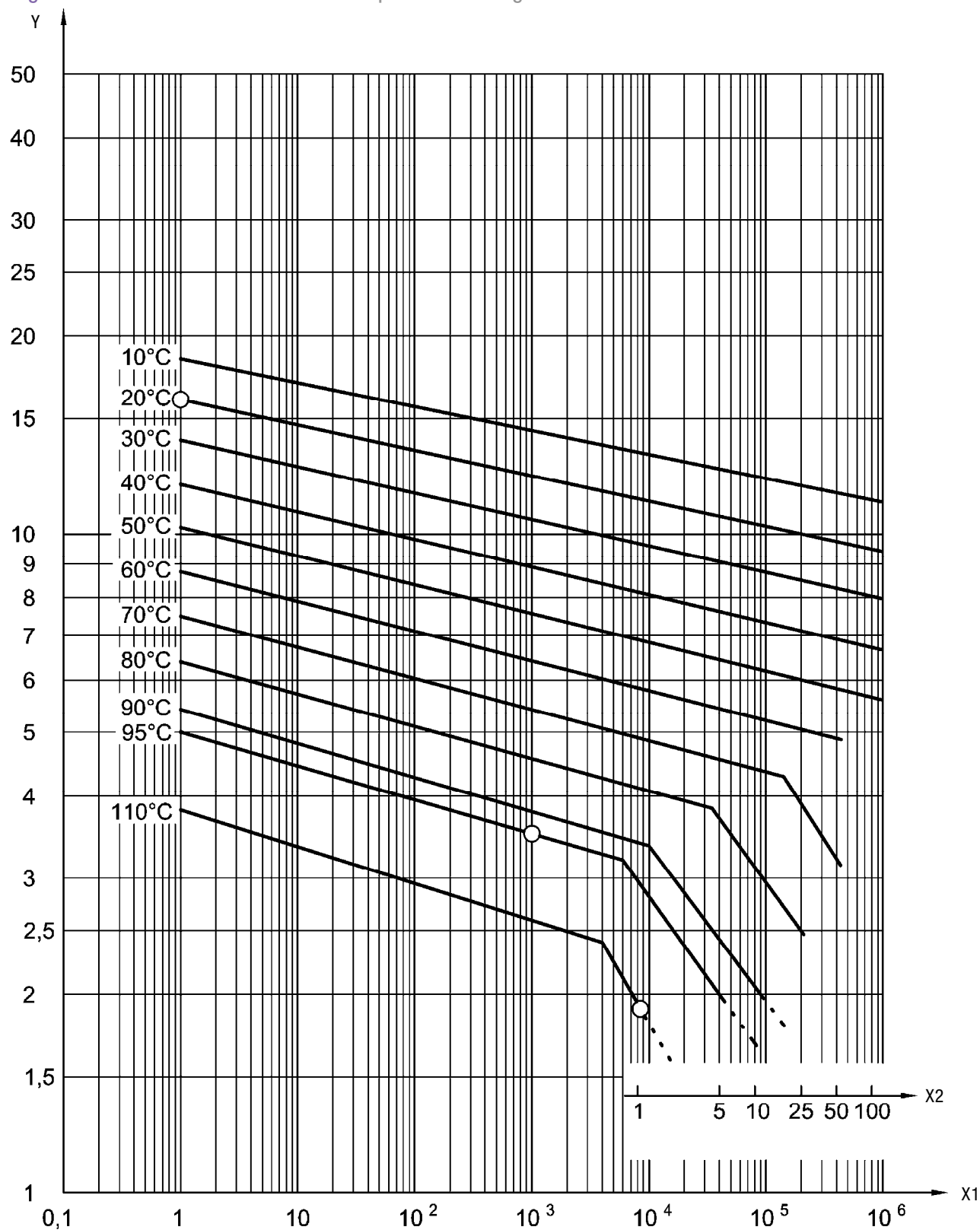
- **K-Aqua** is manufactured with superior German quality.
- **K-Aqua** made of corrosion resistant material, putting an end to old corrosion complications.
- Enjoy some serenity and peacefulness with **K-Aqua** because it is not as noisy as metal pipes.
- Unlike the alternatives, **K-Aqua** is made of opaque polypropylene, which is a nonpolluting material, preventing algae from growing.
- **K-Aqua** is completely recyclable, with no risk of air pollution, making it an environmentally friendly system.

1.12 Material Properties PP-R

Table 1: Physical properties PP-R

Property	Typical	Value	Test Method
Density	905	kg/m ³	ISO 1183
Melt Flow Rate (230° C / 2.16 kg)	0,25	g/10 min	ISO 1133
Flexural Modulus (2 mm/min)	800	MPa	ISO 178
Tensile Modulus (1 mm/min)	900	MPa	ISO 527
Tensile Strain at Yield (50 mm/min)	13,5	%	ISO 527-2
Tensile Strain at Yield (50 mm/min)	25	MPa	ISO 527-2
Thermal Conductivity	0,24	w/(m K)	DIN 52612
Coefficient of Thermal Expansion (0° C/70° C)	1,5*10E-4	1/K	DIN 53752
Charpy Impact Strength unnotched (23° C)	NO break		ISO 179/1eU
Charpy Impact Strength unnotched (0° C)	NO break		ISO 179/1eU
Charpy Impact Strength unnotched (-20° C)	40	kJ/m ²	ISO 179/1eU
Charpy Impact Strength notched (23° C)	20	kJ/m ²	ISO 179/1eA
Charpy Impact Strength notched (0° C)	3,5	kJ/m ²	ISO 179/1eA
Charpy Impact Strength notched (-20° C)	2	kJ/m ²	ISO 179/1eA

Figure 1 - Reference curves for expected strength of PP-R

 DIN EN ISO 15874-2:2013-06
 EN ISO 15874-2:2013 (D/E)


Key

X1 time, t_1 , to fracture, in hoursX2 time, t_2 , to fracture, in yearsY = hoopstress, α , in megapascal

Table 2 - Allowable operation pressures for PP-R pipes, conveying water
Safety factor (SF) = 1,25

DIN 8077:2008-09

Temperature °C	Years of service	Pipe series S									
		20	16	12,5	8,3	8	5	4	3,2	2,5	2
		Diameter/Wall thicknesses SDR									
		41	33	26	17,6	17	11	9	7,4	6	5
		Allowable operation pressure bar									
10	1	5,3	6,6	8,4	12,6	13,3	21,1	26,5	33,4	42,1	53,0
	5	4,9	6,2	7,9	11,9	12,5	19,8	25,0	31,5	39,7	49,9
	10	4,8	6,1	7,7	11,6	12,2	19,3	24,4	30,7	38,6	48,7
	25	4,7	5,9	7,4	11,2	11,8	18,7	23,6	29,7	37,4	47,0
	50	4,5	5,7	7,2	10,9	11,5	18,2	23,0	28,9	36,4	45,9
	100	4,4	5,6	7,0	10,7	11,2	17,8	22,4	28,2	35,5	44,7
20	1	4,5	5,6	7,1	10,8	11,3	18,0	22,6	28,5	35,9	45,2
	5	4,2	5,3	6,7	10,1	10,6	16,9	21,3	26,8	33,7	42,5
	10	4,1	5,2	6,5	9,9	10,4	16,4	20,7	26,1	32,8	41,4
	25	3,9	5,0	6,3	9,5	10,0	15,9	20,0	25,2	31,7	39,9
	50	3,8	4,8	6,1	9,3	9,7	15,4	19,5	24,5	30,9	38,9
	100	3,7	4,7	6,0	9,0	9,5	15,0	18,9	23,9	30,1	37,8
30	1	3,8	4,8	6,1	9,2	9,6	15,3	19,2	24,2	30,5	38,5
	5	3,6	4,5	5,7	8,6	9,0	14,3	18,0	22,7	28,6	36,0
	10	3,5	4,4	5,5	8,4	8,8	13,9	17,5	22,1	27,8	35,0
	25	3,3	4,2	5,3	8,1	8,4	13,4	16,9	21,3	26,8	33,8
	50	3,2	4,1	5,2	7,8	8,2	13,0	16,4	20,7	26,1	32,9
	100	3,1	4,0	5,0	7,6	8,0	12,7	16,0	20,1	25,4	31,9
40	1	3,2	4,1	5,1	7,8	8,2	13,0	16,3	20,6	25,9	32,6
	5	3,0	3,8	4,8	7,3	7,6	12,1	15,3	19,2	24,2	30,5
	10	2,9	3,7	4,7	7,1	7,4	11,8	14,8	18,7	23,5	29,6
	25	2,8	3,5	4,5	6,8	7,1	11,3	14,3	18,0	22,6	28,5
	50	2,7	3,4	4,3	6,6	6,9	11,0	13,9	17,4	22,0	27,7
	100	2,6	3,3	4,2	6,4	6,7	10,7	13,5	16,9	21,4	26,9
50	1	2,7	3,4	4,3	6,6	6,9	11,0	13,8	17,4	21,9	27,6
	5	2,5	3,2	4,0	6,1	6,4	10,2	12,9	16,2	20,4	25,7
	10	2,5	3,1	3,9	5,9	6,2	9,9	12,5	15,7	19,8	25,0
	25	2,4	3,0	3,8	5,7	6,0	9,5	12,0	15,1	19,0	24,0
	50	2,3	2,9	3,6	5,5	5,8	9,2	11,6	14,7	18,5	23,3
	100	2,2	2,8	3,5	5,4	5,6	9,0	11,3	14,2	17,9	22,6

Temperature °C	Years of service	Pipe series S									
		20	16	12,5	8,3	8	5	4	3,2	2,5	2
		Diameter/Wall thicknesses SDR									
		41	33	26	17,6	17	11	9	7,4	6	5
		Allowable operation pressure bar									
60	1	2,3	2,9	3,6	5,5	5,8	9,2	11,6	14,7	18,5	23,3
	5	2,1	2,7	3,4	5,1	5,4	8,6	10,8	13,6	17,2	21,6
	10	2,1	2,6	3,3	5,0	5,2	8,3	10,5	13,2	16,6	21,0
	25	2,0	2,5	3,1	4,8	5,0	8,0	10,1	12,7	16,0	20,1
	50	1,9	2,4	3,0	4,6	4,9	7,7	9,7	12,3	15,5	19,5
70	1	1,9	2,4	3,1	4,6	4,9	7,8	9,8	12,3	15,5	19,6
	5	1,8	2,2	2,8	4,3	4,5	7,2	9,1	11,4	14,4	18,1
	10	1,7	2,2	2,7	4,2	4,4	7,0	8,8	11,1	13,9	17,5
	25	1,5	1,9	2,4	3,6	3,8	6,0	7,6	9,6	12,1	15,2
	50	1,2	1,6	2,0	3,0	3,2	5,1	6,4	8,1	10,2	12,8
80	1	1,6	2,0	2,6	3,9	4,1	6,5	8,2	10,3	13,0	16,4
	5	1,4	1,8	2,3	3,4	3,6	5,7	7,2	9,1	11,5	14,5
	10	1,2	1,5	1,9	2,9	3,0	4,8	6,1	7,7	9,7	12,2
	25	0,9	1,2	1,5	2,3	2,4	3,9	4,9	6,2	7,8	9,8
95	1	1,1	1,4	1,8	2,7	2,9	4,6	5,8	7,3	9,2	11,6
	5	0,7	0,9	1,2	1,8	1,9	3,1	3,9	4,9	6,2	7,8
	(10) ^a	(0,6)	(0,8)	(1,0)	(1,5)	(1,6)	(2,6)	(3,3)	(4,1)	(5,2)	(6,6)
^{a)} The values between parentheses apply in cases where it can be demonstrated that the test was carried out for more than a year at 110° C											

^{a)} The values between parentheses apply in cases where it can be demonstrated that the test was carried out for more than a year at 110° C

Table 3 - Allowable operation pressures for PP-R pipes, conveying water
Safety factor (SF) = 1,5

DIN 8077:2008-09

Temperature °C	Years of service	Pipe series S									
		20	16	12,5	8,3	8	5	4	3,2	2,5	2
		Diameter/Wall thicknesses SDR									
		41	33	26	17,6	17	11	9	7,4	6	5
		Allowable operation pressure bar									
10	1	4,4	5,5	7,0	10,5	11,1	17,5	22,1	27,8	35,1	44,1
	5	4,1	5,2	6,6	9,9	10,4	16,5	20,8	26,2	33,0	41,6
	10	4,0	5,1	6,4	9,7	10,1	16,1	20,3	25,6	32,2	40,5
	25	3,9	4,9	6,2	9,3	9,8	15,6	19,6	24,7	31,1	39,2
	50	3,8	4,8	6,0	9,1	9,6	15,2	19,1	24,1	30,3	38,2
	100	3,7	4,6	5,9	8,9	9,3	14,8	18,6	23,5	29,6	37,2
20	1	3,7	4,7	5,9	9,0	9,4	15,0	18,8	23,7	29,9	37,7
	5	3,5	4,4	5,6	8,4	8,9	14,1	17,7	22,3	28,1	35,4
	10	3,4	4,3	5,4	8,2	8,6	13,7	17,2	21,7	27,4	34,5
	25	3,3	4,1	5,2	7,9	8,3	13,2	16,6	21,0	26,4	33,3
	50	3,2	4,0	5,1	7,7	8,1	12,9	16,2	20,4	25,7	32,4
	100	3,1	3,9	5,0	7,5	7,9	12,5	15,8	19,9	25,0	31,5
30	1	3,2	4,0	5,0	7,6	8,0	12,7	16,0	20,2	25,4	32,0
	5	3,0	3,7	4,7	7,2	7,5	11,9	15,0	18,9	23,8	30,0
	10	2,9	3,6	4,6	7,0	7,3	11,6	14,6	18,4	23,2	29,2
	25	2,8	3,5	4,4	6,7	7,0	11,2	14,1	17,7	22,3	28,1
	50	2,7	3,4	4,3	6,5	6,8	10,9	13,7	17,2	21,7	27,4
	100	2,6	3,3	4,2	6,3	6,6	10,6	13,3	16,8	21,1	26,6
40	1	2,7	3,4	4,3	6,5	6,8	10,8	13,6	17,1	21,6	27,2
	5	2,5	3,2	4,0	6,0	6,3	10,1	12,7	16,0	20,2	25,4
	10	2,4	3,1	3,9	5,9	6,2	9,8	12,3	15,5	19,6	24,7
	25	2,3	2,9	3,7	5,6	5,9	9,4	11,9	15,0	18,8	23,7
	50	2,3	2,9	3,6	5,5	5,8	9,2	11,5	14,5	18,3	23,1
	100	2,2	2,8	3,5	5,3	5,6	8,9	11,2	14,1	17,8	22,4
50	1	2,3	2,8	3,6	5,5	5,7	9,1	11,5	14,5	18,2	23,0
	5	2,1	2,7	3,4	5,1	5,3	8,5	10,7	13,5	17,0	21,4
	10	2,0	2,6	3,3	4,9	5,2	8,2	10,4	13,1	16,5	20,8
	25	2,0	2,5	3,1	4,7	5,0	7,9	10,0	12,6	15,9	20,0
	50	1,9	2,4	3,0	4,6	4,8	7,7	9,7	12,2	15,4	19,4
	100	1,8	2,3	2,9	4,5	4,7	7,5	9,4	11,8	14,9	18,8

Temperature °C	Years of service	Pipe series S									
		20	16	12,5	8,3	8	5	4	3,2	2,5	2
		Diameter/Wall thicknesses SDR									
		41	33	26	17,6	17	11	9	7,4	6	5
		Allowable operation pressure bar									
60	1	1,9	2,4	3,0	4,6	4,8	7,7	9,7	12,2	15,4	19,4
	5	1,8	2,2	2,8	4,3	4,5	7,1	9,0	11,3	14,3	18,0
	10	1,7	2,2	2,7	4,1	4,3	6,9	8,7	11,0	13,9	17,5
	25	1,6	2,1	2,6	4,0	4,2	6,6	8,4	10,5	13,3	16,7
	50	1,6	2,0	2,5	3,8	4,0	6,4	8,1	10,2	12,9	16,2
70	1	1,6	2,0	2,5	3,9	4,1	6,5	8,1	10,3	12,9	16,3
	5	1,5	1,9	2,4	3,6	3,8	6,0	7,5	9,5	12,0	15,1
	10	1,4	1,8	2,3	3,5	3,6	5,8	7,3	9,2	11,6	14,6
	25	1,2	1,5	2,0	3,0	3,1	5,0	6,3	8,0	10,0	12,7
	50	1,0	1,3	1,7	2,5	2,6	4,2	5,3	6,7	8,5	10,7
80	1	1,3	1,7	2,1	3,2	3,4	5,4	6,8	8,6	10,8	13,7
	5	1,2	1,5	1,9	2,9	3,0	4,8	6,0	7,6	9,6	12,1
	10	1,0	1,2	1,6	2,4	2,5	4,0	5,1	6,4	8,1	10,2
	25	0,8	1,0	1,2	1,9	2,0	3,2	4,1	5,1	6,5	8,1
95	1	0,9	1,2	1,5	2,3	2,4	3,8	4,8	6,1	7,6	9,6
	5	0,6	0,8	1,0	1,5	1,6	2,6	3,2	4,1	5,2	6,5
	(10) ^a	(0,5)	(0,6)	(0,8)	(1,3)	(1,3)	(2,2)	(2,7)	(3,4)	(4,3)	(5,5)

^{a)} The values between parentheses apply in cases where it can be demonstrated that the test was carried out for more than a year at 110° C

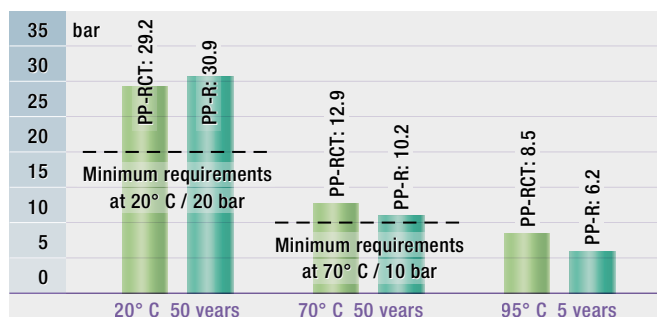
1.13 Material Properties PP-RCT

PP-RCT (Polypropylene-Random Crystallinity Temperature) is a material classification used to describe the second-generation class of PP-R materials. Introduced by Borealis in 2004 through its RA7050 range materials, it sets a milestone in the advancement of PP pressure piping systems. The pipe class has recently been included in EN ISO 15874, the global standard for plastics piping systems for hot and cold water pipe installations. PP-RCT is a polypropylene-random-copolymer with a special crystallinity which provides an improved pressure resistance, especially at elevated temperatures.

Pressure tests according to ISO 9080 on pipes manufactured from PP-RCT materials demonstrate 50 years service life at 70° C of 5 MPa, compared to the 3.2 MPa for standard PP-R materials. These very capabilities allow PP-RCT to increase performance and competitiveness for PP-R producers, and offer advantages for building designers and end-users alike.

Using PP-RCT in your pipe design will allow for increased performance vs. standard PP-R such as:

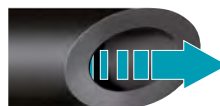
- increased pressure class with the same dimensioning
- higher hydraulic capacity with same outer diameter
- weight reduction (from 14 % up to 25 %) in pipe
- higher percentage of smaller pipes in actual installations (percentage depends upon specific design)
- cost efficient system due to beneficial dimensioning
- easier installation
- greater possibilities to tailor your heating PP-R pipe design
- opening up for special applications such as larger diameter mains for skysrise airconditioning systems
- weldable with known PP welding procedures
- fully established in domestic- as well as global standards incl. ISO and DIN



Higher pressure load through improved crystalline structure

With a special method of treatment of the material even smaller and more crystals are formed on the PP-RCT compared to PP-R. This means even more safety in permanent application because of higher pressure resistance at higher temperatures.

PP-R Pipes



Higher flow

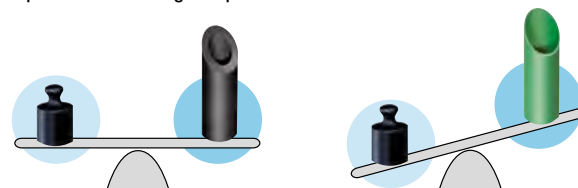
The reduced wall thickness leads to a higher flow rate at same and higher pressure load to increased flow rate.

PP-RCT Pipes



Higher stability at high temperatures

Through the special manufacturing of PP-RCT maximum working temperatures at higher pressure resistance were raised.



Lower weight

Through reduced wall thickness the weight of the PP-RCT pipes is considerably reduced.

Table 4: Physical properties PP-RCT

Property	Typical	Value	Test Method
Density	905	kg/m³	ISO 1183
Melt Flow Rate (230° C / 2.16 kg)	0,3	g/10 min	ISO 1133
Tensile Stress at Yield (50 mm/min)	25	MPa	ISO 527-2
Tensile Strain at Yield (50 mm/min)	10	%	ISO 527-2
Modulus of Elasticity in Tension (1 mm/min)	900	MPa	ISO 527
Charpy Impact Strength notched (+23° C)	40	kJ/m²	ISO 179/1eA
Charpy Impact Strength notched (0° C)	4	kJ/m²	ISO 179/1eA
Charpy Impact Strength notched (-20° C)	2	kJ/m²	ISO 179/1eA
Mean Linear Thermal Coefficient of Expansion from 0° C to 70° C	1,5	*10-4K-1	DIN 53752
Thermal Conductivity	0,24	WK-1m-1	DIN 52612 Part 1
Surface Resistance	> 1012	Ohm	DIN 53482/VDE 0303 Part 2

Figure 2 - Reference curves for expected strength of PP-RCT

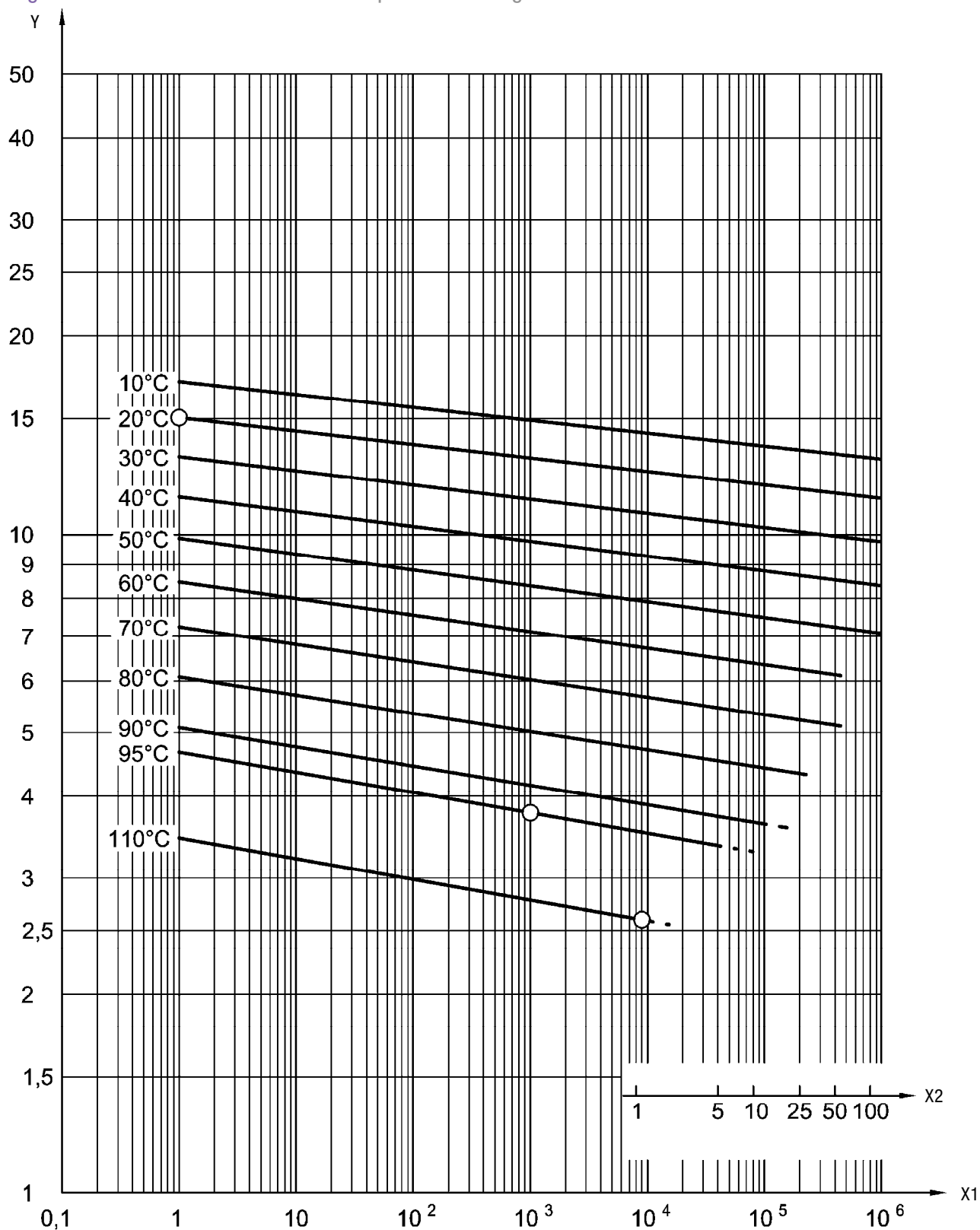
 DIN EN ISO 15874-2:2013-06
 EN ISO 15874-2:2013 (D/E)
**Key**X1 time, t_1 , to fracture, in hoursX2 time, t_2 , to fracture, in yearsY = hoopstress, σ , in megapascal

Table 5 - Allowable operation pressures for PP-RCT, conveying water
Safety factor (SF) = 1,25

DIN 8077:2008-09

Temperature °C	Years of service	Pipe series S									
		20	16	12,5	8,3	8	5	4	3,2	2,5	2
		Diameter/Wall thicknesses SDR									
		41	33	26	17,6	17	11	9	7,4	6	5
		Allowable operation pressure bar									
10	1	5,7	7,2	9,1	13,7	14,4	22,8	28,8	36,2	45,6	57,4
	5	5,5	7,0	8,8	13,3	14,0	22,1	27,9	35,1	44,2	55,7
	10	5,5	6,9	8,7	13,1	13,8	21,9	27,5	34,7	43,7	55,0
	25	5,4	6,8	8,5	12,9	13,5	21,5	27,1	34,1	42,9	54,0
	50	5,3	6,7	8,4	12,7	13,4	21,2	26,7	33,6	42,3	53,3
	100	5,2	6,6	8,3	12,6	13,2	20,9	26,3	33,2	41,8	52,6
20	1	5,0	6,3	7,9	11,9	12,5	19,9	25,0	31,5	39,7	50,0
	5	4,8	6,1	7,6	11,6	12,1	19,3	24,2	30,5	38,5	48,4
	10	4,7	6,0	7,5	11,4	12,0	19,0	23,9	30,1	37,9	47,8
	25	4,6	5,9	7,4	11,2	11,7	18,6	23,5	29,6	37,2	46,9
	50	4,6	5,8	7,3	11,0	11,6	18,4	23,1	29,2	36,7	46,2
	100	4,5	5,7	7,2	10,9	11,4	18,1	22,8	28,8	36,2	45,6
30	1	4,3	5,4	6,8	10,3	10,8	17,2	21,7	27,3	34,4	43,3
	5	4,1	5,2	6,6	10,0	10,5	16,6	20,9	26,4	33,2	41,8
	10	4,1	5,1	6,5	9,8	10,3	16,4	20,6	26,0	32,7	41,2
	25	4,0	5,0	6,4	9,6	10,1	16,1	20,2	25,5	32,1	40,4
	50	3,9	5,0	6,3	9,5	10,0	15,8	19,9	25,1	31,6	39,8
	100	3,9	4,9	6,2	9,4	9,8	15,6	19,7	24,8	31,2	39,3
40	1	3,7	4,6	5,9	8,9	9,3	14,8	18,6	23,5	29,6	37,2
	5	3,5	4,5	5,7	8,6	9,0	14,3	18,0	22,6	28,5	35,9
	10	3,5	4,4	5,6	8,4	8,8	14,1	17,7	22,3	28,1	35,4
	25	3,4	4,3	5,4	8,3	8,7	13,8	17,3	21,8	27,5	34,6
	50	3,4	4,3	5,4	8,1	8,5	13,6	17,1	21,5	27,1	34,1
	100	3,3	4,2	5,3	8,0	8,4	13,3	16,8	21,2	26,7	33,6
50	1	3,1	4,0	5,0	7,6	8,0	12,6	15,9	20,1	25,3	31,8
	5	3,0	3,8	4,8	7,3	7,7	12,2	15,3	19,3	24,3	30,6
	10	3,0	3,7	4,7	7,2	7,5	12,0	15,1	19,0	23,9	30,1
	25	2,9	3,7	4,6	7,0	7,4	11,7	14,7	18,6	23,4	29,5
	50	2,9	3,6	4,6	6,9	7,2	11,5	14,5	18,3	23,0	29,0
	100	2,8	3,5	4,5	6,8	7,1	11,3	14,3	18,0	22,6	28,5

Temperature °C	Years of service	Pipe series S									
		20	16	12,5	8,3	8	5	4	3,2	2,5	2
		Diameter/Wall thicknesses SDR									
		41	33	26	17,6	17	11	9	7,4	6	5
		Allowable operation pressure bar									
60	1	2,7	3,4	4,2	6,4	6,7	10,7	13,5	17,0	21,4	27,0
	5	2,5	3,2	4,1	6,2	6,5	10,3	13,0	16,3	20,6	25,9
	10	2,5	3,2	4,0	6,1	6,4	10,1	12,7	16,0	20,2	25,5
	25	2,4	3,1	3,9	5,9	6,2	9,9	12,4	15,7	19,8	24,9
	50	2,4	3,0	3,8	5,8	6,1	9,7	12,2	15,4	19,4	24,5
70	1	2,2	2,8	3,6	5,4	5,7	9,0	11,3	14,3	18,0	22,7
	5	2,1	2,7	3,4	5,2	5,4	8,6	10,9	13,7	17,3	21,7
	10	2,1	2,6	3,3	5,1	5,3	8,5	10,7	13,5	16,9	21,3
	25	2,0	2,6	3,3	5,0	5,2	8,3	10,4	13,1	16,5	20,8
	50	2,0	2,5	3,2	4,9	5,1	8,1	10,2	12,9	16,2	20,5
80	1	1,8	2,3	3,0	4,5	4,7	7,5	9,5	11,9	15,0	18,9
	5	1,8	2,2	2,8	4,3	4,5	7,2	9,0	11,4	14,4	18,1
	10	1,7	2,2	2,8	4,2	4,4	7,0	8,9	11,2	14,1	17,7
	25	1,7	2,1	2,7	4,1	4,3	6,9	8,6	10,9	13,7	17,2
95	1	1,4	1,7	2,2	3,4	3,5	5,6	7,1	8,9	11,2	14,2
	5	1,3	1,7	2,1	3,2	3,3	5,3	6,7	8,5	10,7	13,5
	(10) ^a	(1,3)	(1,6)	(2,1)	(3,1)	(3,3)	(5,2)	(6,6)	(8,3)	(10,5)	(13,2)

^{a)} The values between parentheses apply in cases where it can be demonstrated that the test was carried out for more than a year at 110° C

Table 5 - Allowable operation pressures for PP-RCT, conveying water
Safety factor (SF) = 1,5

DIN 8077:2008-09

Temperature °C	Years of service	Pipe series S									
		20	16	12,5	8,3	8	5	4	3,2	2,5	2
		Diameter/Wall thicknesses SDR									
		41	33	26	17,6	17	11	9	7,4	6	5
		Allowable operation pressure bar									
10	1	4,7	6,0	7,5	11,4	12,0	19,0	24,0	30,2	38,0	47,9
	5	4,6	5,8	7,3	11,1	11,6	18,4	23,2	29,3	36,9	46,4
	10	4,5	5,7	7,2	10,9	11,5	18,2	22,9	28,9	36,4	45,8
	25	4,5	5,6	7,1	10,7	11,3	17,9	22,5	28,4	35,7	45,0
	50	4,4	5,5	7,0	10,6	11,1	17,7	22,2	28,0	35,3	44,4
	100	4,3	5,5	6,9	10,5	11,0	17,4	21,9	27,6	34,8	43,8
20	1	4,1	5,2	6,6	9,9	10,4	16,6	20,9	26,3	33,1	41,7
	5	4,0	5,0	6,4	9,6	10,1	16,0	20,2	25,4	32,0	40,4
	10	3,9	5,0	6,3	9,5	10,0	15,8	19,9	25,1	31,6	39,8
	25	3,9	4,9	6,1	9,3	9,8	15,5	19,6	24,6	31,0	39,1
	50	3,8	4,8	6,1	9,2	9,6	15,3	19,3	24,3	30,6	38,5
	100	3,8	4,7	6,0	9,1	9,5	15,1	19,0	24,0	30,2	38,0
30	1	3,6	4,5	5,7	8,6	9,0	14,3	18,1	22,7	28,7	36,1
	5	3,4	4,3	5,5	8,3	8,7	13,9	17,4	22,0	27,7	34,9
	10	3,4	4,3	5,4	8,2	8,6	13,6	17,2	21,7	27,3	34,4
	25	3,3	4,2	5,3	8,0	8,4	13,4	16,9	21,2	26,8	33,7
	50	3,3	4,1	5,2	7,9	8,3	13,2	16,6	20,9	26,4	33,2
	100	3,2	4,1	5,1	7,8	8,2	13,0	16,4	20,6	26,0	32,7
40	1	3,1	3,9	4,9	7,4	7,8	12,3	15,5	19,6	24,6	31,0
	5	2,9	3,7	4,7	7,1	7,5	11,9	15,0	18,9	23,8	29,9
	10	2,9	3,7	4,6	7,0	7,4	11,7	14,7	18,6	23,4	29,5
	25	2,8	3,6	4,5	6,9	7,2	11,5	14,4	18,2	22,9	28,9
	50	2,8	3,5	4,5	6,8	7,1	11,3	14,2	17,9	22,6	28,4
	100	2,8	3,5	4,4	6,7	7,0	11,1	14,0	17,6	22,2	28,0
50	1	2,6	3,3	4,2	6,3	6,6	10,5	13,3	16,7	21,0	26,5
	5	2,5	3,2	4,0	6,1	6,4	10,1	12,8	16,1	20,3	25,5
	10	2,5	3,1	3,9	6,0	6,3	10,0	12,6	15,8	19,9	25,1
	25	2,4	3,0	3,8	5,8	6,1	9,7	12,3	15,5	19,5	24,6
	50	2,4	3,0	3,8	5,7	6,0	9,6	12,1	15,2	19,2	24,2
	100	2,3	2,9	3,7	5,7	5,9	9,4	11,9	15,0	18,9	23,8

Temperature °C	Years of service	Pipe series S									
		20	16	12,5	8,3	8	5	4	3,2	2,5	2
		Diameter/Wall thicknesses SDR									
		41	33	26	17,6	17	11	9	7,4	6	5
		Allowable operation pressure bar									
60	1	2,2	2,8	3,5	5,3	5,6	8,9	11,2	14,2	17,8	22,5
	5	2,1	2,7	3,4	5,1	5,4	8,6	10,8	13,6	17,1	21,6
	10	2,1	2,6	3,3	5,0	5,3	8,4	10,6	13,4	16,8	21,2
	25	2,0	2,6	3,2	4,9	5,2	8,2	10,4	13,1	16,5	20,7
	50	2,0	2,5	3,2	4,8	5,1	8,1	10,2	12,8	16,2	20,4
70	1	1,8	2,3	3,0	4,5	4,7	7,5	9,4	11,9	15,0	18,9
	5	1,8	2,2	2,8	4,3	4,5	7,2	9,1	11,4	14,4	18,1
	10	1,7	2,2	2,8	4,2	4,4	7,0	8,9	11,2	14,1	17,8
	25	1,7	2,1	2,7	4,1	4,3	6,9	8,7	10,9	13,8	17,4
	50	1,7	2,1	2,7	4,0	4,2	6,8	8,5	10,7	13,5	17,0
80	1	1,5	1,9	2,5	3,7	3,9	6,2	7,9	9,9	12,5	15,8
	5	1,5	1,9	2,3	3,6	3,7	6,0	7,5	9,5	12,0	15,1
	10	1,4	1,8	2,3	3,5	3,7	5,9	7,4	9,3	11,7	14,8
	25	1,4	1,8	2,2	3,4	3,6	5,7	7,2	9,1	11,4	14,4
95	1	1,1	1,4	1,8	2,8	2,9	4,7	5,9	7,4	9,4	11,8
	5	1,1	1,4	1,7	2,6	2,8	4,4	5,6	7,1	8,9	11,2
	(10) ^a	(1,1)	(1,3)	(1,7)	(2,6)	(2,7)	(4,3)	(5,5)	(6,9)	(8,7)	(11,0)

^{a)} The values between parentheses apply in cases where it can be demonstrated that the test was carried out for more than a year at 110° C

1.14 Table of chemical resistance for PP

Environment	Conc. %	Temperature		
		20° C	60° C	100° C
Acetic acid (glacial)	97	A	B (80° C)	-
Acetic acid	50	A	A (80° C)	-
Acetic acid	40	A	-	-
Acetic acid	10	A	A	-
Acetone	100	A	A	-
Acetophenone	100	B	B	-
Acriflavine (2 % solution in H ₂ O)	2	A	A	- (80° C)
Acrylic emulsions		A	A	-
Aluminum chloride		A	A	-
Aluminum fluoride		A	A	-
Aluminum sulfate		A	A	-
Alums (all types)		A	A	-
Ammonia (aqueous)	30	A	-	-
Ammonia gas (dry)		A	A	-
Ammonium carbonate	Satd.	A	A	-
Ammonium chloride	Satd.	A	A	-
Ammonium fluoride	20	A	A	-
Ammonium hydroxide	10	A	A	-
Ammonium metaphosphate	Satd.	A	A	-
Ammonium nitrate	Satd.	A	A	-
Ammonium persulfate	Satd.	A	A	-
Ammonium sulfate	Satd.	A	A	-
Ammonium sulfide	Satd.	A	A	-
Ammonium thiocyanate	Satd.	A	A	-
Amyl acetate	100	B	C	-
Amyl alcohol	100	A	B	-
Amyl chloride	100	C	C	-
Aniline	100	A	A	-
Anisole	100	B	B	-
Antimony chloride		A	A	-
Aviation fuel (115/145 octane)	100	B	C	-
Aviation turbine fuel	100	B	C	-
Barium carbonate	Satd.	A	A	-
Barium chloride	Satd.	A	A	-
Barium hydroxide		A	A	-
Barium sulfate	Satd.	A	A	-
Barium sulfide	Satd.	A	A	-
Beer		A	A	-
Benzene	100	B	C	C
Benzoic acid	A	A	-	-
Benzyl alcohol		A	A (80° C)	-
Bismuth carbonate	Satd.	A	A	-
Borax		A	A	-
Boric acid		A	A	-
Brine	Satd.	A	A	-
Bromine liquid	100	D	-	-
Bromine water	(a)	C	-	-
Butyl acetate	100	C	C	-
Butyl alcohol	100	A	A	-
Calcium carbonate	Stad.	A	A	-
Calcium chlorate	Satd.	A	A	-
Calcium chloride	50	A	A	-
Calcium hydroxide		A	A	-
Calcium hypochlorite bleach	20(a)	A	B	-
Calcium nitrate		A	A	-
Calcium phosphate	50	A	-	-

Environment	Conc. %	Temperature		
		20° C	60° C	100° C
Calcium sulfate		A	A	-
Calcium sulfite		A	A	-
Carbon dioxide (dry)		A	A	-
Carbon dioxide (wet)		A	A	-
Carbon disulfide	100	B	C	-
Carbon monoxide		A	A	-
Carbon tetrachloride	100	C	C	C
Carbonic acid		A	A	-
Castor oil		A	-	-
Cetyl alcohol	100	A	-	-
Chlorine (gas)	100	D	D	-
Chlorobenzene	100	C	C	-
Chloroform	100	C	D	D
Chlorosulfonic acid	100	D	D	D
Chrome alum		A	A	-
Chromic acid	80(a)	A	-	-
Chromic acid	50(a)	A	A	-
Chromic acid	10(a)	A	A	-
Chromic/sulfuric acid		D	D	-
Cider		A	A	-
Citric acid	10	A	A	-
Copper chloride	Satd.	A	A	-
Copper cyanide	Satd.	A	A	-
Copper fluoride	Satd.	A	A	-
Copper nitrate	Satd.	A	A	-
Copper sulfate	Satd.	A	A	-
Cottonseed oil		A	A	-
Cuprous chloride	Satd.	A	A	-
Cyclohexanol	100	A	B	-
Cyclohexanone	100	B	C	-
Decalin	100	C	C	C
Detergents	2	A	A	A
Developers (photographic)		A	A	-
Dibutyl phthalate	100	A	B	D
Dichloroethylene	100	A	-	-
Diethanolamine	100	A	A	-
Diisooctyl phthalate	100	A	A	-
Emulsifiers		A	A	-
Ethanolamine	100	A	A	-
Ethyl acetate	100	B	B	-
Ethyl alcohol	96	A	A (80° C)	-
Ethyl chloride	100	C	C	-
Ethylene dichloride	100	B	-	-
Ethylene glycol		A	A	-
Ethylene oxide	100	B (10° C)	-	-
Ethyl ether	100	B	-	-
Fatty acids (C ₆)	100	A	A	-
Ferric chloride	Satd.	A	A	-
Ferric nitrate	Satd.	A	A	-
Ferric sulfate	Satd.	A	A	-
Ferrous chloride	Satd.	A	A	-
Ferrous sulfate	Satd.	A	A	-
Fluorosilicic acid		A	A	-
Formaldehyde	40	A	A	-
Formic acid	100	A	-	-
Formic acid	10	A	A	-
Fructose		A	A	-
Fruit juices		A	A	-

Environment	Conc. %	Temperature		
		20° C	60° C	100° C
Furfural	100	C	C	-
Gas liquor		C	-	-
Gasoline	100	B	C	C
Gearbox oil	100	A	B	-
Gelatin		A	A	-
Glucose	20	A	A	-
Glycerin	100	A	A	A
Glycol		A	A	-
Hexane	100	A	B	-
Hydrobromic acid	50(a)	A	A	-
Hydrobromic acid	30(a)	A	B	D
Hydrobromic acid	20	A	A (80° C)	-
Hydrobromic acid	10	A	A (80° C)	B
Hydrobromic acid	2	A	A	A
50-50 HCl-HNO ₃	(a)	B	D (80° C)	-
Hydrofluoric acid	40	A	-	-
Hydrofluoric acid	60(a)	A	A (40° C)	-
Hydrogen chloride gas (dry)	100	A	A	-
Hydrogen peroxide	30	A	-	D
Hydrogen peroxide	10	A	B	-
Hydrogen peroxide	3	A	-	-
Hydrogen sulfide		A	A	-
Hydroquinone		A	A	-
Inks		A	A	-
Iodine tincture		A	-	-
Isooctane	100	C	C	-
Isopropyl alcohol	100	A	A	-
Ketones		A	-	-
Lactic acid	20	A	A	-
Lanolin	100	A	A	-
Lead acetate	Satd.	A	A	-
Linseed oil	100	A	A	-
Lubricating oil	100	A	B	-
Magenta dye (aqueous solution)	2	A	A some staining	-
Magnesium carbonate	Satd.	A	A	-
Magnesium chloride	Satd.	A	A	-
Magnesium hydroxide	Satd.	A	A	-
Magnesium nitrate	Satd.	A	A	-
Magnesium sulfate	Satd.	A	A	-
Magnesium sulfite	Satd.	A	A	-
Meat juices		A	A	-
Mercuric chloride	40	A	A	-
Mercuric cyanide	Satd.	A	A	-
Mercurous nitrate	Satd.	A	A	-
Mercury	100	A	A	-
Methyl alcohol	100	A	A	-
Methylene chloride	100	A	-	-
Methyl ethyl ketone	100	A	B	-
Milk and its products		A	A	A
Mineral oil	100	A	B	-
Molasses		A	A	-
Motor oil	100	A	B	-
Naphthalene	100	A	A	A
Nickel chloride	Satd.	A	A	-
Nickel nitrate	Satd.	A	A	-
Nickel sulfate	Satd.	A	A	-
Nitric acid	fuming	D	D	D

Environment	Conc. %	Temperature		
		20° C	60° C	100° C
Nitric acid	70(a)	C	D	-
Nitric acid	60	A	D (80° C)	-
Nitric acid	10	A	A	A
50-50 HNO ₃ -HCl	(a)	B	D (80° C)	-
50-50 HNO ₃ -H ₂ SO ₄	(a)	C	D (80° C)	-
Nitrobenzene	100	A	A	-
Oleic acid		A	B	-
Oleum			-	D
Olive oil	100	A	A	-
Oxalic acid (aqueous)	50	A	B	-
Paraffin	100	A	B	-
Paraffin wax	100	A	A	-
Petrol	100	B	C	-
Petroleum ether (boiling point 100°-140° C)	100	C	C	-
Phenol	100	A	A	-
Phosphoric acid	95	A	A	-
Plating solutions, brass		A	A	-
Plating solutions, cadmium		A	A	-
Plating solutions, chromium		A	A	-
Plating solutions, copper		A	A	-
Plating solutions, gold		A	A	-
Plating solutions, indium		A	A	-
Plating solutions, lead		A	A	-
Plating solutions, nickel		A	A	-
Plating solutions, rhodium		A	A	-
Plating solutions, silver		A	A	-
Plating solutions, tin		A	A	-
Plating solutions, zinc		A	A	-
Potassium bicarbonate	Satd.	A	A	-
Potassium borate	1	A	A	-
Potassium bromate	10	A	A	-
Potassium bromide	Satd.	A	A	-
Potassium carbonate	Satd.	A	A	-
Potassium chlorate	Satd.	A	A	-
Potassium chloride	Satd.	A	A	-
Potassium chromate	40	A	A	-
Potassium cyanide	Satd.	A	A	-
Potassium dichromate	40	A	A	-
Potassium ferri-/ferrocyanide		A	A	-
Potassium fluoride		A	A	-
Potassium hydroxide	50	A	A	-
Potassium hydroxide	10	A	A	A
Potassium nitrate	Satd.	A	A	-
Potassium perborate	Satd.	A	A	-
Potassium perchlorate	10	A	A	-
Potassium permanganate	20	A	A	-
Potassium sulfate		A	A	-
Potassium sulfide		A	A	-
Potassium sulfite		A	A	-
Propyl alcohol	100	A	A	-
Pyridine	100	A	-	-
Silicone oil	100	A	A	-
Soap solution (concentrated)		A	A	-
Sodium acetate		A	A	-
Sodium bicarbonate	Satd.	A	A	-
Sodium bisulfate	Satd.	A	A	-
Sodium bisulfite	Satd.	A	A	-

Environment	Conc. %	Temperature		
		20° C	60° C	100° C
Sodium borate		A	A	-
Sodium bromide oil solution		A	A	-
Sodium carbonate	Satd.	A	A	-
Sodium chlorate	Satd.	A	A	-
Sodium chloride	Satd.	A	A	A
Sodium chlorite	2	A	A (80° C)	-
Sodium chlorite	5	A (80° C)	A	-
Sodium chlorite	10	A (80° C)	A	-
Sodium chlorite	20	A (80° C)	A	-
Sodium cyanide	Satd.	A	A	-
Sodium dichromate	Satd.	A	A	-
Sodium ferricyanide	Satd.	A	A	-
Sodium ferrocyanide	Satd.	A	A	-
Sodium fluoride	Satd.	A	A	-
Sodium hydroxide	50	A	A	-
Sodium hydroxide	10	A	A	A
Sodium hypochlorite	20	A	B	B
Sodium nitrate		A	A	-
Sodium nitrite		A	A	-
Sodium silicate		A	A	-
Sodium sulfate	Satd.	A	A	-
Sodium sulfide	25	A	A	-
Sodium sulfite	Satd.	A	A	-
Stannic chloride	Satd.	A	A	-
Stannous chloride	Satd.	A	A	-
Starch		A	A	-
Sugars and syrups		A	A	-
Sulfamic acid		A	A (80° C)	-
Sulfates of Calcium and magnesium	Satd.	A	A	-
Sulfates of potassium and sodium	Satd.	A	A	-
Sulfur		A	A	-
Sulfuric acid	98(a)	C	-	D
Sulfuric acid	60	A	B (80° C)	-
Sulfuric acid	50	A	B	-
Sulfuric acid	10	A	A	A
50-50 H ₂ SO ₄ /HNO ₃	(a)	C	D (80° C)	-
Tallow		A	A	-
Tannic acid	10	A	A	-
Tartaric acid		A	A	-
Tetrahydrofuran	100	C	C	C
Tetralin	100	C	C	C
Toluene	100	C	C	-
Transformer oil	100	A	C	-
Trichloroacetic acid	10	A	A	-
Trichloroethylene	100	A	A (80° C)	-
Turpentine	100	C	C	C
Urea		A	A	-
Urine		A	A	-
Water (distilled, soft, hard and vapor)		A	A	A
Wet chlorine gas		-	D (70° C)	-
Whiskey		A	A	A
White Paraffin	100	A	B (80° C)	-
White spirit	100	B	C	-
Wines		A	A	-
Xylene	100	C	C	C

Environment	Conc. %	Temperature		
		20° C	60° C	100° C
Yeast		A	A	-
Zinc chloride	Satd.	A	A	-
Zinc oxide		A	A	-
Zinc sulfate	Satd.	A	A	-

(a) May produce cracking in material under stress

This chart shows the chemical resistance of polypropylene resin under static conditions and not under pressure.

Note: the user is advised to carry out his/her own tests to determine the suitability of polypropylene in a particular environment.

A = Negligible effect

The material should be suitable for all applications where these environmental conditions exist.

B = Limited absorption or attack

The material should be suitable for most applications but the user is advised to carry out his/her own tests to determine the suitability of polypropylene in a particular environment.

C = Extensive absorption and/or rapid permeation

The material should be suitable for applications where only intermittent service is involved, or where the swelling produced has no detrimental effect on the part. The user should carry out his/her own tests to determine the suitability of polypropylene in a particular environment.

D = Extensive attack

The specimen dissolves or disintegrates. Polypropylene is not recommended.

Note:

This table should be used only as a guide. Polypropylene subjected to mechanical stress may behave different and show different result. For any doubts we recommend to contact the K-Aqua Customer Service.

1.15 Material Properties Transition/Threaded Parts

Transition/threaded parts:

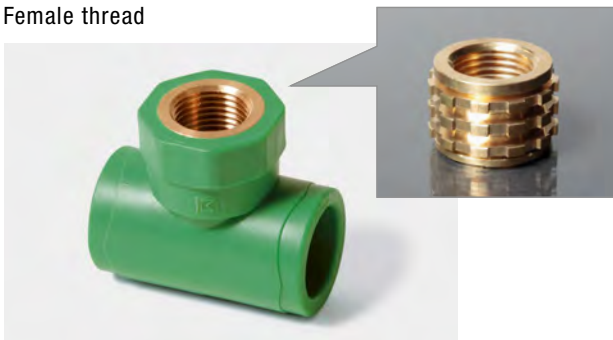


Male thread



R = Conical male thread acc. to ISO 7 / DIN EN 10226

Female thread



Rp = Cylindrical female thread acc. to ISO 7 / DIN EN 10226



Rc = Conical female thread acc. to ISO 7 / DIN EN 10226

Note: Threaded parts acc. to ISO 7 / DIN EN 10226 additional seal recommended (e.g. PTFE tape for sealing)

Threaded parts acc. to EN – ISO 228 additional seal required (gasket or O-ring)

The K-Aqua threaded parts have a special designed profile that avoids the turning of the insert once it is injected!





2

Quality Assurance

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2.1 Quality Assurance Systems

Various national and international authorities and institutions provide regulatory supervision on quality control measurements. The external supervision for pipes & fittings for potable cold & hot water system certificates from abroad are given by SKZ or IMA, who is

authorized by the DVGW (German institute for gas and water) as controlling organization. External supervision aims to audit **quality assurance systems** by which test equipments and procedures verify installation standards, as well as hygienic and toxicity tests.

2.2 Standards

ISO 15874	Plastic piping system for hot and cold water installations – Polypropylene (PP)
Part 1	General
Part 2	Pipes
Part 3	Fittings
Part 5	Fitness for purpose of the system
Part 7	Recommendations for the assessment of conformity
DIN 8077	Polypropylene (PP) pipes – PP-H, PP-B, PP-R, PP-RCT – dimensions
DIN 8078	Polypropylene (PP) pipes – PP-H, PP-B, PP-R, PP-RCT – general quality requirements and testing
DIN 16962	Pipe joints and components for pressure systems of Polypropylene (PP)
DIN 16928	Pipes of Thermoplastic Materials, Pipe Joints, Elements for Pipes, Laying; General Directions
DVGW	
W544	Plastic pipe systems for drinking water – pipes
W534	Plastic pipe systems for drinking water – fittings
W270	Assesment of microbiological growth
DVS 2207-11	Socket welding, butt-welding and electrofusion welding of Polypropylene pipes and components
DVS 2208-1	Machines and tools for the welding of Polypropylene pipes and components
ISO 7 / EN 10226	Pipe threads where pressure tight joints are done on the thread
EN - ISO 228	Pipe threads where pressure tight joints are not made on the threads

2.3 System Control

The production of **K-Aqua** superior German quality piping system calls for the regulation and control of all areas of the operations. All results are documented and archived:

- Testing and accepting incoming goods
- Process control
- In-process inspection
- Final inspection tests

Regulations for the quality control of **K-Aqua** sanitary piping system include:

- DIN-guidelines
- DVGW working sheets
- Supervisory Regulations of IMA Dresden

These standards and guidelines feature the minimum requirements for internal control.

Conformance to the superior quality standards is verified through independent authorities, by internal audits and laboratory tests.

K-Aqua highest quality standard is controlled by IMA Dresden.

KWT GmbH is a highly qualified and experienced manufacturer in extrusion and injection moulding.

This is reflected in our internal quality standards and procedures, which are illustrated by the constant quality of our products.

2.4 Internal Control

A team of highly trained and qualified QC engineers, equipped with a state of art laboratory, ensures that all tests are carried out in compliance with our quality control policies, which includes:

- Testing all raw material
- Measuring and inspecting our production equipments
- Auditing our production procedures
- A final inspection for the quality of our finished products
- All internal quality audits are documented and archived in accordance with the highest standard quality control policies

2.5 Testing and Accepting Incoming Goods

All incoming goods are carefully inspected, to ensure that the raw material conforms with the set requirements. Goods that have not been tested do not get released for production.

The incoming raw material is tested according to DIN EN ISO 1133.

2.6 In-Process Inspection

The quality plan requires that all inspections are carried out at the beginning as well as during production. As production starts all relevant data are checked by the Quality Assurance Department. Pre-production samples are tested by the laboratory technicians for:

- Surface finish
- Dimensional accuracy
- Data from extrusion and injection moulding machines
- The product is only released if optimal test results are achieved

2.7 Process Control

We have an extensive quality control process in the field of extrusion and injection. This enables constant observation and control of production. This ensures that only perfect quality products are packed and stored. All data received during production is studied and analyzed, in details.

2.8 Final inspection

QC requires that inspections and tests are carried out on all finished products. The results are all documented. Finished products are only released to storage when all tests and inspections have conformed to the authorized procedures and specifications. The final inspection test includes a time laps procedure. This measures the usability of the products in their field of application, as well as removing production weaknesses. These inspections are the method for quality assurance during production and for design tests. The results document the system quality and serve to optimize the manufacturing process.

The final inspection covers the following procedures:

- Dimensional control
- Surface finish
- Measurement of the melt flow index
- Impact bending test
- Heat reversion test
- Homogeneity of the material
- Behavior under long period of stress

2.9 External Control

External supervision consists of measuring the fixed scope at fixed intervals. The respective supervising institutions appoint the appropriate authorized inspection organization to carry out external supervision. Inspection includes:

- External tests of products
- Internal audit of **K-Aqua** quality assurance system and test procedures
- Calibration of the test equipment
- Hygienic and toxicity tests

2.10 Guarantee

When **K-Aqua** components are correctly installed and used, the trouble free and satisfactory operation of the installed system can be expected on the long run. Should a fault ever occur, however, our after sales service is ready to help you.

Purchases of **K-Aqua** components are covered by the guarantee rights stipulated in the purchase contract with your dealer. In accordance with the provisions of law (product liability) the manufacturer is responsible within the scope of the German product liability legislation for damage and injury caused to objects and persons respectively by faulty **K-Aqua** components. A guarantee of 10 years is granted on all the pipes and fittings manufactured and distributed by **K-Aqua**, within the scope of the German product liability legislation and the manufacturer's liability insurance.

Product liability insurance

For the coverage of our liability for product damage (personal injury and material damage and especially dismantling and installation costs), we have concluded an extended business and product liability insurance.

2.11 Certificates/External supervision

TEST CERTIFICATE
According to the Guidelines for Hygienic Assessment of Organic Materials in Contact with Drinking Water (KTW Guideline, 7 March 2016)

Kessel Wassertechnologie GmbH
Auweg 3
35647 Waldsolms-Brandenberndorf
Germany

Test Certificate No.: 17-K-111
Issued: 12.08.2017 till 12.08.2022
Product: K-Aqua Filter Pipe made from PP-RCT (PP-R) (Pipesystem)
Trademark: K-Aqua Pipe 25 x 3,5 mm

The product K-Aqua Filter Pipe made from PP-RCT (PP-R) (Pipesystem), 25 x 3,5 mm has been tested in accordance with the Guidelines on the hygienic assessment of organic materials in contact with drinking water of the German Environment Agency and has passed the test (according to test report 03101217-0002 Rev1 dated 12.08.2017) for the following application and temperature range:

Product group	Test water	Drinking water	Test water
	20-70 °C	10-20 °C	10-20 °C
Pipes and their joints (20 x 30 mm (vertical and horizontal))	passed	passed	passed
Accessories			
Pipe 80 mm x 4 mm x 300 mm (distribution pipes)			
Pipe 20 x 3,5 mm (single connection and fittings)			
Fittings, endfitting 20 x 3,5 mm			
Fittings, endfitting 80 mm x 3,5 mm x 300 mm			
Fittings, endfitting 20 x 3,5 mm			
Component of fittings, endfitting 20 x 3,5 mm			
Component of fittings, endfitting 80 mm x 3,5 mm			
Component of fittings, endfitting 80 mm x 3,5 mm			

Compliance to the requirements of the KTW Guideline, 7 March 2016, for pipes, fittings, or filter manufacturing processes. This test certificate is not applicable to other products.

11/20 Kessel Wassertechnologie GmbH

Signature: [Signature]

Stamp: [Stamp]

DVGW

IMA DRESDEN

SKZ

kiwa

**TÜVRheinland®
DIN CERTCO**

Certificate
The SKZ-Cert GmbH Zertifizierungen certifies to the company

Kessel Wassertechnologie GmbH
Auweg 3
35647 Waldsolms-Brandenberndorf

the introduction and application of a management system according to
ISO 9001:2015
ISO 14001:2015
ISO 50001:2018

scope:
Manufacturing and selling of pipes, fittings and accessories for water supply and the manufacturing, trading and selling of pipes and fittings for drainage

The proof has been furnished by an audit.
This certificate is valid from **05.10.2022** to **22.11.2024**
Certificate No.: **000522 EQU77 N**

DVGW type examination certificate
DVGW-Baumusterprüfzertifikat

Field of Application: products of water supply
Produkte der Wasserversorgung

Owner of Certificate: Kessel Wassertechnologie GmbH
Auweg 3, D-35647 Waldsolms-Brandenberndorf

Distributor: Kessel Wassertechnologie GmbH
Auweg 3, D-35647 Waldsolms-Brandenberndorf

Product Category: Installation systems and system joints: pipe joint for pipes in drinking water installation systems (8511)

Product Description: welding joints made of plastic (PP-R) for pipes made of plastic (PP-R) for the drinking water installation, type: S-RS

Model: K-Aqua

Test Reports: laboratory control test: V090/20-2 from 26.11.2020 (IMA)
supplement test: V090/18-2 from 26.10.2018 (IMA)
type testing: BA0718 from 26.12.2019 (IMA)
KTW testing: KR 094/20 from 18.06.2020 (TZW)
hygienic testing: MO 12917 from 02.11.2017 (TZW)
KTW testing: 031521137-0001 from 17.02.2016 (IMA)
hygienic testing: MO 17016 from 25.09.2016 (TZW)

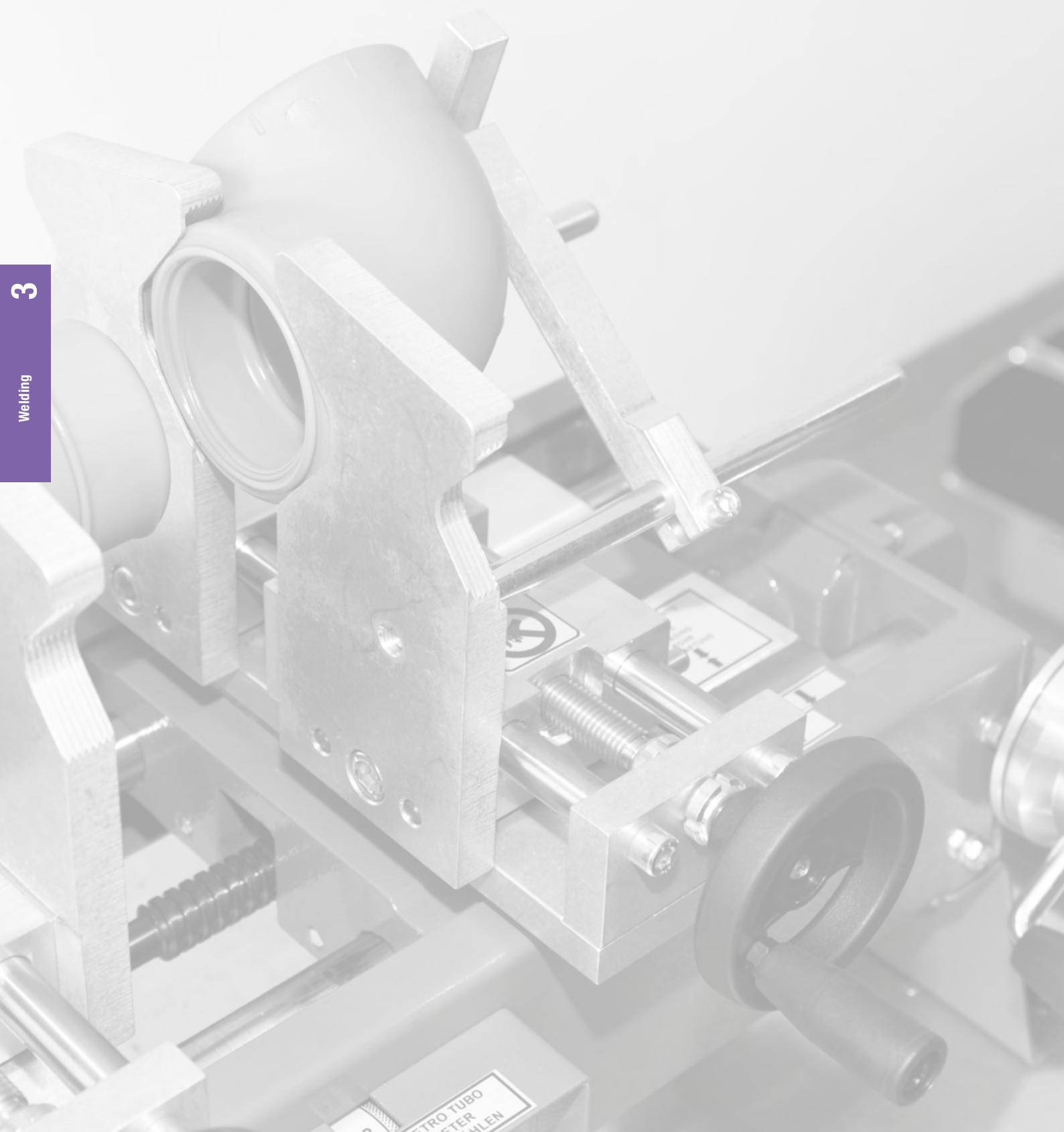
Test Basis: DVGW W 534-01 (01.07.2015)
USA KTW 02 (03.2016)
DVGW W 276 (01.11.2007)

Date of Expiry / File No.: 26.12.2025 / 20-0977-WNV

11.01.2021 LEA-1/2
Date, location, time, name of official or official
Date, location, time, name of official or official
DVGW CERT Center is an approved body for issuing certificates to DIN CERTCO
DIN CERTCO is a certification body for issuing certificates to DIN CERTCO
DIN CERTCO is a certification body for issuing certificates to DIN CERTCO
DIN CERTCO is a certification body for issuing certificates to DIN CERTCO

Product liability insurance:

For the coverage of our liability for product damage (personal injury and material damage and especially dismantling and installation costs), we have conducted an external business and product liability insurance which is valid for 10 years.





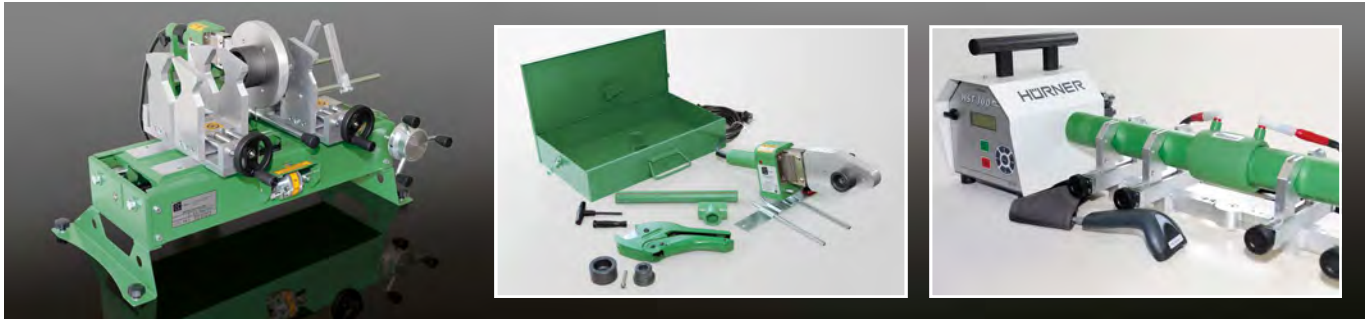
3

Welding

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3.1 Welding Tools



3.2 Hand Welding Machine Type AQ98063/AQ98032

Our polyfusion devices are delivered as a kit in a metal box. The thermoelement has an electronical thermostat which controls the temperature automatically. The polyfusion device runs at 230 V. Depending on the type that you need we provide the kits in two versions: Type AQ98032 consists of Polywelder and bushes dim 20-32 Type AQ98063 consists of Polywelder and bushes dim 20-63

3.3 Welding Machine Type AQ988125

These are welding devices for on-site use, suitable for polyfusion socket welding pipes and fittings according to the regulations currently in force. These devices consist of:

- A self-centering clamp system, which achieves perfect welding
- A manual selector switch to ensure correct welding
- A thermo-element which controls temperature electronically
- A series of coated bushes with non-stick material
- A tripod to support the pipe
- A transport case

Note: This device requires a 220 V power supply and features 700 W power. The presses and their accessories are supplied in a metal box.

3.4 Welding Procedure

The **K-Aqua** pipework is coupled by socket fusion. The welded pipes and fittings have a longitudinally overlapping connection. The heating of the pipes' ends and fitting faucets is done by a heating element with bushes. After the necessary welding temperature is reached, the joining process is done. The pipe and fitting faucets diameters, as well as the respective heating bush diameters, are matched to build up the necessary pressure during the jointing process. The heating element is electrically heated. It complies with DVS Directive 2208 part 1 in construction and accuracy.

3.4.1 Preparations

Cut the pipe using a pipe cutter that is suitable for plastic pipes (Fig. 1). The pipe end and fitting faucet, to be thoroughly cleaned with absorbent paper. Second, marking the bush depth on the pipe (Fig. 2) while bringing the heating element to 260° C (remember that the temperature tolerance is $\pm 10^\circ$ C) by checking the integrated thermometer on the heating element. Otherwise the temperature must be controlled and measured by an appropriate measuring device.

Note: Must not start heating the joint parts before reaching the set temperature of 260° C. Also cleaning the mandrel and bush before each use.

3.4.2 Welding

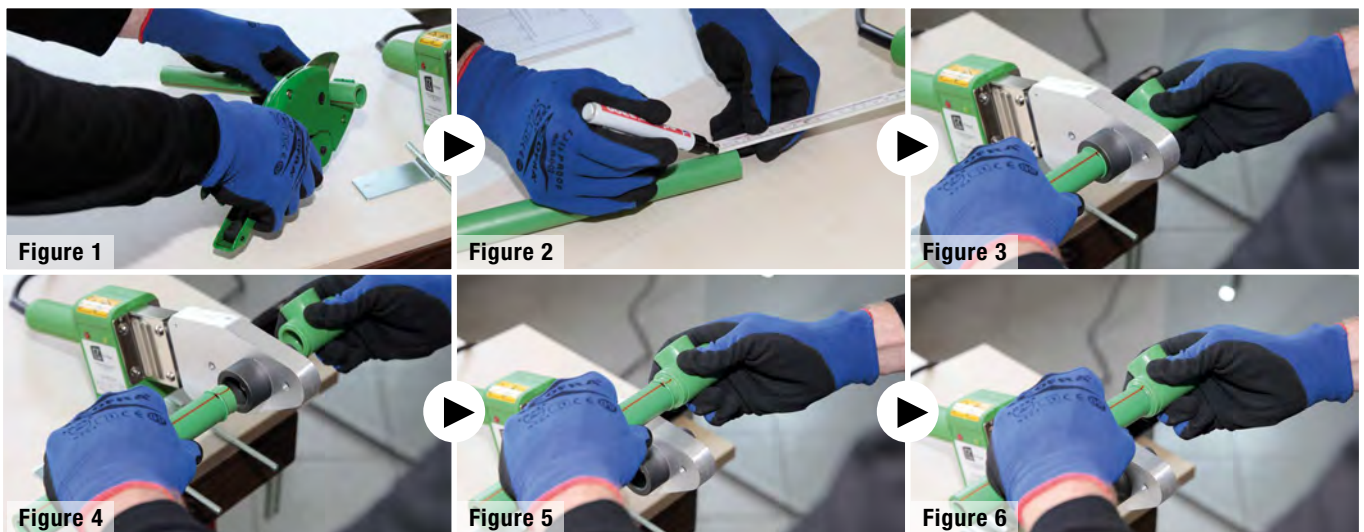
Starting with pushing the pipe and fitting ends, quickly and axially, up to the stop of the mandrel and the marked insertion depth (Fig. 3), respectively fast without torsion. The heating of the joint faces is done according to the table A. When the heating period is up, the pipe and fitting ends are pulled abruptly from the heating element (Fig. 4) and joined immediately without torsion, minding the correct insertion depth (Fig. 5 and 6).

Note: We recommend fixing the two joint parts again for a certain time (the heating period). Do not expose the welded joint to mechanical stress until the cooling period is done.

Pipe Ø	Heating-up time (sec)	Processing time (sec)	Cooling-down time (min.)	Welding depth (mm)
20	5	4	2	14
25	7	4	2	15
32	8	6	4	17
40	12	6	4	18
50	18	6	4	20
63	24	8	6	26
75	30	8	6	29
90	40	8	6	32
110	50	10	8	35
125	60	10	8	41

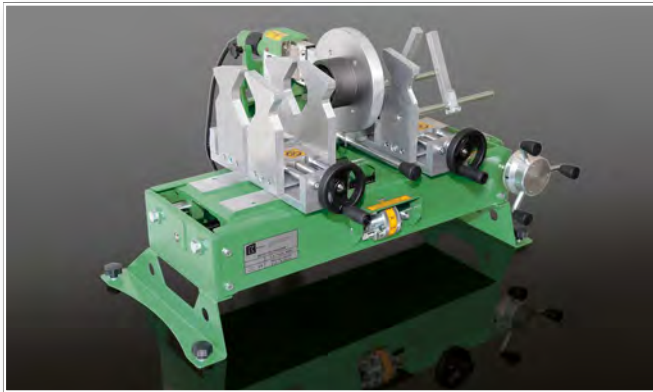
Table A

Hand welding - working steps welding together



Note: If welding is to be carried out outdoors when the temperature is below +5° C, the heating up time in acc. with DVS 2207 Part 11 should be increased by 50 %.

3.5 Socket Welding Machine Article AQ988125 - working steps welding together



Socket welding machine equipped with movement slides on which prismatic clamps are placed to allow an automatic self-centering of any kinds of pipes and fittings. The socket welding machine is delivered in a metal transport box.



Figure 1



Figure 2



Figure 3



Figure 4



Figure 5



Figure 6



Figure 7

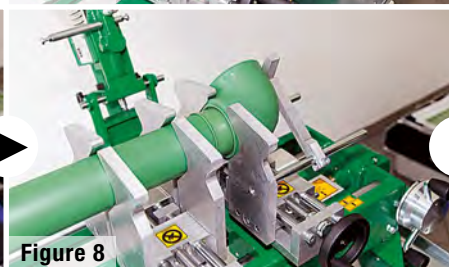


Figure 8

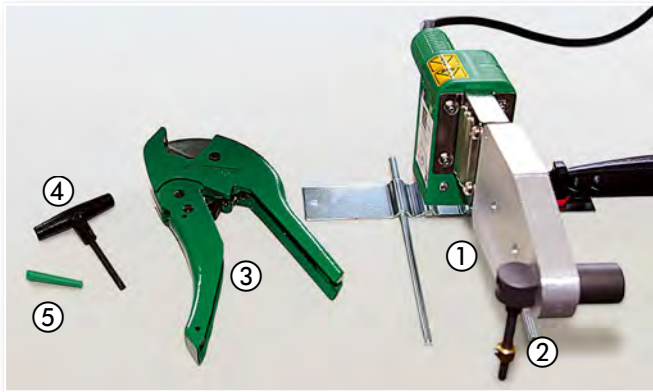


Figure 9

Select the pipe/fittings diameter (Fig. 1) and adjust the position of the slides (Fig. 2). Press the fitting into the clamping tool up to the stop and fix it (Fig. 3). Adjust the stop to hold the fitting. Align the pipe axially into the fitting (Fig. 4). Check the welding plate temperature and adjust it if necessary. Move the welding plate between pipe and fitting (Fig. 5). Slide the pipe and fitting at the same time into the heating tools (Fig. 6) up to the stop.

Hold this position for the heating-up time acc. to Table A (see page 41). After the heating-up time is reached move the slide back, move out the welding plate and join the pipe and fitting rapidly together (Fig. 7, 8, 9) moving the slide up to the stop. Remove the welding joint after the end of the cooling time from the clamping jaws.

3.6 Hole Repair in case of Damaged Pipe



In case a **K-Aqua** pipe is damaged, it is possible to be repaired, using a special tool, mounted on the polyfusion device with a special repair patch (type AQ5937/AQ59311).

Note: The repaired part can work again under pressure.

Tool set hole repair:

- | | |
|-----------------------------------|------------------|
| ① Hand welding machine | ④ Imbus key |
| ② Welding tool for repairing plug | ⑤ Repairing plug |
| ③ Pipe cutter | |



Figure 1

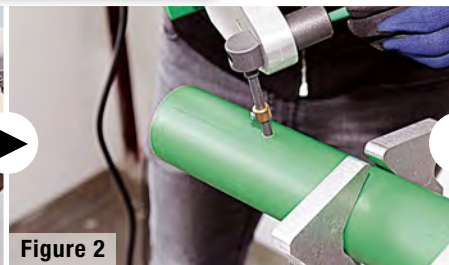


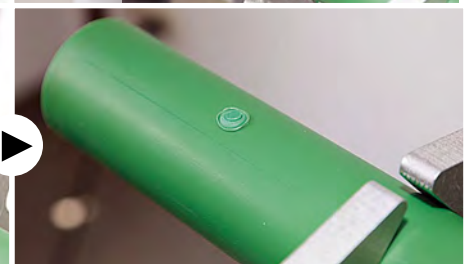
Figure 2



Figure 3



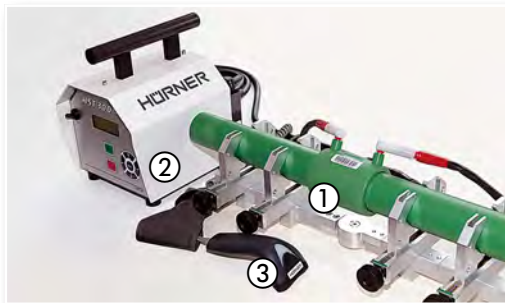
Figure 4



Mark the degree of the push-in depth (wall thickness) on the repair plug (Fig. 1). Distance tool to be fixed according to the wall thickness of the pipe and tighten the screw. Heat up the borehole and the welding plugs with the repair set for 15 seconds (Fig. 2).

After removing the welding device, set in the repair plug precisely without twisting it (Fig. 3). After a cooling time of 5 minutes, remove the protruding end of the repair plug (Fig. 4).

3.7 Electrofusion Welding with Electrofusion Machine Type AQ990



- ① Electrofusion socket
- ② Electrofusion machine
- ③ Scanner



Figure 1



Figure 2



Figure 3



Figure 4



Figure 5



Figure 6



Figure 7



Figure 8



Figure 9



Figure 10



Figure 11



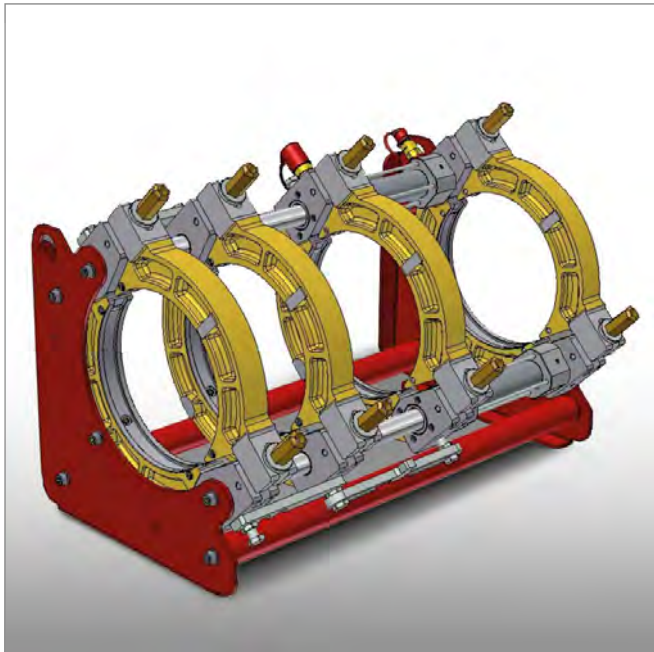
Figure 12

Cut the pipe properly in rectangularly position. Afterwards remove the oxidation layer in the welding area. Use a hand scraper (code AQ974) or a rotary scraper (Fig. 1). Once the surface is scraped properly (Fig. 2) clean the pipe surface and the inside coupler with absorbent, lint-free and non-dyed paper (Fig. 3). Mark the depth of the coupler (Fig. 4 and 5). Slide in the coupler up to the marked position on the pipe. It's recommended to secure the pipe against dislocation, e.g. with a pipe clamp (Fig. 6). Connect the cables to the contact pins of

the coupler (Fig. 7) and start welding process (Fig. 8). You can neither insert the datas manually acc. to the datas mentioned on the barcode label of the socket or you can use the barcode reader (Fig. 9). Afterwards proceed with the welding following the steps mentioned on the welding machine (Fig. 10 and 11). At the end of the welding cycle (Fig. 12) wait for the cooling time. After the cooling time you can stress the electrofusion joint to the permissible operation pressure.

3.8 Butt Welding

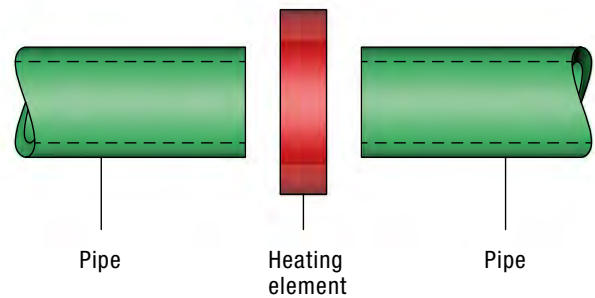
The welding machine part no. AQ989250 is suitable for butt welding of pipes and/or fittings made of PP-R / PP-RCT from dia = 90 mm up to dia = 250 mm



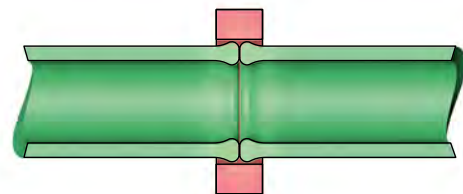
The butt welding machine with heating element part no. AQ989250 includes:

- Basic machine with movable slide
- Heating element
- Hydraulic aggregate
- Electrical planing tool
- Flexible hydraulic hoses
- Metal box for electrical heating elements and planing tool.

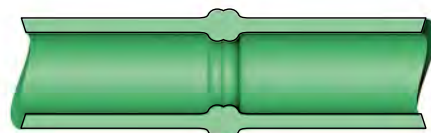
Preparation



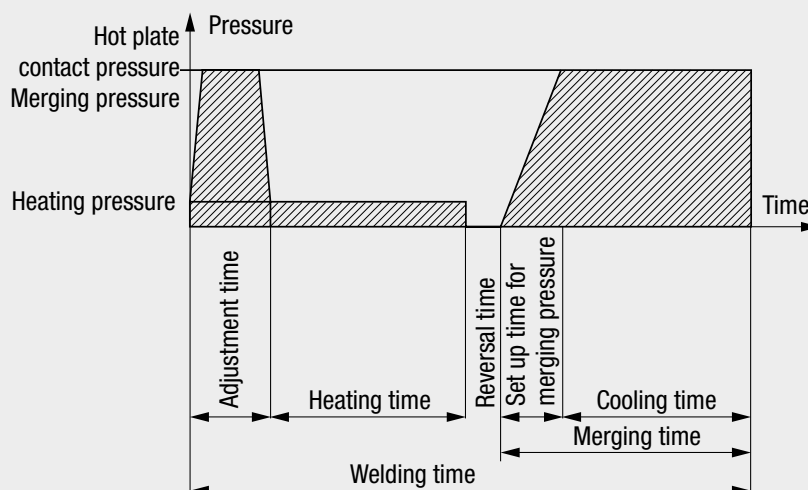
Heating up



Finished connection



Process steps in heated tool butt welding



Butt Fusion welding procedure

Create permissible working conditions.

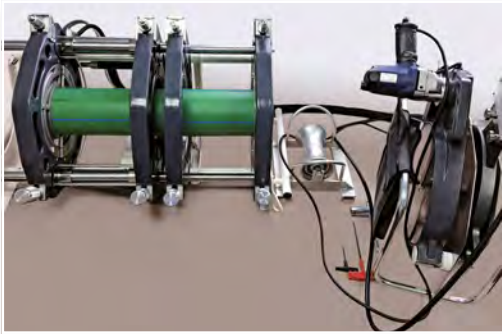


Figure 1



Figure 2

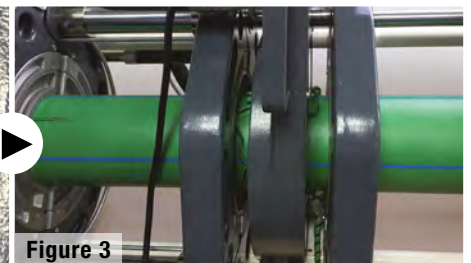


Figure 3

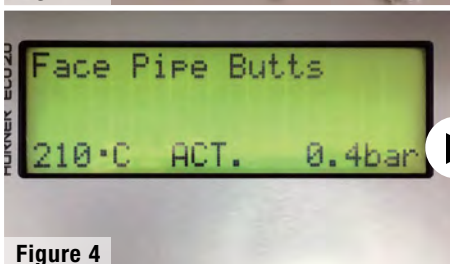


Figure 4

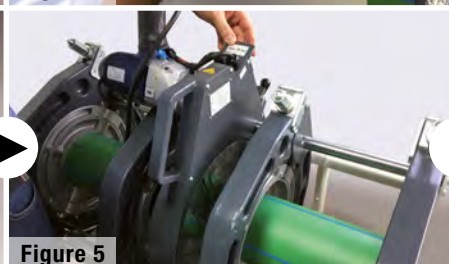


Figure 5

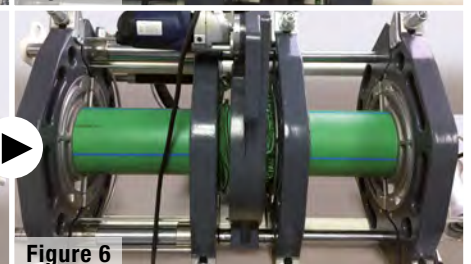


Figure 6



Figure 7



Figure 8

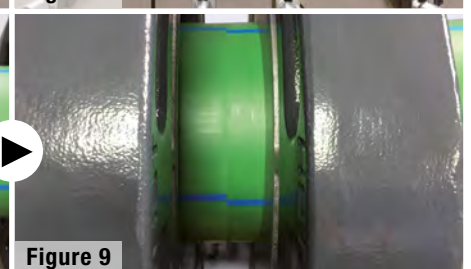


Figure 9

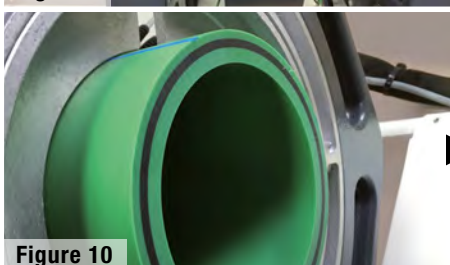


Figure 10

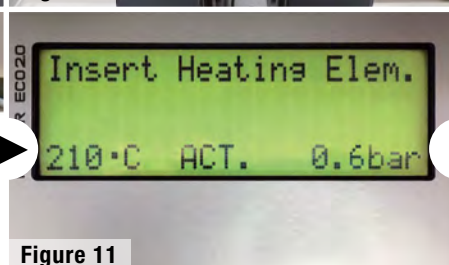


Figure 11



Figure 12



Figure 13



Figure 14

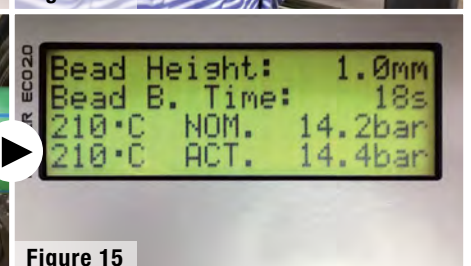


Figure 15

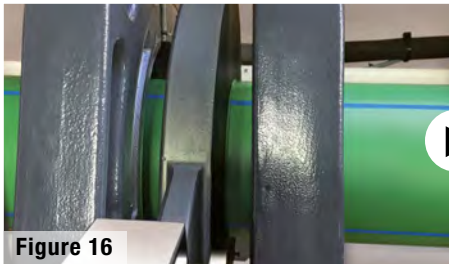


Figure 16



Figure 17

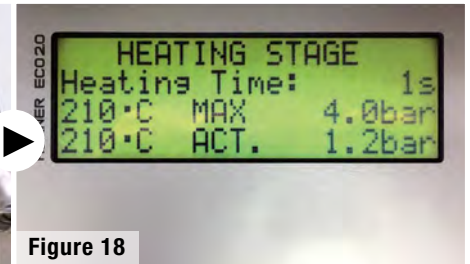


Figure 18

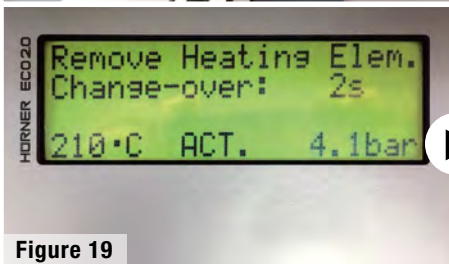


Figure 19



Figure 20

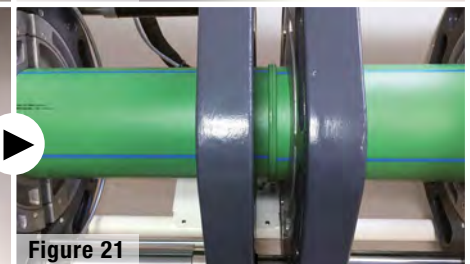


Figure 21



Figure 22

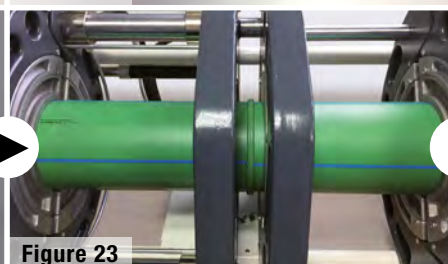


Figure 23



Figure 24



Figure 25

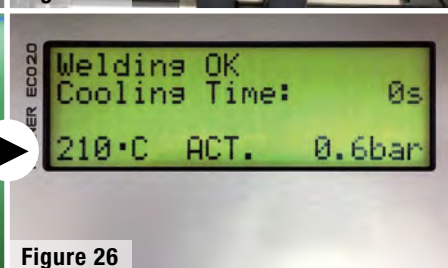


Figure 26

Align and clamp the parts to be welded, e. g. with dollies (Fig. 1). Clean the joining faces beyond the welding area with a cleaning agent with unused, absorbent, non-fraying and non-dyed paper. Machine the joining faces, e.g. in the case of pipes using a plane (Fig. 2, 3, 4). After planing the pipe ends take out the plane (Fig. 5) and remove the chips from the welding area without touching the joining faces (Fig. 6, 7). Check the plane parallelism by moving the joining faces together (max. gap width according to Table 1), (Fig. 8). Check the misalignment (max. $0.1 \times$ wall thickness), (Fig. 9, 10). Check the heated tool temperature ($210 \pm 10^\circ \text{C}$), (Fig. 11). Clean the heated tool with a cleaning agent with unused, absorbent, non-fraying and non-dyed paper and ensure extraction. Determine the movement pressure or the movement force before every welding operation and make a note of it on the welding record sheet. Determine the setting values for the alignment, heating-up and joining pressures. Stipulate the guide values

according to Table 2 (Fig. 12). Move the heated tool into the welding position (Fig. 13). Align the faces to the heated tool until a bead arises (according to Table 2) (Fig. 14, 15). Heating-up at a reduced pressure $\leq 0.01 \text{ N/mm}^2$, heating-up time according to Table 2 (Fig. 16). At the end of the heating-up, detach the joining faces to be welded from the heated tool and move this out of the welding position (Fig. 17, 18, 19). Within the changeover time, quickly move together the faces to be welded until they almost touch. The faces must come into contact at a speed of nearly zero. Immediately afterwards, build up the joining pressure with a linear rise in the build-up time (Table 2), (Fig. 20). A bead must exist after the joining at a pressure of 0.10 N/mm^2 . According to Figure 1, K must be > 0 at every point (Fig. 21, 22). Cooling under the joining pressure according to Table 2 (Fig. 23). Unclamping of the welded parts when the cooling time has elapsed (Fig. 24, 25). Complete the welding record sheet (can be provided upon request), (Fig. 26).

Table 1:

Maximum Gap Widths between the Machined Welding Faces

Pipe outside diameter d mm	Gap width mm	Panel width mm
≤ 355	0.5	
400... < 630	1.0	$\leq 1,500$
630... < 800	1.3	$> 1,500 \leq 2,000$
800... $\leq 1,000$	1.5	$> 2,000 \leq 2,300$
$> 1,000$	2.0	$> 2,300 \leq 3,000$

Table 2: Parameters for Welding Machines Type OMISA SP

dia	SDR	Merging pressure bar	Bead height mm	Heating time sec.	Welding pressure bar	Cooling time min.
160	17	8	1	147	8	16
	11	13	1	225	13	24
200	17	13	1	180	13	20
	11	20	1	290	20	30
250	17	21	1	217	21	24
	11	32	1,5	313	32	35

Permissible misalignment of wall 0,1 x wall thickness (s)

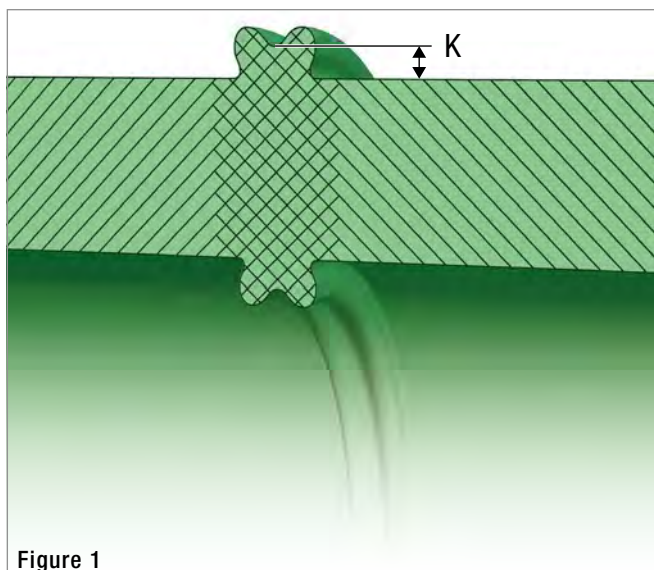


Figure 1



4

Installation

Page

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4.1 Installation

K-Aqua unrivalled as it is a unique poly-fusion connection system. It can be installed in no time; for instance, a 20 mm external diameter pipe takes only 9 seconds to install.

4.2 Thermal Installation of Cold Water Pipes

As stipulated in section 2 of the DIN 1988 Standard, drinkable water facilities must be protected from heat and condensation. The diagram below reports the standard values for minimum insulation thickness. Those figures can be applied to all types of pipes, including **K-Aqua**. (Fig. 1)

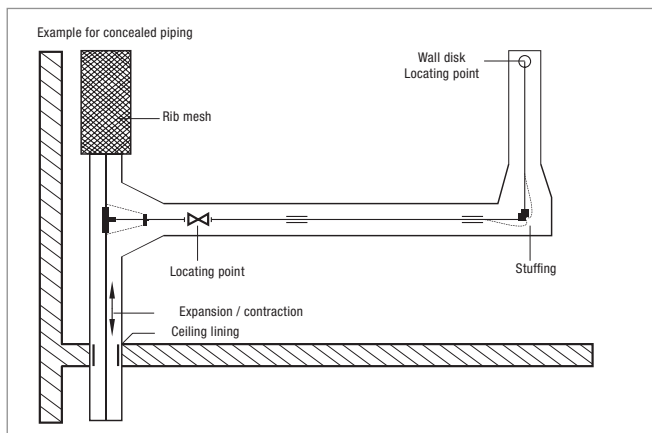
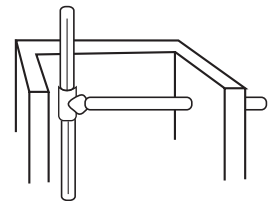


Figure 1

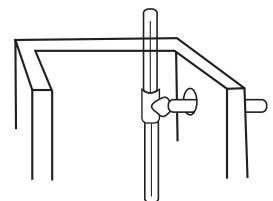
4.3 Applications in Sanitary Installation Shaft

When installing an apartment's pipe connections from the main pipe, the following techniques can be applied to compensate for the pipe's thermal expansions: (Fig. 2)

Pipe connection can be made at some distance, away from the wall.



Connecting pipe can be passed through a hole much larger than the pipe diameter.



Apartment connection can be made through a branch pipe to provide flexibility.

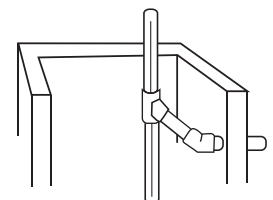


Figure 2

4.4 Design and Realization

Realization and installation must be performed according to the specific standards of the countries of application.

4.5 Operation Conditions

Classification of operating conditions to DIN EN ISO 15874-1

The selection of a particular application class according to the following table should be agreed among the contracting parties.

For each application class an allowable operating pressure p_0 of 4 bar¹⁾, 6 bar, 8 bar or 10 bar applies, depending on the application.

Applicatio class	Calculation-temperature TD °C	Service life ^{b)} at T _D Years	T _{max} °C	Service life at T _{max} Year(s)	T _{mal} °C	Service life at T _{mal} h	Typical application area	PP-R pipe system SDR 6	PP-RCT pipe system SDR 7,4
1 ^{a)}	60	49	80	1	95	100	Hot water supply (60° C)	10 bar	10 bar
2 ^{a)}	70	49	80	1	95	100	Hot water supply (70° C)	8 bar	10 bar
4 ^{b)}	20 Followed by 40 Followed by 60 Followed by (see next column)	2,5 20 25 (see next column)	70 Followed by (see next column)	2,5 (see next column)	100	100	Floor heating and low temperature radiator connections	10 bar	10 bar
5 ^{b)}	20 Followed by 60 Followed by 80 Followed by (see next column)	14 25 10 (see next column)	90 Followed by (see next column)	1 (see next column)	100	100	High temperature radiator connections	6 bar	8 bar

^{a)} Pertinent to the national regulations either application class 1 or application class 2 may be selected.

^{b)} If there is more than one operational temperature for one application area the corresponding service life time should be summed.
For example the temperature collective for class 5 for a period of 50 years consists of:

- 20° C over 14 years followed by
- 60° C over 25 years followed by
- 80° C over 10 years followed by
- 90° C over 1 year followed by
- 100° C over 100 h

Explanation:

The column **T_{mal}** indicates the highest allowed temperature (for example at disruption of the controlling), max 100° C.
The column **Service life at T_{mal}** shows that this breakdown temperature allows a max period of 100 h (over 50 years) whereas single breakdown segments should not exceed 3 hours.

REMARK:

This norm does not apply when higher values are assigned to T_D, T_{max} and T_{mal} than those quoted on the table.

Figure 3

¹⁾ 1 bar = 10⁵ N/m² = 0,01 MPa

4.6 Pressure loss pipes

Pressure loss for **SDR 6 pipes** due to pipe resistance **R** and flow rate **v** depending on flow **V**

Temperature 20 °C

Roughness: 0.007 mm

Density: 998.29 kg/m³

Kin. viscosity: 1.004E-06 m²/s

		Dimension	20 mm	25 mm	32 mm	40 mm	50 mm	63 mm	75 mm	90 mm	110 mm	125 mm
		Wall thickness	3.4 mm	4.2 mm	5.4 mm	6.7 mm	8.3 mm	10.5 mm	12.5 mm	15.0 mm	18.3 mm	20.8 mm
l/s	m ³ /h											
0.01	0.04	R in mbar/m v in m/s	0.15 0.07	0.05 0.05	0.02 0.03	0.01 0.02						
0.02	0.07	R	0.45	0.16	0.05	0.02	0.01					
		v	0.15	0.09	0.06	0.04	0.02					
0.03	0.11	R	0.87	0.30	0.10	0.03	0.01					
		v	0.22	0.14	0.08	0.05	0.03					
0.04	0.14	R	1.39	0.48	0.15	0.05	0.02	0.01				
		v	0.29	0.18	0.11	0.07	0.05	0.03				
0.05	0.18	R	2.02	0.69	0.22	0.08	0.03	0.01				
		v	0.37	0.23	0.14	0.09	0.06	0.04				
0.06	0.22	R	2.74	0.94	0.30	0.10	0.04	0.01	0.01			
		v	0.44	0.28	0.17	0.11	0.07	0.04	0.03			
0.07	0.25	R	3.55	1.21	0.39	0.13	0.05	0.02	0.01			
		v	0.51	0.32	0.20	0.13	0.08	0.05	0.04			
0.08	0.29	R	4.46	1.51	0.48	0.17	0.06	0.02	0.01			
		v	0.58	0.37	0.23	0.14	0.09	0.06	0.04			
0.09	0.32	R	5.45	1.85	0.59	0.20	0.07	0.02	0.01			
		v	0.66	0.42	0.25	0.16	0.10	0.06	0.05			
0.10	0.36	R	6.52	2.21	0.70	0.24	0.08	0.03	0.01	0.01		
		v	0.73	0.46	0.28	0.18	0.11	0.07	0.05	0.04		
0.12	0.43	R	8.92	3.01	0.95	0.33	0.11	0.04	0.02	0.01		
		v	0.88	0.55	0.34	0.22	0.14	0.09	0.06	0.04		
0.14	0.50	R	11.66	3.92	1.23	0.42	0.15	0.05	0.02	0.01		
		v	1.02	0.65	0.40	0.25	0.16	0.10	0.07	0.05		
0.16	0.58	R	14.71	4.93	1.55	0.53	0.18	0.06	0.03	0.01		
		v	1.17	0.74	0.45	0.29	0.18	0.12	0.08	0.06		
0.18	0.65	R	18.07	6.05	1.89	0.65	0.22	0.08	0.03	0.01	0.01	
		v	1.32	0.83	0.51	0.32	0.21	0.13	0.09	0.06	0.04	
0.20	0.72	R	21.75	7.26	2.27	0.78	0.27	0.09	0.04	0.02	0.01	
		v	1.46	0.92	0.57	0.36	0.23	0.14	0.10	0.07	0.05	
0.30	1.08	R	44.65	14.77	4.58	1.56	0.53	0.18	0.08	0.03	0.01	0.01
		v	2.19	1.39	0.85	0.54	0.34	0.22	0.15	0.11	0.07	0.05
0.40	1.44	R	74.89	24.60	7.58	2.56	0.87	0.29	0.13	0.05	0.02	0.01
		v	2.92	1.85	1.13	0.72	0.46	0.29	0.20	0.14	0.09	0.07
0.50	1.80	R	112.32	36.68	11.24	3.78	1.28	0.43	0.19	0.08	0.03	0.02
		v	3.65	2.31	1.42	0.90	0.57	0.36	0.25	0.18	0.12	0.09
0.60	2.16	R	156.82	50.97	15.55	5.21	1.76	0.59	0.26	0.11	0.04	0.02
		v	4.38	2.77	1.70	1.08	0.68	0.43	0.31	0.21	0.14	0.11
0.70	2.52	R	208.34	67.43	20.49	6.85	2.30	0.77	0.34	0.14	0.05	0.03
		v	5.12	3.23	1.98	1.26	0.80	0.51	0.36	0.25	0.17	0.13
0.80	2.88	R	266.84	86.05	26.05	8.68	2.91	0.97	0.42	0.18	0.07	0.04
		v	5.85	3.70	2.27	1.44	0.91	0.58	0.41	0.28	0.19	0.15
0.90	3.24	R	332.29	106.80	32.22	10.72	3.58	1.20	0.52	0.22	0.08	0.05
		v	6.58	4.16	2.55	1.62	1.03	0.65	0.46	0.32	0.21	0.16
1.00	3.60	R	404.65	129.67	39.01	12.94	4.32	1.44	0.63	0.26	0.10	0.06
		v	7.31	4.62	2.83	1.80	1.14	0.72	0.51	0.35	0.24	0.18
1.10	3.96	R	483.92	154.66	46.40	15.36	5.11	1.70	0.74	0.31	0.12	0.07
		v	8.04	5.08	3.12	1.98	1.26	0.79	0.56	0.39	0.26	0.20
1.20	4.32	R	570.09	181.75	54.40	17.97	5.97	1.98	0.86	0.36	0.14	0.08
		v	8.77	5.54	3.40	2.16	1.37	0.87	0.61	0.42	0.28	0.22
1.30	4.68	R	663.13	210.95	62.99	20.77	6.89	2.29	0.99	0.42	0.16	0.09
		v	9.50	6.01	3.68	2.34	1.48	0.94	0.66	0.46	0.31	0.24
1.40	5.04	R	763.06	242.24	72.18	23.75	7.86	2.61	1.13	0.47	0.18	0.10
		v	10.23	6.47	3.97	2.52	1.60	1.01	0.71	0.50	0.33	0.26
1.60	5.76	R		311.09	92.33	30.28	10.00	3.31	1.43	0.60	0.23	0.12
		v		7.39	4.53	2.88	1.83	1.15	0.81	0.57	0.38	0.29
1.80	6.48	R		388.29	114.85	37.56	12.37	4.08	1.76	0.74	0.28	0.15
		v		8.32	5.10	3.24	2.05	1.30	0.92	0.64	0.43	0.33
2.00	7.20	R		473.81	139.72	45.56	14.97	4.93	2.13	0.89	0.34	0.18
		v		9.24	5.67	3.60	2.28	1.44	1.02	0.71	0.47	0.37
2.20	7.92	R		567.64	166.93	54.30	17.80	5.85	2.52	1.05	0.40	0.22
		v		10.17	6.23	3.96	2.51	1.59	1.12	0.78	0.52	0.40
2.40	8.64	R			196.48	63.77	20.85	6.84	2.94	1.22	0.47	0.25
		v			6.80	4.32	2.74	1.73	1.22	0.85	0.57	0.44

		Dimension	20 mm	25 mm	32 mm	40 mm	50 mm	63 mm	75 mm	90 mm	110 mm	125 mm
		Wall thickness	3.4 mm	4.2 mm	5.4 mm	6.7 mm	8.3 mm	10.5 mm	12.5 mm	15.0 mm	18.3 mm	20.8 mm
l/s	m³/h											
2.60	9.36	R			228.36	73.96	24.14	7.90	3.40	1.41	0.54	0.29
		v			7.37	4.68	2.97	1.88	1.32	0.92	0.61	0.48
2.80	10.08	R			262.57	84.87	27.65	9.04	3.88	1.61	0.61	0.33
		v			7.93	5.04	3.20	2.02	1.43	0.99	0.66	0.51
3.00	10.80	R			299.11	96.51	31.38	10.24	4.39	1.82	0.69	0.38
		v			8.50	5.40	3.42	2.17	1.53	1.06	0.71	0.55
3.50	12.60	R			400.59	128.74	41.70	13.56	5.80	2.40	0.91	0.49
		v			9.92	6.30	3.99	2.53	1.78	1.24	0.83	0.64
4.00	14.40	R			516.57	165.44	53.41	17.31	7.39	3.05	1.15	0.62
		v			11.33	7.20	4.57	2.89	2.04	1.41	0.95	0.73
4.50	16.20	R				206.60	66.50	21.49	9.16	3.78	1.42	0.77
		v				8.10	5.14	3.25	2.29	1.59	1.06	0.82
5.00	18.00	R				252.22	80.96	26.09	11.10	4.57	1.72	0.93
		v				9.00	5.71	3.61	2.55	1.77	1.18	0.92
5.50	19.80	R				302.28	96.79	31.12	13.22	5.43	2.04	1.10
		v				9.90	6.28	3.97	2.80	1.95	1.30	1.01
6.00	21.60	R				356.78	113.99	36.57	15.51	6.36	2.39	1.29
		v				10.80	6.85	4.33	3.06	2.12	1.42	1.10
6.50	23.40	R					132.56	42.44	17.97	7.36	2.76	1.49
		v					7.42	4.69	3.31	2.30	1.54	1.19
7.00	25.20	R					152.49	48.73	20.61	8.43	3.16	1.70
		v					7.99	5.05	3.57	2.48	1.65	1.28
7.50	27.00	R					173.78	55.44	23.42	9.57	3.58	1.93
		v					8.56	5.41	3.82	2.65	1.77	1.37
8.00	28.80	R					196.43	62.57	26.39	10.77	4.03	2.17
		v					9.13	5.77	4.07	2.83	1.89	1.46
8.50	30.60	R					220.43	70.11	29.54	12.04	4.50	2.42
		v					9.70	6.14	4.33	3.01	2.01	1.56
9.00	32.40	R					245.80	78.06	32.86	13.38	4.99	2.68
		v					10.27	6.50	4.58	3.18	2.13	1.65
9.50	34.20	R						86.43	36.34	14.79	5.51	2.96
		v						6.86	4.84	3.36	2.25	1.74
10.00	36.00	R						95.22	40.00	16.26	6.05	3.25
		v						7.22	5.09	3.54	2.36	1.83
10.50	37.80	R						104.42	43.82	17.80	6.62	3.55
		v						7.58	5.35	3.71	2.48	1.92
11.00	39.60	R						114.03	47.82	19.40	7.21	3.87
		v						7.94	5.60	3.89	2.60	2.01
11.50	41.40	R						124.06	51.98	21.07	7.83	4.19
		v						8.30	5.86	4.07	2.72	2.11
12.00	43.20	R						134.49	56.31	22.81	8.46	4.53
		v						8.66	6.11	4.24	2.84	2.20
12.50	45.00	R						145.34	60.80	24.61	9.12	4.88
		v						9.02	6.37	4.42	2.95	2.29
13.00	46.80	R						156.61	65.47	26.48	9.81	5.25
		v						9.38	6.62	4.60	3.07	2.38
13.50	48.60	R						168.28	70.30	28.41	10.52	5.62
		v						9.74	6.88	4.77	3.19	2.47
14.00	50.40	R						180.37	75.30	30.41	11.25	6.01
		v						10.11	7.13	4.95	3.31	2.56
14.50	52.20	R							80.47	32.48	12.00	6.41
		v							7.38	5.13	3.43	2.65
15.00	54.00	R							85.80	34.61	12.78	6.83
		v							7.64	5.31	3.54	2.75
16.00	57.60	R							96.97	39.06	14.41	7.69
		v							8.15	5.66	3.78	2.93
17.00	61.20	R							108.80	43.78	16.13	8.60
		v							8.66	6.01	4.02	3.11
18.00	64.80	R							121.31	48.76	17.94	9.56
		v							9.17	6.37	4.25	3.29
19.00	68.40	R							134.47	54.00	19.85	10.57
		v							9.68	6.72	4.49	3.48
20.00	72.00	R							148.31	59.49	21.84	11.63
		v							10.19	7.07	4.73	3.66
21.00	75.60	R								65.25	23.93	12.73
		v								7.43	4.96	3.84
22.00	79.20	R								71.27	26.12	13.88
		v								7.78	5.20	4.03
23.00	82.80	R								77.54	28.39	15.08
		v								8.13	5.44	4.21
24.00	86.40	R								84.08	30.75	16.33
		v								8.49	5.67	4.39
25.00	90.00	R								90.87	33.21	17.63
		v								8.84	5.91	4.58

Pressure loss for SDR 7,4 pipes due to pipe resistance R and flow rate v depending on flow V

Temperature 20 °C

Roughness: 0.007 mm

Density: 998.29 kg/m³Kin. viscosity: 1.004E-06 m²/s

		Dimension	20 mm	25 mm	32 mm	40 mm	50 mm	63 mm	75 mm	90 mm	110 mm	125 mm	160 mm	200 mm	250 mm
		Wall thickness	2.8 mm	3.5 mm	4.4 mm	5.5 mm	6.9 mm	8.6 mm	10.3 mm	12.3 mm	15.1 mm	17.1 mm	21.6 mm	27.4 mm	34.2 mm
l/s	m ³ /h	R in mbar/m v in m/s													
0.01	0.04	R	0.10	0.04	0.01										
		v	0.06	0.04	0.02										
0.02	0.07	R	0.30	0.11	0.03	0.01									
		v	0.12	0.08	0.05	0.03									
0.03	0.11	R	0.58	0.21	0.06	0.02									
		v	0.18	0.12	0.07	0.05									
0.04	0.14	R	0.93	0.33	0.10	0.04	0.01								
		v	0.25	0.16	0.09	0.06	0.04								
0.05	0.18	R	1.34	0.47	0.15	0.05	0.02								
		v	0.31	0.20	0.12	0.08	0.05								
0.06	0.22	R	1.82	0.64	0.20	0.07	0.03	0.01							
		v	0.37	0.24	0.14	0.09	0.06	0.04							
0.07	0.25	R	2.36	0.83	0.25	0.09	0.03	0.01							
		v	0.43	0.28	0.17	0.11	0.07	0.04							
0.08	0.29	R	2.96	1.04	0.32	0.11	0.04	0.01							
		v	0.49	0.31	0.19	0.12	0.08	0.05							
0.09	0.32	R	3.61	1.26	0.38	0.14	0.05	0.02							
		v	0.55	0.35	0.21	0.14	0.09	0.05							
0.10	0.36	R	4.32	1.51	0.46	0.16	0.06	0.02	0.01						
		v	0.61	0.39	0.24	0.15	0.10	0.06	0.04						
0.12	0.43	R	5.90	2.05	0.62	0.22	0.08	0.03	0.01						
		v	0.74	0.47	0.28	0.18	0.12	0.07	0.05						
0.14	0.50	R	7.70	2.67	0.81	0.28	0.10	0.03	0.02						
		v	0.86	0.55	0.33	0.21	0.14	0.08	0.06						
0.16	0.58	R	9.70	3.36	1.01	0.35	0.13	0.04	0.02						
		v	0.98	0.63	0.38	0.24	0.16	0.10	0.07						
0.18	0.65	R	11.91	4.11	1.24	0.43	0.15	0.05	0.02	0.01					
		v	1.11	0.71	0.43	0.27	0.17	0.11	0.08	0.05					
0.20	0.72	R	14.32	4.94	1.48	0.52	0.18	0.06	0.03	0.01					
		v	1.23	0.79	0.47	0.30	0.19	0.12	0.09	0.06					
0.30	1.08	R	29.30	10.01	2.98	1.03	0.36	0.12	0.05	0.02	0.01				
		v	1.84	1.18	0.71	0.45	0.29	0.18	0.13	0.09	0.06				
0.40	1.44	R	49.02	16.64	4.92	1.70	0.59	0.20	0.09	0.04	0.01	0.01			
		v	2.46	1.57	0.95	0.61	0.39	0.24	0.17	0.12	0.08	0.06			
0.50	1.80	R	73.35	24.77	7.29	2.50	0.87	0.29	0.13	0.05	0.02	0.01			
		v	3.07	1.96	1.18	0.76	0.49	0.30	0.22	0.15	0.10	0.08			
0.60	2.16	R	102.21	34.36	10.06	3.45	1.20	0.39	0.17	0.07	0.03	0.02			
		v	3.68	2.36	1.42	0.91	0.58	0.36	0.26	0.18	0.12	0.09			
0.70	2.52	R	135.57	45.40	13.24	4.52	1.57	0.51	0.23	0.09	0.04	0.02	0.01		
		v	4.30	2.75	1.66	1.06	0.68	0.42	0.30	0.21	0.14	0.11	0.07		
0.80	2.88	R	173.38	57.86	16.82	5.73	1.98	0.64	0.28	0.12	0.05	0.03	0.01		
		v	4.91	3.14	1.89	1.21	0.78	0.49	0.34	0.24	0.16	0.12	0.07		
0.90	3.24	R	215.63	71.73	20.78	7.06	2.43	0.79	0.35	0.15	0.06	0.03	0.01		
		v	5.53	3.54	2.13	1.36	0.87	0.55	0.39	0.27	0.18	0.14	0.08		
1.00	3.60	R	262.30	87.00	25.14	8.52	2.93	0.95	0.42	0.17	0.07	0.04	0.01		
		v	6.14	3.93	2.37	1.51	0.97	0.61	0.43	0.30	0.20	0.15	0.09		
1.10	3.96	R	313.36	103.67	29.87	10.11	3.47	1.12	0.49	0.21	0.08	0.04	0.01		
		v	6.75	4.32	2.60	1.67	1.07	0.67	0.47	0.33	0.22	0.17	0.10		
1.20	4.32	R	368.81	121.73	34.99	11.82	4.05	1.31	0.58	0.24	0.09	0.05	0.02	0.01	
		v	7.37	4.72	2.84	1.82	1.17	0.73	0.52	0.36	0.24	0.19	0.11	0.07	
1.30	4.68	R	428.65	141.17	40.48	13.65	4.67	1.51	0.66	0.28	0.11	0.06	0.02	0.01	
		v	7.98	5.11	3.08	1.97	1.26	0.79	0.56	0.39	0.26	0.20	0.12	0.08	
1.40	5.04	R	492.86	162.00	46.35	15.60	5.33	1.72	0.76	0.31	0.12	0.07	0.02	0.01	
		v	8.60	5.50	3.31	2.12	1.36	0.85	0.60	0.42	0.28	0.22	0.13	0.08	
1.60	5.76	R	634.39	207.77	59.21	19.86	6.77	2.18	0.96	0.40	0.15	0.08	0.03	0.01	
		v	9.82	6.29	3.78	2.42	1.55	0.97	0.69	0.48	0.32	0.25	0.15	0.10	
1.80	6.48	R	793.36	259.03	73.57	24.61	8.37	2.69	1.18	0.49	0.19	0.10	0.03	0.01	
		v	11.05	7.07	4.26	2.73	1.75	1.09	0.77	0.54	0.36	0.28	0.17	0.11	
2.00	7.20	R		315.77	89.40	29.83	10.12	3.24	1.42	0.59	0.23	0.12	0.04	0.01	
		v		7.86	4.73	3.03	1.94	1.21	0.86	0.60	0.40	0.31	0.19	0.12	
2.20	7.92	R		377.96	106.70	35.52	12.02	3.85	1.68	0.69	0.27	0.14	0.04	0.02	0.01
		v		8.65	5.20	3.33	2.14	1.34	0.95	0.65	0.44	0.34	0.21	0.13	0.08
2.40	8.64	R		445.60	125.47	41.67	14.08	4.50	1.96	0.81	0.31	0.17	0.05	0.02	0.01
		v		9.43	5.68	3.63	2.33	1.46	1.03	0.71	0.48	0.37	0.22	0.14	0.09
2.60	9.36	R		518.69	145.71	48.30	16.29	5.19	2.26	0.93	0.36	0.19	0.06	0.02	0.01
		v		10.22	6.15	3.94	2.53	1.58	1.12	0.77	0.52	0.40	0.24	0.16	0.10

		Dimension	20 mm	25 mm	32 mm	40 mm	50 mm	63 mm	75 mm	90 mm	110 mm	125 mm	160 mm	200 mm	250 mm
		Wall thickness	2.8 mm	3.5 mm	4.4 mm	5.5 mm	6.9 mm	8.6 mm	10.3 mm	12.3 mm	15.1 mm	17.1 mm	21.6 mm	27.4 mm	34.2 mm
l/s	m³/h														
2.80	10.08	R			167.40	55.38	18.64	5.93	2.58	1.06	0.41	0.22	0.07	0.02	0.01
		v			6.62	4.24	2.72	1.70	1.20	0.83	0.56	0.43	0.26	0.17	0.11
3.00	10.80	R			190.56	62.93	21.15	6.72	2.92	1.20	0.46	0.25	0.08	0.03	0.01
		v			7.10	4.54	2.91	1.82	1.29	0.89	0.60	0.46	0.28	0.18	0.12
3.50	12.60	R			254.82	83.82	28.07	8.89	3.86	1.58	0.61	0.33	0.10	0.03	0.01
		v			8.28	5.30	3.40	2.12	1.51	1.04	0.70	0.54	0.33	0.21	0.14
4.00	14.40	R			328.14	107.58	35.90	11.33	4.91	2.01	0.77	0.42	0.12	0.04	0.02
		v			9.46	6.06	3.89	2.43	1.72	1.19	0.80	0.62	0.37	0.24	0.15
4.50	16.20	R			410.53	134.19	44.65	14.06	6.07	2.49	0.95	0.51	0.15	0.05	0.02
		v			10.65	6.81	4.37	2.73	1.94	1.34	0.90	0.69	0.42	0.27	0.17
5.00	18.00	R			163.65	54.32	17.05	7.36	3.01	1.15	0.62	0.18	0.07	0.02	0.01
		v			7.57	4.86	3.03	2.15	1.49	1.00	0.77	0.47	0.30	0.19	0.12
5.50	19.80	R			195.95	64.88	20.32	8.75	3.57	1.36	0.73	0.22	0.08	0.03	0.01
		v			8.33	5.34	3.34	2.37	1.64	1.10	0.85	0.51	0.33	0.21	0.14
6.00	21.60	R			231.09	76.36	23.86	10.26	4.18	1.60	0.86	0.26	0.09	0.03	0.01
		v			9.08	5.83	3.64	2.58	1.79	1.20	0.93	0.56	0.36	0.23	0.15
6.50	23.40	R			269.06	88.73	27.68	11.89	4.84	1.84	0.99	0.29	0.10	0.04	0.02
		v			9.84	6.32	3.95	2.80	1.93	1.30	1.00	0.61	0.39	0.25	0.17
7.00	25.20	R			309.86	102.00	31.76	13.62	5.54	2.11	1.13	0.34	0.12	0.04	0.02
		v			10.60	6.80	4.25	3.01	2.08	1.40	1.08	0.65	0.42	0.27	0.17
7.50	27.00	R				116.17	36.10	15.47	6.28	2.39	1.28	0.38	0.13	0.05	0.02
		v				7.29	4.55	3.23	2.23	1.50	1.16	0.70	0.45	0.29	0.19
8.00	28.80	R				131.24	40.72	17.42	7.07	2.68	1.44	0.43	0.15	0.05	0.02
		v				7.77	4.86	3.44	2.38	1.60	1.24	0.75	0.48	0.31	0.20
8.50	30.60	R				147.20	45.60	19.49	7.90	3.00	1.60	0.48	0.17	0.06	0.02
		v				8.26	5.16	3.66	2.53	1.70	1.31	0.79	0.51	0.33	0.21
9.00	32.40	R				164.05	50.75	21.67	8.77	3.32	1.78	0.53	0.18	0.06	0.02
		v				8.74	5.46	3.87	2.68	1.80	1.39	0.84	0.54	0.35	0.22
9.50	34.20	R				181.80	56.16	23.96	9.69	3.67	1.96	0.58	0.20	0.07	0.02
		v				9.23	5.77	4.09	2.83	1.90	1.47	0.89	0.57	0.37	0.24
10.00	36.00	R				200.45	61.84	26.35	10.65	4.03	2.15	0.64	0.22	0.08	0.02
		v				9.72	6.07	4.30	2.98	2.00	1.54	0.93	0.60	0.39	0.25
10.50	37.80	R				219.98	67.78	28.86	11.65	4.40	2.35	0.69	0.24	0.08	0.02
		v				10.20	6.37	4.52	3.13	2.10	1.62	0.98	0.63	0.41	0.26
11.00	39.60	R					73.99	31.48	12.70	4.79	2.56	0.76	0.26	0.09	0.02
		v					6.68	4.73	3.27	2.20	1.70	1.03	0.66	0.42	0.27
11.50	41.40	R					80.46	34.20	13.78	5.20	2.77	0.82	0.29	0.10	0.02
		v					6.98	4.95	3.42	2.30	1.78	1.07	0.69	0.44	0.29
12.00	43.20	R					87.20	37.04	14.91	5.62	3.00	0.88	0.31	0.11	0.02
		v					7.28	5.16	3.57	2.40	1.85	1.12	0.72	0.46	0.30
12.50	45.00	R					94.20	39.98	16.09	6.06	3.23	0.95	0.33	0.11	0.02
		v					7.59	5.38	3.72	2.50	1.93	1.17	0.75	0.48	0.31
13.00	46.80	R					101.46	43.03	17.30	6.51	3.47	1.02	0.36	0.12	0.02
		v					7.89	5.59	3.87	2.60	2.01	1.21	0.79	0.50	0.32
13.50	48.60	R					108.99	46.19	18.56	6.98	3.71	1.09	0.38	0.13	0.02
		v					8.19	5.81	4.02	2.70	2.08	1.26	0.82	0.52	0.33
14.00	50.40	R					116.78	49.46	19.86	7.46	3.97	1.17	0.41	0.14	0.02
		v					8.50	6.02	4.17	2.80	2.16	1.31	0.85	0.54	0.34
14.50	52.20	R					124.83	52.84	21.20	7.96	4.23	1.24	0.43	0.15	0.02
		v					8.80	6.24	4.32	2.90	2.24	1.35	0.88	0.56	0.35
15.00	54.00	R					133.14	56.33	22.59	8.48	4.50	1.32	0.46	0.16	0.02
		v					9.10	6.45	4.47	3.00	2.32	1.40	0.91	0.58	0.36
16.00	57.60	R					150.56	63.62	25.48	9.55	5.07	1.49	0.52	0.18	0.02
		v					9.71	6.88	4.76	3.20	2.47	1.49	0.97	0.62	0.37
17.00	61.20	R					169.03	71.35	28.54	10.69	5.67	1.66	0.58	0.20	0.02
		v					10.32	7.31	5.06	3.40	2.63	1.59	1.03	0.66	0.39
18.00	64.80	R						79.51	31.77	11.88	6.30	1.84	0.64	0.22	0.02
		v						7.74	5.36	3.60	2.78	1.68	1.09	0.69	0.40
19.00	68.40	R						88.09	35.16	13.14	6.96	2.03	0.71	0.24	0.02
		v						8.17	5.66	3.80	2.93	1.77	1.15	0.73	0.41
20.00	72.00	R						97.11	38.73	14.45	7.65	2.23	0.78	0.26	0.02
		v						8.60	5.95	4.00	3.09	1.87	1.21	0.77	0.42
21.00	75.60	R						106.56	42.46	15.83	8.38	2.44	0.85	0.29	0.02
		v						9.04	6.25	4.20	3.24	1.96	1.27	0.81	0.43
22.00	79.20	R						116.44	46.35	17.26	9.13	2.66	0.92	0.31	0.02
		v						9.47	6.55	4.40	3.40	2.05	1.33	0.85	0.44
23.00	82.80	R						126.75	50.41	18.76	9.92	2.89	1.00	0.34	0.02
		v						9.90	6.85	4.60	3.55	2.15	1.39	0.89	0.45
24.00	86.40	R						137.49	54.64	20.31	10.73	3.12	1.08	0.37	0.02
		v						10.33	7.14	4.80	3.71	2.24	1.45	0.93	0.46
25.00	90.00	R							59.03	21.93	11.58	3.36	1.17	0.39	0.02
		v							7.44	5.00	3.86	2.33	1.51	0.97	0.47
26.00	93.60	R							63.59	23.60	12.46	3.62	1.25	0.42	0.02
		v							7.74	5.20	4.02	2.43	1.57	1.00	0.48

		Dimension	20 mm	25 mm	32 mm	40 mm	50 mm	63 mm	75 mm	90 mm	110 mm	125 mm	160 mm	200 mm	250 mm
		Wall thickness	2.8 mm	3.5 mm	4.4 mm	5.5 mm	6.9 mm	8.6 mm	10.3 mm	12.3 mm	15.1 mm	17.1 mm	21.6 mm	27.4 mm	34.2 mm
l/s	m³/h														
27.00	97.20	R								68.31	25.34	13.37	3.88	1.34	0.45
		v								8.04	5.40	4.17	2.52	1.63	1.04
28.00	100.80	R								73.20	27.13	14.31	4.15	1.43	0.48
		v								8.34	5.60	4.32	2.61	1.69	1.08
29.00	104.40	R								78.26	28.98	15.28	4.42	1.53	0.52
		v								8.63	5.80	4.48	2.71	1.75	1.12
30.00	108.00	R								83.48	30.90	16.28	4.71	1.63	0.55
		v								8.93	6.00	4.63	2.80	1.81	1.16
32.00	115.20	R								94.42	34.90	18.37	5.31	1.83	0.62
		v								9.53	6.40	4.94	2.99	1.93	1.24
34.00	122.40	R								106.01	39.14	20.59	5.94	2.05	0.69
		v								10.12	6.80	5.25	3.17	2.05	1.31
36.00	129.60	R									43.61	22.93	6.60	2.27	0.76
		v									7.20	5.56	3.36	2.17	1.39
38.00	136.80	R									48.32	25.38	7.30	2.51	0.84
		v									7.60	5.87	3.55	2.29	1.47
40.00	144.00	R									53.27	27.96	8.03	2.76	0.93
		v									8.00	6.18	3.73	2.42	1.54
42.00	151.20	R									58.45	30.67	8.80	3.02	1.01
		v									8.40	6.49	3.92	2.54	1.62
44.00	158.40	R									63.87	33.49	9.60	3.29	1.10
		v									8.80	6.80	4.11	2.66	1.70
46.00	165.60	R									69.53	36.43	10.43	3.57	1.20
		v									9.20	7.10	4.29	2.78	1.78
48.00	172.80	R									75.42	39.50	11.30	3.86	1.29
		v									9.60	7.41	4.48	2.90	1.85
50.00	180.00	R									81.54	42.68	12.19	4.17	1.39
		v									10.00	7.72	4.67	3.02	1.93
52.00	187.20	R										45.99	13.13	4.48	1.50
		v										8.03	4.85	3.14	2.01
54.00	194.40	R										49.41	14.09	4.81	1.61
		v										8.34	5.04	3.26	2.08
56.00	201.60	R										52.96	15.09	5.15	1.72
		v										8.65	5.23	3.38	2.16
58.00	208.80	R										56.63	16.12	5.49	1.83
		v										8.96	5.41	3.50	2.24
60.00	216.00	R										60.41	17.18	5.85	1.95
		v										9.27	5.60	3.62	2.32
62.00	223.20	R										64.32	18.28	6.22	2.07
		v										9.57	5.79	3.74	2.39
64.00	230.40	R										68.35	19.41	6.60	2.20
		v										9.88	5.97	3.87	2.47
66.00	237.60	R										72.50	20.57	6.99	2.33
		v										10.19	6.16	3.99	2.55
68.00	244.80	R											21.77	7.39	2.46
		v											6.35	4.11	2.63
70.00	252.00	R											22.99	7.81	2.60
		v											6.53	4.23	2.70
72.00	259.20	R											24.26	8.23	2.73
		v											6.72	4.35	2.78
74.00	266.40	R											25.55	8.66	2.88
		v											6.91	4.47	2.86
78.00	280.80	R											28.24	9.56	3.17
		v											7.28	4.71	3.01
80.00	288.00	R											29.63	10.03	3.33
		v											7.47	4.83	3.09
82.00	295.20	R											31.05	10.51	3.48
		v											7.65	4.95	3.17
84.00	302.40	R											32.51	11.00	3.64
		v											7.84	5.07	3.24
86.00	309.60	R											34.00	11.49	3.81
		v											8.03	5.19	3.32
88.00	316.80	R											35.52	12.00	3.97
		v											8.21	5.31	3.40
90.00	324.00	R											37.08	12.52	4.14
		v											8.40	5.44	3.47

Pressure loss for SDR 11 pipes due to pipe resistance R and flow rate v depending on flow V

Temperature 20 °C

Roughness: 0.007 mm

Density: 998.29 kg/m³

Kin. viscosity: 1.004E-06 m²/s

		Dimension	20 mm	25 mm	32 mm	40 mm	50 mm	63 mm	75 mm	90 mm	110 mm	125 mm	160 mm	200 mm	250 mm
		Wall thickness	1.9 mm	2.3 mm	2.9 mm	3.7 mm	4.6 mm	5.8 mm	6.8 mm	8.2 mm	10.0 mm	11.4 mm	14.6 mm	18.2 mm	22.7 mm
I/s	m³/h	R in mbar/m v in m/s													
0.01	0.04	R	0.06	0.02	0.01										
		v	0.05	0.03	0.02										
0.02	0.07	R	0.18	0.06	0.02	0.01									
		v	0.10	0.06	0.04	0.02									
0.03	0.11	R	0.34	0.12	0.04	0.01									
		v	0.15	0.09	0.06	0.04									
0.04	0.14	R	0.54	0.18	0.06	0.02	0.01								
		v	0.19	0.12	0.07	0.05	0.03								
0.05	0.18	R	0.78	0.27	0.08	0.03	0.01								
		v	0.24	0.15	0.09	0.06	0.04								
0.06	0.22	R	1.05	0.36	0.11	0.04	0.01	0.01							
		v	0.29	0.18	0.11	0.07	0.05	0.03							
0.07	0.25	R	1.36	0.46	0.14	0.05	0.02	0.01							
		v	0.34	0.21	0.13	0.08	0.05	0.03							
0.08	0.29	R	1.70	0.58	0.18	0.06	0.02	0.01							
		v	0.39	0.24	0.15	0.10	0.06	0.04							
0.09	0.32	R	2.07	0.70	0.22	0.08	0.03	0.01							
		v	0.44	0.28	0.17	0.11	0.07	0.04							
0.10	0.36	R	2.48	0.84	0.26	0.09	0.03	0.01	0.01						
		v	0.49	0.31	0.19	0.12	0.08	0.05	0.03						
0.12	0.43	R	3.38	1.14	0.35	0.13	0.04	0.02	0.01						
		v	0.58	0.37	0.22	0.14	0.09	0.06	0.04						
0.14	0.50	R	4.40	1.48	0.46	0.16	0.06	0.02	0.01						
		v	0.68	0.43	0.26	0.17	0.11	0.07	0.05						
0.16	0.58	R	5.54	1.86	0.57	0.20	0.07	0.02	0.01						
		v	0.78	0.49	0.30	0.19	0.12	0.08	0.05						
0.18	0.65	R	6.79	2.27	0.70	0.25	0.09	0.03	0.01	0.01					
		v	0.87	0.55	0.33	0.22	0.14	0.09	0.06	0.04					
0.20	0.72	R	8.16	2.72	0.83	0.30	0.10	0.04	0.02	0.01					
		v	0.97	0.61	0.37	0.24	0.15	0.10	0.07	0.05					
0.30	1.08	R	16.61	5.50	1.67	0.59	0.21	0.07	0.03	0.01	0.01				
		v	1.46	0.92	0.56	0.36	0.23	0.14	0.10	0.07	0.05				
0.40	1.44	R	27.68	9.11	2.75	0.97	0.34	0.11	0.05	0.02	0.01				
		v	1.94	1.22	0.74	0.48	0.31	0.19	0.14	0.09	0.06				
0.50	1.80	R	41.30	13.53	4.07	1.43	0.49	0.17	0.07	0.03	0.01	0.01			
		v	2.43	1.53	0.93	0.60	0.38	0.24	0.17	0.12	0.08	0.06			
0.60	2.16	R	57.42	18.73	5.61	1.97	0.68	0.23	0.10	0.04	0.02	0.01			
		v	2.91	1.84	1.11	0.72	0.46	0.29	0.20	0.14	0.09	0.07			
0.70	2.52	R	75.99	24.69	7.37	2.58	0.89	0.30	0.13	0.05	0.02	0.01			
		v	3.40	2.14	1.30	0.84	0.54	0.34	0.24	0.16	0.11	0.09			
0.80	2.88	R	97.01	31.41	9.34	3.27	1.12	0.37	0.16	0.07	0.03	0.01			
		v	3.88	2.45	1.48	0.96	0.61	0.39	0.27	0.19	0.13	0.10			
0.90	3.24	R	120.44	38.87	11.53	4.02	1.37	0.46	0.20	0.08	0.03	0.02	0.01		
		v	4.37	2.75	1.67	1.08	0.69	0.43	0.30	0.21	0.14	0.11	0.07		
1.00	3.60	R	146.28	47.08	13.93	4.85	1.65	0.55	0.24	0.10	0.04	0.02	0.01		
		v	4.85	3.06	1.85	1.20	0.76	0.48	0.34	0.24	0.16	0.12	0.07		
1.10	3.96	R	174.52	56.03	16.53	5.74	1.96	0.65	0.28	0.12	0.05	0.02	0.01		
		v	5.34	3.37	2.04	1.32	0.84	0.53	0.37	0.26	0.17	0.13	0.08		
1.20	4.32	R	205.14	65.70	19.34	6.71	2.28	0.76	0.32	0.14	0.05	0.03	0.01		
		v	5.82	3.67	2.23	1.44	0.92	0.58	0.41	0.28	0.19	0.15	0.09		
1.30	4.68	R	238.15	76.11	22.36	7.74	2.63	0.87	0.37	0.16	0.06	0.03	0.01		
		v	6.31	3.98	2.41	1.56	0.99	0.63	0.44	0.31	0.20	0.16	0.10		
1.40	5.04	R	273.54	87.24	25.57	8.84	3.00	0.99	0.42	0.18	0.07	0.04	0.01		
		v	6.79	4.28	2.60	1.68	1.07	0.67	0.47	0.33	0.22	0.17	0.10		
1.60	5.76	R	351.43	111.67	32.61	11.25	3.80	1.25	0.54	0.23	0.09	0.05	0.01		
		v	7.76	4.90	2.97	1.92	1.22	0.77	0.54	0.38	0.25	0.20	0.12		
1.80	6.48	R	438.78	138.97	40.45	13.91	4.69	1.54	0.66	0.28	0.11	0.06	0.02	0.01	
		v	8.73	5.51	3.34	2.16	1.38	0.87	0.61	0.42	0.28	0.22	0.13	0.09	
2.00	7.20	R	535.58	169.14	49.09	16.84	5.67	1.86	0.79	0.33	0.13	0.07	0.02	0.01	
		v	9.70	6.12	3.71	2.40	1.53	0.96	0.68	0.47	0.31	0.24	0.15	0.10	
2.20	7.92	R	641.81	202.17	58.51	20.03	6.73	2.21	0.94	0.39	0.15	0.08	0.03	0.01	
		v	10.67	6.73	4.08	2.64	1.68	1.06	0.74	0.52	0.35	0.27	0.16	0.10	
2.40	8.64	R		238.06	68.72	23.48	7.87	2.58	1.10	0.46	0.18	0.10	0.03	0.01	
		v		7.34	4.45	2.88	1.84	1.16	0.81	0.56	0.38	0.29	0.18	0.11	
2.60	9.36	R		276.78	79.71	27.18	9.10	2.97	1.26	0.53	0.20	0.11	0.03	0.01	
		v		7.95	4.82	3.11	1.99	1.25	0.88	0.61	0.41	0.32	0.19	0.12	

		Dimension	20 mm	25 mm	32 mm	40 mm	50 mm	63 mm	75 mm	90 mm	110 mm	125 mm	160 mm	200 mm	250 mm
		Wall thickness	1.9 mm	2.3 mm	2.9 mm	3.7 mm	4.6 mm	5.8 mm	6.8 mm	8.2 mm	10.0 mm	11.4 mm	14.6 mm	18.2 mm	22.7 mm
l/s	m³/h														
2.80	10.08	R		318.35	91.49	31.14	10.40	3.40	1.44	0.60	0.23	0.13	0.04	0.01	
		v		8.57	5.19	3.35	2.14	1.35	0.95	0.66	0.44	0.34	0.21	0.13	
3.00	10.80	R		362.76	104.04	35.35	11.79	3.84	1.63	0.68	0.26	0.14	0.04	0.02	0.01
		v		9.18	5.56	3.59	2.29	1.45	1.01	0.71	0.47	0.37	0.22	0.14	0.09
3.50	12.60	R		486.16	138.82	47.00	15.62	5.07	2.15	0.90	0.34	0.19	0.06	0.02	0.01
		v		10.71	6.49	4.19	2.68	1.69	1.18	0.82	0.55	0.43	0.26	0.17	0.11
4.00	14.40	R			178.44	60.21	19.95	6.46	2.73	1.14	0.43	0.24	0.07	0.03	0.01
		v			7.42	4.79	3.06	1.93	1.35	0.94	0.63	0.49	0.30	0.19	0.12
4.50	16.20	R			222.89	74.99	24.77	8.01	3.38	1.41	0.53	0.29	0.09	0.03	0.01
		v			8.35	5.39	3.44	2.17	1.52	1.06	0.71	0.55	0.33	0.21	0.14
5.00	18.00	R			272.15	91.32	30.09	9.70	4.08	1.70	0.64	0.35	0.11	0.04	0.01
		v			9.27	5.99	3.82	2.41	1.69	1.18	0.79	0.61	0.37	0.24	0.15
5.50	19.80	R			326.21	109.21	35.90	11.55	4.85	2.02	0.76	0.41	0.13	0.04	0.02
		v			10.20	6.59	4.21	2.65	1.86	1.29	0.86	0.67	0.41	0.26	0.17
6.00	21.60	R				128.65	42.20	13.55	5.69	2.36	0.89	0.48	0.15	0.05	0.02
		v				7.19	4.59	2.89	2.03	1.41	0.94	0.73	0.45	0.29	0.18
6.50	23.40	R				149.64	48.99	15.70	6.58	2.73	1.03	0.56	0.17	0.06	0.02
		v				7.79	4.97	3.13	2.20	1.53	1.02	0.79	0.48	0.31	0.20
7.00	25.20	R				172.17	56.26	17.99	7.53	3.12	1.18	0.64	0.20	0.07	0.02
		v				8.39	5.35	3.37	2.36	1.65	1.10	0.85	0.52	0.33	0.21
7.50	27.00	R				196.24	64.02	20.44	8.55	3.53	1.33	0.72	0.22	0.08	0.03
		v				8.99	5.74	3.61	2.53	1.76	1.18	0.91	0.56	0.36	0.23
8.00	28.80	R				221.85	72.27	23.03	9.62	3.97	1.50	0.81	0.25	0.08	0.03
		v				9.58	6.12	3.86	2.70	1.88	1.26	0.98	0.60	0.38	0.24
8.50	30.60	R				249.01	80.99	25.78	10.76	4.44	1.67	0.90	0.28	0.09	0.03
		v				10.18	6.50	4.10	2.87	2.00	1.34	1.04	0.63	0.40	0.26
9.00	32.40	R					90.20	28.67	11.95	4.93	1.85	1.00	0.31	0.10	0.04
		v					6.88	4.34	3.04	2.12	1.41	1.10	0.67	0.43	0.27
9.50	34.20	R					99.89	31.70	13.20	5.44	2.04	1.10	0.34	0.11	0.04
		v					7.27	4.58	3.21	2.23	1.49	1.16	0.71	0.45	0.29
10.00	36.00	R					110.06	34.89	14.52	5.97	2.24	1.21	0.37	0.13	0.04
		v					7.65	4.82	3.38	2.35	1.57	1.22	0.74	0.48	0.30
10.50	37.80	R					120.71	38.22	15.89	6.53	2.45	1.32	0.40	0.14	0.05
		v					8.03	5.06	3.55	2.47	1.65	1.28	0.78	0.50	0.32
11.00	39.60	R					131.84	41.69	17.32	7.12	2.67	1.44	0.44	0.15	0.05
		v					8.41	5.30	3.72	2.59	1.73	1.34	0.82	0.52	0.33
11.50	41.40	R					143.45	45.32	18.81	7.72	2.89	1.56	0.47	0.16	0.06
		v					8.80	5.54	3.88	2.70	1.81	1.40	0.86	0.55	0.35
12.00	43.20	R					155.54	49.08	20.36	8.35	3.13	1.69	0.51	0.17	0.06
		v					9.18	5.78	4.05	2.82	1.89	1.46	0.89	0.57	0.36
12.50	45.00	R					168.11	53.00	21.96	9.00	3.37	1.82	0.55	0.19	0.06
		v					9.56	6.02	4.22	2.94	1.96	1.52	0.93	0.59	0.38
13.00	46.80	R					181.16	57.06	23.63	9.68	3.62	1.95	0.59	0.20	0.07
		v					9.94	6.27	4.39	3.06	2.04	1.58	0.97	0.62	0.40
13.50	48.60	R					194.69	61.26	25.35	10.38	3.88	2.09	0.63	0.22	0.07
		v					10.33	6.51	4.56	3.17	2.12	1.65	1.00	0.64	0.41
14.00	50.40	R						65.61	27.13	11.10	4.14	2.23	0.68	0.23	0.08
		v						6.75	4.73	3.29	2.20	1.71	1.04	0.67	0.43
14.50	52.20	R						70.10	28.97	11.84	4.42	2.38	0.72	0.24	0.08
		v						6.99	4.90	3.41	2.28	1.77	1.08	0.69	0.44
15.00	54.00	R						74.74	30.87	12.61	4.70	2.53	0.76	0.26	0.09
		v						7.23	5.07	3.53	2.36	1.83	1.12	0.71	0.46
16.00	57.60	R						84.46	34.84	14.22	5.30	2.85	0.86	0.29	0.10
		v						7.71	5.40	3.76	2.52	1.95	1.19	0.76	0.49
17.00	61.20	R						94.75	39.04	15.91	5.92	3.18	0.96	0.33	0.11
		v						8.19	5.74	4.00	2.67	2.07	1.27	0.81	0.52
18.00	64.80	R						105.62	43.47	17.70	6.58	3.53	1.06	0.36	0.12
		v						8.67	6.08	4.23	2.83	2.19	1.34	0.86	0.55
19.00	68.40	R						117.06	48.13	19.58	7.27	3.90	1.17	0.40	0.14
		v						9.16	6.42	4.47	2.99	2.32	1.41	0.90	0.58
20.00	72.00	R						129.09	53.03	21.55	7.99	4.29	1.29	0.44	0.15
		v						9.64	6.75	4.70	3.14	2.44	1.49	0.95	0.61
21.00	75.60	R						141.69	58.15	23.61	8.75	4.69	1.41	0.48	0.16
		v						10.12	7.09	4.94	3.30	2.56	1.56	1.00	0.64
22.00	79.20	R							63.51	25.77	9.54	5.11	1.53	0.52	0.18
		v							7.43	5.17	3.46	2.68	1.64	1.05	0.67
23.00	82.80	R							69.09	28.01	10.36	5.55	1.66	0.56	0.19
		v							7.77	5.41	3.62	2.80	1.71	1.09	0.70
24.00	86.40	R							74.90	30.34	11.21	6.00	1.80	0.61	0.21
		v							8.11	5.64	3.77	2.93	1.79	1.14	0.73
25.00	90.00	R							80.95	32.76	12.10	6.47	1.94	0.65	0.22
		v							8.44	5.88	3.93	3.05	1.86	1.19	0.76
26.00	93.60	R							87.22	35.28	13.02	6.96	2.08	0.70	0.24
		v							8.78	6.11	4.09	3.17	1.93	1.24	0.79

		Dimension	20 mm	25 mm	32 mm	40 mm	50 mm	63 mm	75 mm	90 mm	110 mm	125 mm	160 mm	200 mm	250 mm
		Wall thickness	1.9 mm	2.3 mm	2.9 mm	3.7 mm	4.6 mm	5.8 mm	6.8 mm	8.2 mm	10.0 mm	11.4 mm	14.6 mm	18.2 mm	22.7 mm
l/s	m³/h														
27.00	97.20	R							93.72	37.88	13.97	7.46	2.23	0.75	0.25
		v							9.12	6.35	4.24	3.29	2.01	1.28	0.82
28.00	100.80	R							100.46	40.57	14.95	7.99	2.38	0.80	0.27
		v							9.46	6.58	4.40	3.41	2.08	1.33	0.85
29.00	104.40	R							107.42	43.36	15.96	8.52	2.54	0.86	0.29
		v							9.79	6.82	4.56	3.54	2.16	1.38	0.88
30.00	108.00	R							114.61	46.23	17.01	9.08	2.71	0.91	0.31
		v							10.13	7.05	4.72	3.66	2.23	1.43	0.91
32.00	115.20	R								52.25	19.20	10.24	3.05	1.02	0.35
		v								7.52	5.03	3.90	2.38	1.52	0.97
34.00	122.40	R								58.62	21.51	11.46	3.41	1.14	0.39
		v								7.99	5.34	4.14	2.53	1.62	1.03
36.00	129.60	R								65.36	23.96	12.76	3.79	1.27	0.43
		v								8.46	5.66	4.39	2.68	1.71	1.09
38.00	136.80	R								72.45	26.53	14.12	4.19	1.40	0.47
		v								8.93	5.97	4.63	2.83	1.81	1.16
40.00	144.00	R								79.90	29.22	15.54	4.60	1.54	0.52
		v								9.40	6.29	4.88	2.98	1.90	1.22
42.00	151.20	R								87.71	32.05	17.03	5.04	1.68	0.57
		v								9.87	6.60	5.12	3.13	2.00	1.28
44.00	158.40	R								95.87	35.00	18.59	5.49	1.84	0.62
		v								10.34	6.92	5.36	3.27	2.09	1.34
46.00	165.60	R									38.08	20.21	5.97	1.99	0.67
		v									7.23	5.61	3.42	2.19	1.40
48.00	172.80	R									41.28	21.90	6.46	2.15	0.72
		v									7.55	5.85	3.57	2.28	1.46
50.00	180.00	R									44.61	23.66	6.97	2.32	0.78
		v									7.86	6.10	3.72	2.38	1.52
52.00	187.20	R									48.07	25.48	7.50	2.50	0.84
		v									8.17	6.34	3.87	2.47	1.58
54.00	194.40	R									51.65	27.37	8.05	2.68	0.90
		v									8.49	6.58	4.02	2.57	1.64
56.00	201.60	R									55.36	29.32	8.61	2.86	0.96
		v									8.80	6.83	4.17	2.66	1.70
58.00	208.80	R									59.20	31.34	9.20	3.06	1.02
		v									9.12	7.07	4.32	2.76	1.76
60.00	216.00	R									63.16	33.42	9.80	3.25	1.09
		v									9.43	7.31	4.47	2.85	1.82
62.00	223.20	R									67.24	35.57	10.42	3.46	1.16
		v									9.75	7.56	4.61	2.95	1.89
64.00	230.40	R									71.46	37.78	11.06	3.67	1.23
		v									10.06	7.80	4.76	3.04	1.95
66.00	237.60	R										40.06	11.72	3.88	1.30
		v										8.05	4.91	3.14	2.01
68.00	244.80	R										42.40	12.40	4.11	1.37
		v										8.29	5.06	3.23	2.07
70.00	252.00	R										44.81	13.10	4.33	1.45
		v										8.53	5.21	3.33	2.13
75.00	270.00	R										51.12	14.91	4.93	1.64
		v										9.14	5.58	3.57	2.28
80.00	288.00	R										57.84	16.85	5.56	1.85
		v										9.75	5.95	3.81	2.43
85.00	306.00	R										64.96	18.90	6.23	2.07
		v										10.36	6.33	4.04	2.59
90.00	324.00	R											21.06	6.93	2.30
		v											6.70	4.28	2.74
95.00	342.00	R											23.33	7.67	2.55
		v											7.07	4.52	2.89
100.00	360.00	R											25.72	8.45	2.80
		v											7.44	4.76	3.04
110.00	396.00	R											30.85	10.11	3.35
		v											8.19	5.23	3.35
120.00	432.00	R											36.42	11.92	3.94
		v											8.93	5.71	3.65
130.00	468.00	R											42.45	13.87	4.58
		v											9.67	6.18	3.95
140.00	504.00	R											48.94	15.96	5.26
		v											10.42	6.66	4.26
150.00	540.00	R											55.87	18.20	5.99
		v											11.16	7.14	4.56
160.00	576.00	R											63.26	20.58	6.76
		v											11.91	7.61	4.87
170.00	612.00	R											71.10	23.10	7.58
		v											12.65	8.09	5.17

4.7 Pressure Loss Fitting

Description	Symbol	Coefficient of resistance (r)
Coupling pipe		0,25
90° Elbow		2,00
45° Elbow		0,60
Identical 90° Tee		1,80
Reduced 90° Tee		3,60
Identical 90° Tee		1,30
Reduced 90° Tee		2,60
Identical 90° Tee		4,20
Reduced 90° Tee		9,00
Identical 90° Tee		2,20
Reduced 90° Tee		5,00
Male threaded 90° Tee		0,80
Concentric reduction pipe up 2 dim.		0,55
Concentric reduction pipe up 3 dim.		0,85
Male threaded joint		0,40
Reduced male threaded joint		0,85
Male threaded elbow		2,20
Reduced male threaded elbow		3,50

Figure 4

4.8 Linear Deformation under Heat Influence

The calculation of the linear deformation is based on the laying temperature. The following example gives you an idea of how to calculate.

Example for a pipe length of 8 m:

1. Lowest pipe wall temperature + 9° C (cold water pipe)

Difference 7° K

2. Laying temperature + 16° C

Difference 54° K

3. Highest pipe wall temperature + 70° C (hot water pipe)

To 1. Shortening of the pipe: $8 \text{ m} \times 7^\circ \times 0,03 = 1,68 \text{ mm}$

To 3. Extension of the pipe: $8 \text{ m} \times 54^\circ \times 0,03 = 12,96 \text{ mm}$

Diagram and chart to determine the temperature-dependent linear expansion of PP-R and PP-RCT monolayer pipes.

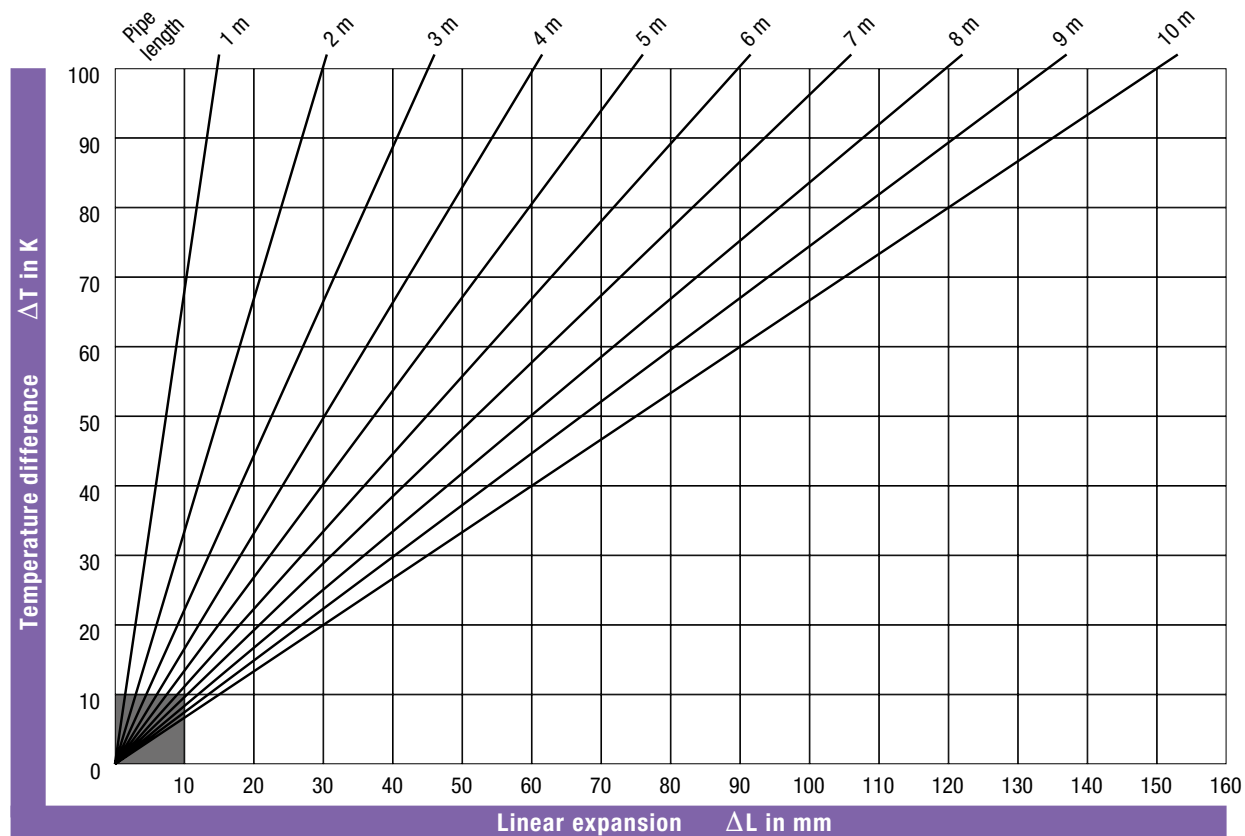


Figure 5

Pipe length	Temperature difference ΔT in K									
	10	20	30	40	50	60	70	80	90	100
0,1 m	0,15	0,30	0,45	0,60	0,75	0,90	1,05	1,20	1,35	1,50
0,2 m	0,30	0,60	0,90	1,20	1,50	1,80	2,10	2,40	2,70	3,00
0,3 m	0,45	0,90	1,35	1,80	2,25	2,70	3,15	3,60	4,05	4,50
0,4 m	0,60	1,20	1,80	2,40	3,00	3,60	4,20	4,80	5,40	6,00
0,5 m	0,75	1,50	2,25	3,00	3,75	4,50	5,25	6,00	6,75	7,50
0,6 m	0,90	1,80	2,70	3,60	4,50	5,40	6,30	7,20	8,10	9,00
0,7 m	1,05	2,10	3,15	4,20	5,25	6,30	7,35	8,40	9,45	10,50
0,8 m	1,20	2,40	3,60	4,80	6,00	7,20	8,40	9,60	10,80	12,00
0,9 m	1,35	2,70	4,05	5,40	6,75	8,10	9,45	10,80	12,15	13,50
1,0 m	1,50	3,00	4,50	6,00	7,50	9,00	10,50	12,00	13,50	15,00
2,0 m	3,00	6,00	9,00	12,00	15,00	18,00	21,00	24,00	27,00	30,00
3,0 m	4,50	9,00	13,50	18,00	22,50	27,00	31,50	36,00	40,50	45,00
4,0 m	6,00	12,00	18,00	24,00	30,00	36,00	42,00	48,00	54,00	60,00
5,0 m	7,50	15,00	22,50	30,00	37,50	45,00	52,50	60,00	67,50	75,00
6,0 m	9,00	18,00	27,00	36,00	45,00	54,00	63,00	72,00	81,00	90,00
7,0 m	10,50	21,00	31,50	42,00	52,50	63,00	73,50	84,00	94,50	105,00
8,0 m	12,00	24,00	36,00	48,00	60,00	72,00	84,00	96,00	108,00	120,00
9,0 m	13,50	27,00	40,50	54,00	67,50	81,00	94,50	108,00	121,50	135,00
10,0 m	15,00	30,00	45,00	60,00	75,00	90,00	105,00	120,00	135,00	150,00

Figure 5/1

Diagram and chart to determine the temperature dependent linear expansion of multilayer pipes with inlayed fiberglass layer

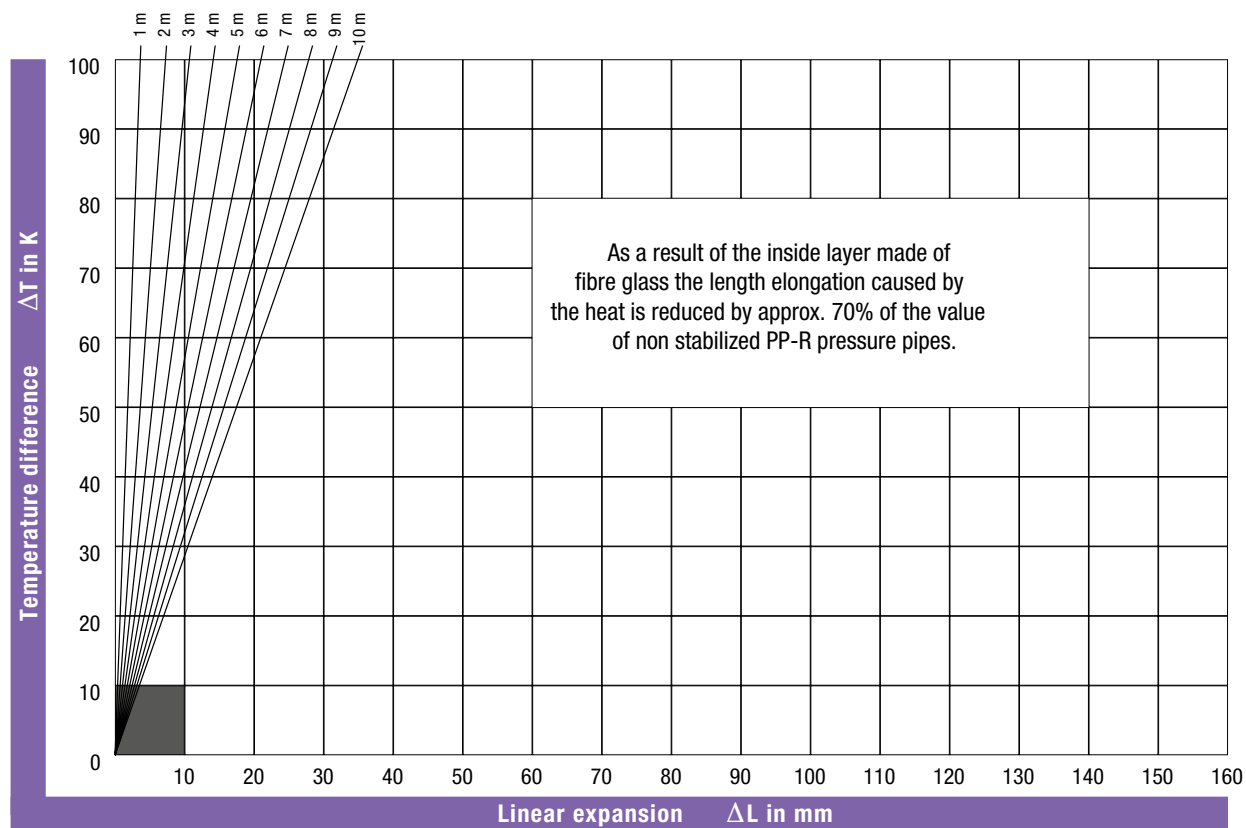


Figure 6

Pipe length	Temperature difference ΔT in K									
	10	20	30	40	50	60	70	80	90	100
0,1 m	0,04	0,07	0,11	0,14	0,18	0,21	0,25	0,28	0,32	0,35
0,2 m	0,07	0,14	0,21	0,28	0,35	0,42	0,49	0,56	0,63	0,70
0,3 m	0,11	0,21	0,32	0,42	0,53	0,63	0,74	0,84	0,95	1,05
0,4 m	0,14	0,28	0,42	0,56	0,70	0,84	0,98	1,12	1,26	1,40
0,5 m	0,18	0,35	0,53	0,70	0,88	1,05	1,23	1,40	1,58	1,75
0,6 m	0,21	0,42	0,63	0,84	1,05	1,26	1,47	1,68	1,89	2,10
0,7 m	0,25	0,49	0,74	0,98	1,23	1,47	1,72	1,96	2,21	2,45
0,8 m	0,28	0,56	0,84	1,12	1,40	1,68	1,96	2,24	2,52	2,80
0,9 m	0,32	0,63	0,95	1,26	1,58	1,89	2,21	2,52	2,84	3,15
1,0 m	0,35	0,70	1,05	1,40	1,75	2,10	2,45	2,80	3,15	3,50
2,0 m	0,70	1,40	2,10	2,80	3,50	4,20	4,90	5,60	6,30	7,00
3,0 m	1,05	2,10	3,15	4,20	5,25	6,30	7,35	8,40	9,45	10,50
4,0 m	1,40	2,80	4,20	5,60	7,00	8,40	9,80	11,20	12,60	14,00
5,0 m	1,75	3,50	5,25	7,00	8,75	10,50	12,25	14,00	15,75	17,50
6,0 m	2,10	4,20	6,30	8,40	10,50	12,60	14,70	16,80	18,90	21,00
7,0 m	2,45	4,90	7,35	9,80	12,25	14,70	17,15	19,60	22,05	24,50
8,0 m	2,80	5,60	8,40	11,20	14,00	16,80	19,60	22,40	25,20	28,00
9,0 m	3,15	6,30	9,45	12,60	15,75	18,90	22,05	25,20	28,35	31,50
10,0 m	3,50	7,00	10,50	14,00	17,50	21,00	24,50	28,00	31,50	35,00

Figure 6/1

4.9 Distance Between the Supporting Points

The type and number of the pipe fixings, depend on the type of structure and on the longitudinal expansion. The fixed points must divide the pipe into sections in which contraction or expansion would take place; at any rate expansion must never occur on the inserts or on the fillings. The different pipe sections are maintained by sliding hinges. The distance between those fixings, i.e. the

distance between the supporting points, depend on the working conditions and on the weight of the pipes (including that of the fluid which is conveyed). On a practical point of view, the distance between the supporting points reported in the following tables have been found to be valid.

Distance between the supporting points in cm. at different temperatures

Ø mm.	20° C	30° C	40° C	50° C	60° C	70° C	80° C
16	60	60	60	55	45	45	40
20	65	65	60	60	60	55	50
25	75	75	70	70	65	60	55
32	90	90	85	85	80	75	70
40	110	110	105	100	95	90	85
50	125	120	115	110	105	100	90
63	140	135	130	125	120	115	105
75	155	150	145	135	130	125	115
90	165	160	155	145	140	130	120
110	185	180	170	165	155	150	140
125	190	185	180	170	160	155	150

Figure 7 Distance for K-Pipe 2,0 MPa

Ø mm.	20° C	30° C	40° C	50° C	60° C		
20	60	55	50	45	40		
25	75	70	65	60	55		
32	90	85	75	70	65		
40	100	95	90	85	75		
50	120	115	105	100	90		
63	140	130	120	110	100		
75	150	145	135	125	115		
90	160	155	150	145	130		
110	180	170	160	155	140		
125	190	185	175	165	150		
160	200	195	185	175	160		
200	245	235	225	245	205		
250	275	265	255	245	235		
315	290	280	270	260	250		

Figure 8 Distance for K-Pipe 1,0 MPa

Distance between the supporting points in cm. at different temperatures

Ø mm.	20° C	30° C	40° C	50° C	60° C	70° C	80° C
20	100	90	85	85	80	70	65
25	105	100	95	90	85	80	75
32	120	115	110	105	100	95	90
40	130	125	120	115	110	105	100
50	150	145	140	135	130	125	120
63	160	155	150	145	140	135	130
75	180	175	170	165	160	155	145
90	190	185	180	175	170	165	150
110	200	195	190	180	175	170	160
125	220	210	205	195	185	175	165
160	220	210	205	195	185	175	165
200	245	235	230	220	210	200	190
250	275	265	255	245	235	225	210
315	310	300	290	275	265	260	250

Figure 9 Distance for K-Fiber Pipe 2,0 MPa

Ø mm.	20° C	30° C	40° C	50° C	60° C	70° C	80° C
20	80	80	75	75	70	60	55
25	95	90	85	80	75	70	65
32	110	105	100	95	90	85	80
40	120	115	110	105	100	95	90
50	140	135	130	125	120	115	110
63	150	145	140	135	130	125	120
75	165	160	155	150	145	140	130
90	175	170	165	160	155	150	135
110	185	180	175	165	160	155	145
125	205	195	190	180	170	160	150
160	205	195	190	180	170	160	150
200	230	220	210	200	190	180	170
250	250	240	230	220	210	200	185
315	290	285	275	265	255	250	240

Figure 10 Distance for K-Fiber Pipe 1,6 MPa

Linear extension compension of pp-r pipes

The linear extension of a PP-R pipe can in most of the cases be compensated by a change in direction. Attention has to be paid to the fact that the pipeline can easily move in axial direction. Should linear extension compensation by directational change not be possible the fitting in of an expansion bend is required. Axial bellow expansion joints are mostly unfit and uneconomical for optimum resiliency of the pipe. The size bending limb is important this calculated by the

opposite formula. The figures 11 and 12 show the effects of the linear deformation and its compension with regard bending limbs L_s make sure to chose the correct locating points.

$$L_s = C \times \sqrt{d \times \Delta L} \quad (\text{mm})$$

L_s = Length of bending limb (mm)
 d = Outside pipe diameter (mm)
 ΔL = Linear deformation (mm)
 C = Material depending constant
 for pp-r = 20

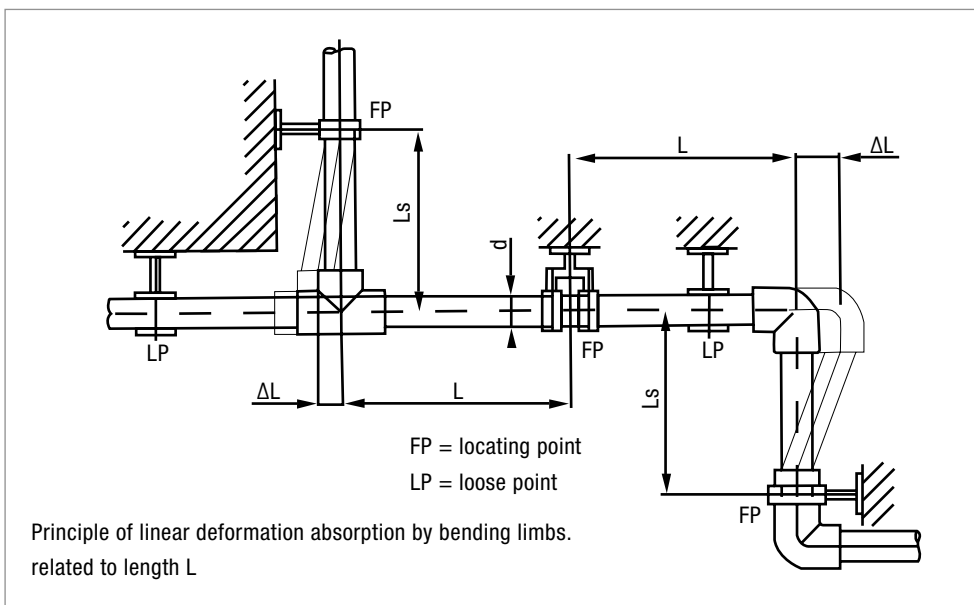


Figure 11

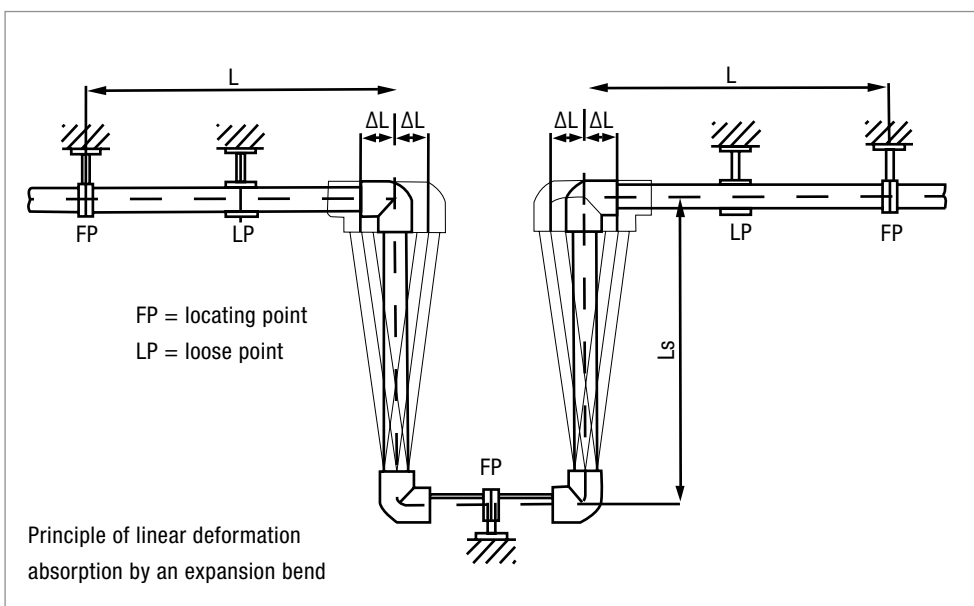


Figure 12

Construction of expansion bends:

Expansion bends can easily be made right at the site.

Besides the required pipe length 4 elbows (AQ090)

To construct an expansion bend, the bending limb L is calculated in dependence on the linear deformation ΔL . As standard value, the L value given in the fig. 14 diagram can be used.

Spacing B should be at least $10 \times d$.

Expansion bend, made of **K-Aqua** pipe and 90° elbow. (Fig. 13)

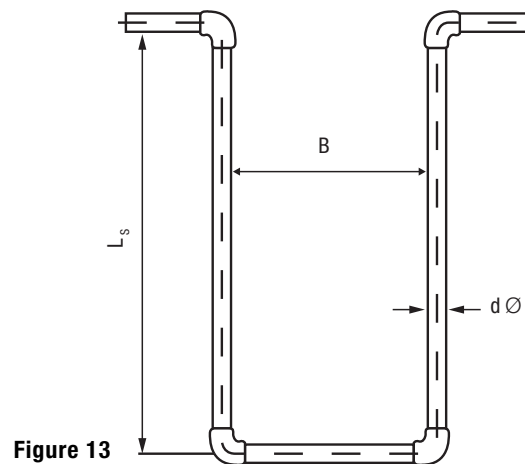


Figure 13

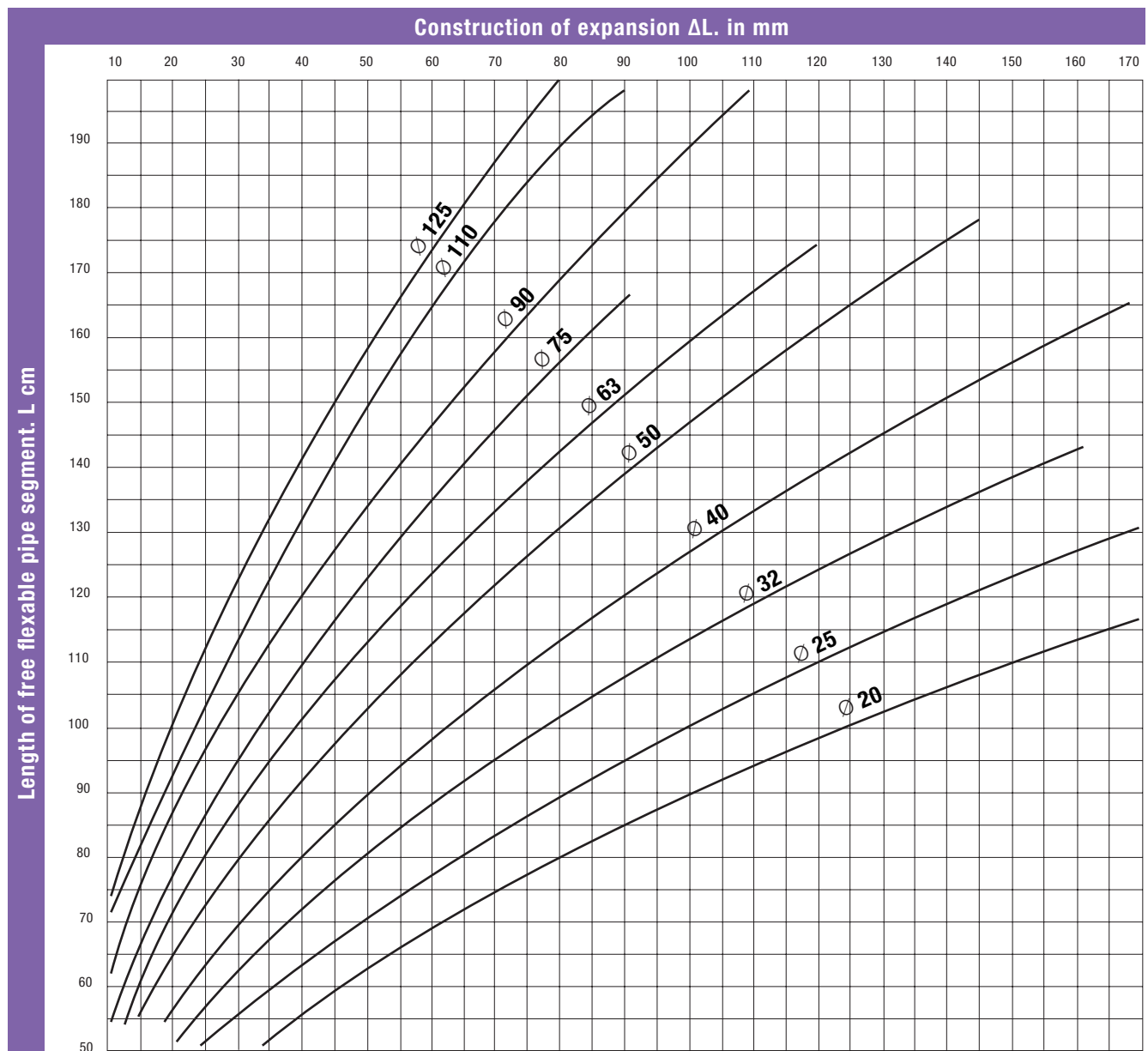


Figure 14

4.10 Length Variation of K-Aqua Pipes Due to Heat (Thermal Stress)

K-Aqua pipes are subject to a relatively high thermal expansion whenever they are exposed to temperature variations.

The longitudinal thermal expansion of these pipes is about 11 times more important than in steel pipes.

This fact must be taken into account during the starting-up phase. Therefore, as early as in the design phase, all the alternatives regarding the position or the path of the pipes must be thoroughly examined in order to compensate thermal expansion in the various pipe sections.

The coefficient of longitudinal thermal expansion for **K-Aqua** monolayer pipes is:

$$\epsilon t = 1,5 \times 10^{-4} \text{ (K}^{-1}\text{)}.$$

K-Fiber pipes

$$\epsilon t = 0.35 \times 10^{-4} \text{ (K}^{-1}\text{)}$$

Δl = Linear extension in (mm)

ϵt = Thermal expansion (mm)
coefficient in (m °C)

L = Pipe length (m)

Δt = Temperature difference (°K)

The linear deformation of a pipe is thus calculated according to the following formula:

$$\Delta l = \epsilon t \times L \times \Delta t \text{ (mm)}$$

4.11 System Testing

The testing of the water supply system is carried out through tests and checks during the installation (for the parts that are no longer accessible once the work is completed) and tests and final checks for the implementation of contractual obligations.

The cold hydraulic test pressure prescribed by European Standards CEN TR 12108 and EN 806-4 (Procedure C) is carried out according to the following procedure.

1. Fill the system slowly to vent it (do not fully tighten the highest plugs that will be closed when the water will come out with a continuous jet).
2. Bring the pressure up to 15 bar and repeat the process 2 more times every 10 minutes.
3. Measure the pressure after the first 30 minutes.
4. Read the pressure after another 30 minutes (1 hour). If the difference is less than 0,6 bar, there is no pressure drop and the test can be continued with the same pressure for 2 more hours.
5. In these last 2 hours the pressure must not drop more than 0,2 bar.
6. The results of this test shall be recorded.

It is also important to perform:

- Cold water supply test with cold water bleed from all utilities to check the flow rate and pressure.
- Hot water supply test with hot water bleed from all utilities to check the flow rate and pressure.
- Noise level check in accordance with the regulations in force.

N.B.:

Testing pressure shall be reduced if radiators, taps or valves are present.





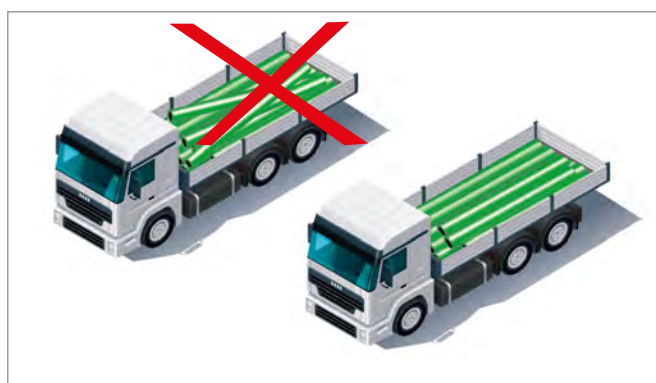
5 Storage & Handling Page

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5.3	Handling	71
5.4	Dispose of waste materials	71



5.1 Transportation

During the various working phases, the surface of **K-Aqua** may be exposed nicks or cuts. In order to avoid dangerous situations due to possible cuts, all handling must be made with the upmost care. It is strongly prohibited to install damaged pipes or fittings.



5.2.2 Fittings

Fittings are packed in plastic bags, plastic bags are packed in carton boxes:

Carton box	Box dimensions LxBxH (cm)	Boxes/pallet
1	40x30x22	32 or 40
2	40x20x20	48 or 60



5.2 Packaging

5.2.1 Pipes

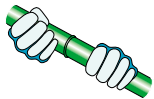
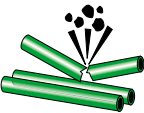
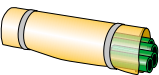
Pipes are packed in bundles, bundles are packed in foil. Pipes can be identified by the marking on the pipe.

5.3 Handling

Thanks to the material properties of polypropylene, the pipes and fittings can be stored for a long time under varying temperatures. The storage of pipes and fittings must be in accordance with the following conditions:

- 1. The pipes should be supported along their full length.
- 2. Bending of the pipes is to be avoided.
- 3. The material becomes sensitive to impact at low temperatures and in particular at temperatures below 0° C. For this reason knocks and similar impacts are to be avoided under these conditions.
- 4. High-polymer materials are sensitive to UV radiation. For this reason the **K-Aqua** material should also be protected against the effects of UV radiation.

Suggestions for the correct treatment of pipe systems

	
 Avoid hard knocks or impacts at the end of a pipe	 Put pipes down carefully
 Do not use pipes which are damaged or cracked at the interface	 Cut pipes only with sharp tools
 Do not twist either pipe or fitting after joining together	 Alignment corrections of up to 5 relative to the axis of the pipe can be carried out immediately after joining
 Do not expose to UV-radiation for any extended period of time	 Store protected from sun and rain
 Provide protection against hard impacts and falling stones on the site	 Cover pipes at risk to prevent damage

5.4 Dispose of Waste Materials

Dispose waste material separated according to the regulations. PP pipes and fittings are recyclable.

Transition fittings	Recyclable, after separation of PP and brass
Gaskets	general waste
Cardboard boxes	recyclable
Plastic bags	recyclable
Chips	general waste
Wipes	general waste

International regulation on drinking water for human consumption prevents to apply recycled material in the production process for water supply systems.



6

Products

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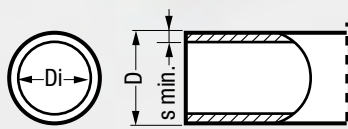
6.1	Pipes	75 - 82
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Pipes

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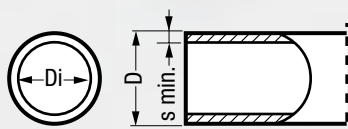
K-Pipe PP-R SDR 6	76
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K-Pipe „purple“ PP-R SDR 11	77
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K-Fiber UV Pipe PP-R SDR 7,4	82

„K Pipe PP-R” SDR 6 – S 2,5 (20° C/2,0 MPa – 70° C/1,0 MPa), length 4 meter


Material: PP-R
 Colour: green with 1 red stripe
 Standards: DIN EN ISO 15874
 DIN 8077 / 8078

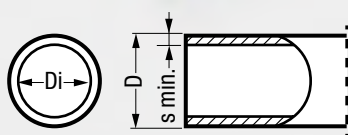
Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
AQ200P20*	20	12	13,2	3,4	100	0,18	0,14
AQ200P25*	25	15	16,6	4,2	100	0,28	0,22
AQ200P32*	32	20	21,2	5,4	60	0,46	0,35
AQ200P40*	40	25	26,6	6,7	40	0,68	0,56
AQ200P50*	50	32	33,2	8,3	20	1,09	0,87
AQ200P63*	63	40	42,0	10,5	20	1,60	1,39
AQ200P75	75	50	50,0	12,5	12	2,50	1,96
AQ200P90	90	-	60,0	15,0	8	3,30	2,83
AQ200P110	110	65	73,2	18,3	4	5,00	4,21
AQ200P125	125	80	83,2	20,8	4	6,50	5,46

* pipe can be delivered in 5,80 meter length on special request with product code AQ258P+dimension

„K Pipe PP-RCT” SDR 7,4 – S 3,2 (20° C/2,0 MPa – 70° C/1,0 MPa), length 4 meter


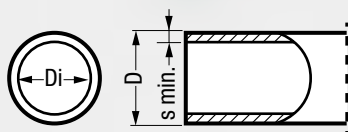
Material: PP-RCT
 Colour: green with 1 red stripe
 Standards: DIN EN ISO 15874
 DIN 8077 / 8078

Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
AQ20020	20	15	14,4	2,8	100	0,16	0,16
AQ20025	25	20	18,0	3,5	100	0,23	0,25
AQ20032	32	25	23,2	4,4	60	0,37	0,42
AQ20040	40	32	29,0	5,5	40	0,57	0,66
AQ20050	50	40	36,2	6,9	20	0,88	1,03
AQ20063	63	50	45,8	8,6	20	1,39	1,63
AQ20075	75	-	54,4	10,3	12	1,98	2,31
AQ20090	90	65	65,4	12,3	8	2,83	3,32
AQ200110	110	80	79,8	15,1	4	4,25	4,97
AQ200125	125	90	90,8	17,1	4	5,41	6,47
AQ200160	160	115	116,2	21,9	4	9,09	10,60
AQ200200	200	145	145,2	27,4	4	13,70	16,55
AQ200250	250	180	181,6	34,2	4	21,25	25,89
AQ200315	315	250	232,80	41,1	4	33,69	39,39

„K Pipe PP-R” SDR 11 – S 5 (20° C/1,2 MPa – 60° C/0,6 MPa), length 4 meter


Material: PP-R
 Colour: green with 1 blue stripe
 Standards: DIN EN ISO 15874
 DIN 8077 / 8078

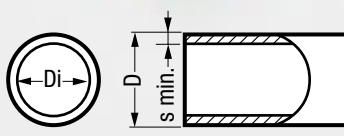
Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
AQ111P20	20	15	16,2	1,9	100	0,11	0,21
AQ111P25	25	20	20,4	2,3	100	0,16	0,33
AQ111P32	32	25	26,2	2,9	60	0,26	0,54
AQ111P40	40	32	32,6	3,7	40	0,41	0,83
AQ111P50	50	40	40,8	4,6	20	0,64	1,31
AQ111P63	63	50	51,4	5,8	20	1,01	2,07
AQ111P75	75	-	61,4	6,8	12	1,41	2,96
AQ111P90	90	65	73,6	8,2	8	2,03	4,25
AQ111P110	110	80	90,0	10,0	4	3,01	6,36
AQ111P125	125	100	102,2	11,4	4	3,91	8,20
AQ111P160	160	125	130,8	14,6	4	6,38	13,43
AQ111P200	200	160	163,6	18,2	4	9,76	21,01
AQ111P250	250	200	204,6	22,7	4	15,00	32,86
AQ111P315	315	250	257,8	28,6	4	24,67	52,17
AQ111P355	355	300	290,60	32,2	4	31,93	66,29
AQ111P400	400	-	327,40	36,3	4	40,31	84,14
AQ111P450	450	350	368,20	40,9	4	49,96	106,28
AQ111P500	500	400	409,20	45,4	4	61,49	131,32

„K Pipe purple PP-R” SDR 11 – S 5 (20° C/1,2 MPa – 60° C/0,6 MPa), length 4 meter


Material: PP-R
 Colour: outside layer purple
 inside layer green
 Standards: DIN EN ISO 15874
 DIN 8077 / 8078

Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
AQ111PL20	20	15	16,2	1,9	100	0,11	0,21
AQ111PL25	25	20	20,4	2,3	100	0,16	0,33
AQ111PL32	32	25	26,2	2,9	60	0,26	0,54
AQ111PL40	40	32	32,6	3,7	40	0,41	0,83
AQ111PL50	50	40	40,8	4,6	20	0,64	1,31
AQ111PL63	63	50	51,4	5,8	20	1,01	2,07
AQ111PL75	75	-	61,4	6,8	12	1,41	2,96
AQ111PL90	90	65	73,6	8,2	8	2,03	4,25
AQ111PL110	110	80	90	10,0	4	3,01	6,36
AQ111PL125	125	100	102,2	11,4	4	3,91	8,20
AQ111PL160	160	125	130,8	14,6	4	6,38	13,43

„K-Fiber Pipe PP-RCT” SDR 7,4 – S 3,2 (20° C/2,0 MPa – 70° C/1,0 MPa), length 4 meter

Material: PP-RCT GF

Colour: green with 4 red stripes

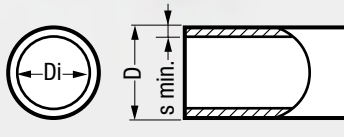
Standards: DIN EN ISO 15874
DIN 8077 / 8078

Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
AQ200F20*	20	15	14,4	2,8	100	0,16	0,16
AQ200F25*	25	20	18,0	3,5	100	0,25	0,25
AQ200F32*	32	25	23,2	4,4	60	0,39	0,42
AQ200F40*	40	32	29,0	5,5	40	0,59	0,66
AQ200F50*	50	40	36,2	6,9	20	0,91	1,03
AQ200F63*	63	50	45,8	8,6	20	1,45	1,63
AQ200F75*	75	-	54,4	10,3	12	2,06	2,31
AQ200F90*	90	65	65,4	12,3	8	2,94	3,32
AQ200F110*	110	80	79,8	15,1	4	4,36	4,97
AQ200F125*	125	90	90,8	17,1	4	5,61	6,47
AQ200F160*	160	115	116,2	21,9	4	9,09	10,60
AQ200F200*	200	145	145,2	27,4	4	14,23	16,55
AQ200F250*	250	180	181,6	34,2	4	22,08	25,89
AQ200F315*	315	230	229,8	43,1	4	34,89	39,39

* pipe can be delivered in 5,80 meter length on special request with product code AQ258F+dimension

„K-Fiberclima Pipe PP-RCT” SDR 11 – S 5 (20° C/1,6 MPa – 70° C/0,8 MPa), length 4 meter

	Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
	AQ160F20	20	15	14,4	2,8*	100	0,16	0,16
	AQ160F25	25	20	18,0	3,5*	100	0,25	0,25
	AQ160F32	32	25	26,2	2,9	60	0,28	0,54
	AQ160F40	40	32	32,6	3,7	40	0,43	0,83
	AQ160F50	50	40	40,8	4,6	20	0,67	1,31
	AQ160F63	63	50	51,4	5,8	20	1,04	2,07
	AQ160F75	75	-	61,4	6,8	12	1,44	2,96
	AQ160F90	90	65	73,6	8,2	8	2,08	4,25
	AQ160F110	110	80	90,0	10,0	4	3,10	6,36
	AQ160F125	125	100	102,2	11,4	4	4,02	8,20
	AQ160F160	160	125	130,8	14,6	4	6,50	13,43
	AQ160F200	200	160	163,6	18,2	4	10,09	21,01
	AQ160F250	250	200	204,6	22,7	4	15,01	32,86
	AQ160F315	315	250	257,8	28,6	4	24,67	52,17

	AQ160F355	355	300	290,60	32,2	4	31,93	66,29
	AQ160F400	400	-	327,40	36,3	4	40,31	84,14
	AQ160F450	450	350	368,20	40,9	4	49,96	106,28
	AQ160F500	500	400	409,20	45,4	4	61,49	131,32

Material:

PP-RCT GF

Colour:

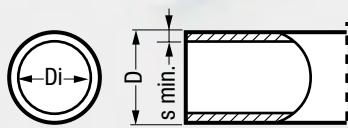
green with 4 blue stripes

Standards:

DIN EN ISO 15874

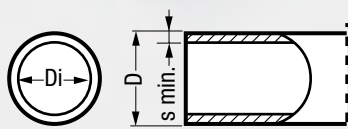
DIN 8077 / 8078

* SDR 7,4 - S 3,2

„K-Fiber Pipe PP-R” SDR 6 – S 2,5 (20° C/2,0 MPa – 70° C/1,0 MPa), length 4 meter


Material: PP-R GF
 Colour: green with 4 red stripes
 Standards: DIN EN ISO 15874
 DIN 8077 / 8078

Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
AQ206PF20*	20	12	13,2	3,4	100	0,18	0,14
AQ206PF25*	25	15	16,6	4,2	100	0,28	0,22
AQ206PF32*	32	20	21,2	5,4	60	0,46	0,35
AQ206PF40*	40	25	26,6	6,7	40	0,68	0,56
AQ206PF50*	50	32	33,2	8,3	20	1,09	0,87
AQ206PF63*	63	40	42,0	10,5	20	1,60	1,39
AQ206PF75	75	50	50,0	12,5	12	2,50	1,96
AQ206PF90	90	-	60,0	15,0	8	3,30	2,83
AQ206PF110	110	65	73,2	18,3	4	5,00	4,21
AQ206PF125	125	80	83,2	20,8	4	6,50	5,46

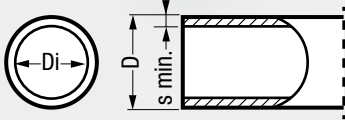
„K-Fiber Pipe PP-R” SDR 7,4 – S 3,2 (20° C/2,0 MPa – 60° C/1,0 MPa), length 4 meter


Material: PP-R GF
 Colour: green with 4 red stripes
 Standards: DIN EN ISO 15874
 DIN 8077 / 8078

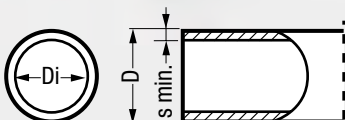
Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
AQ207PF20*	20	15	14,4	2,8	100	0,16	0,16
AQ207PF25*	25	20	18,0	3,5	100	0,25	0,25
AQ207PF32*	32	25	23,2	4,4	60	0,39	0,42
AQ207PF40*	40	32	29,0	5,5	40	0,59	0,66
AQ207PF50*	50	40	36,2	6,9	20	0,91	1,03
AQ207PF63*	63	50	45,8	8,6	20	1,45	1,63
AQ207PF75*	75	-	54,4	10,3	12	2,06	2,31
AQ207PF90*	90	65	65,4	12,3	8	2,94	3,32
AQ207PF110*	110	80	79,8	15,1	4	4,36	4,97
AQ207PF125*	125	90	90,8	17,1	4	5,61	6,47
AQ207PF160*	160	115	116,2	21,9	4	9,09	10,60
AQ207PF200*	200	145	145,2	27,4	4	14,23	16,55
AQ207PF250*	250	180	181,6	34,2	4	22,08	25,89
AQ207PF315*	315	230	229,8	43,1	4	34,89	39,39

* pipe can be delivered in 5,80 meter length on special request with product code AQ258PF+dimension

„K-Fiber Pipe PP-R” SDR 9 – S 4 (20° C/1,6 MPa – 60° C/0,8 MPa), length 4 meter

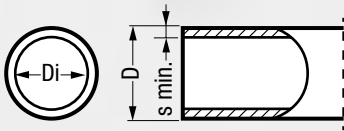
  Material: PP-R GF Colour: green with 4 grey stripes Standards: DIN EN ISO 15874 DIN 8077 / 8078	Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
	AQ169PF32	32	25	24,80	3,6	60	0,33	0,48
	AQ169PF40	40	32	31,00	4,5	40	0,52	0,75
	AQ169PF50	50	40	38,80	5,6	20	0,80	1,18
	AQ169PF63	63	50	48,80	7,1	20	1,25	1,87
	AQ169PF75	75	-	58,20	8,4	12	1,77	2,66
	AQ169PF90	90	65	69,80	10,1	8	2,55	3,83
	AQ169PF110	110	80	85,40	12,3	4	3,78	5,73
	AQ169PF125	125	100	97,00	14,0	4	4,89	7,39
	AQ169PF160	160	125	124,20	17,9	4	7,70	12,12
	AQ169PF200	200	160	155,20	22,4	4	12,35	18,92
	AQ169PF250	250	200	194,20	27,9	4	19,08	29,62
	AQ169PF315	315	250	244,60	35,2	4	30,10	46,99
	AQ169PF355	355	300	275,60	39,7	4	38,14	59,66

„K-Fiber Pipe PP-R” SDR 11 – S 5 (20° C/1,2 MPa – 60° C/0,6 MPa), length 4 meter

  Material: PP-R GF Colour: green with 4 blue stripes Standards: DIN EN ISO 15874 DIN 8077 / 8078	Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
	AQ111PF20	20	15	14,4	2,8*	100	0,16	0,16
	AQ111PF25	25	20	18,0	3,5*	100	0,25	0,25
	AQ111PF32	32	25	26,2	2,9	60	0,28	0,54
	AQ111PF40	40	32	32,6	3,7	40	0,43	0,83
	AQ111PF50	50	40	40,8	4,6	20	0,67	1,31
	AQ111PF63	63	50	51,4	5,8	20	1,04	2,07
	AQ111PF75	75	-	61,4	6,8	12	1,44	2,96
	AQ111PF90	90	65	73,6	8,2	8	2,08	4,25
	AQ111PF110	110	80	90,0	10,0	4	3,10	6,36
	AQ111PF125	125	100	102,2	11,4	4	4,02	8,20
	AQ111PF160	160	125	130,8	14,6	4	6,50	13,43
	AQ111PF200	200	160	163,6	18,2	4	10,09	21,01
	AQ111PF250	250	200	204,6	22,7	4	15,01	32,86
	AQ111PF315	315	250	257,8	28,6	4	24,67	52,17
	AQ111PF355	355	300	290,60	32,2	4	31,93	66,29
	AQ111PF400	400	350	327,40	36,3	4	40,31	84,14
	AQ111PF450	450	400	368,20	40,9	4	49,96	106,28
	AQ111PF500	500	450	409,20	45,4	4	61,49	131,32

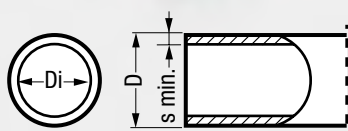
* SDR 7,4 - S 3,2

„K-Fiber Pipe PP-R” SDR 17 – S 8 (20° C/0,8 MPa – 60° C/0,4 MPa), length 4 meter



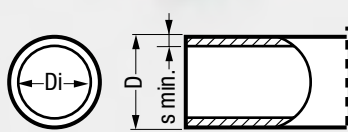
Material: PP-R GF
 Colour: green with 4 red stripes
 Standards: DIN EN ISO 15874
 DIN 8077 / 8078

Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
AQ117PF90	90	65	79,20	5,4	8	1,46	4,93
AQ117PF110	110	80	96,80	6,6	4	2,16	7,36
AQ117PF125	125	100	110,20	7,4	4	2,77	9,54
AQ117PF160	160	125	141,00	9,5	4	4,52	15,61
AQ117PF200	200	160	176,20	11,9	4	7,05	24,38
AQ117PF250	250	200	220,40	14,8	4	10,90	38,15
AQ117PF315	315	250	277,60	18,7	4	17,26	60,52
AQ117PF355	355	300	312,80	21,1	4	21,83	76,85
AQ117PF400	400	-	352,60	23,7	4	27,61	97,65
AQ117PF450	450	400	396,60	26,7	4	34,16	123,54
AQ117PF500	500	450	440,60	29,7	4	42,07	152,47
AQ117PF560	560	-	493,60	33,2	4	52,51	191,36
AQ117PF630	630	500	555,20	37,4	4	66,34	242,10

„K-Fiber UV Pipe PP-RCT” SDR 7,4 – S 3,2 (20° C/2,0 MPa – 70° C/1,0 MPa), length 4 meter


Material: PP-RCT GF
 Colour: outside layer black
 inside layer green
 Standards: DIN EN ISO 15874
 DIN 8077 / 8078

Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
AQ200FUV20	20	15	14,4	2,8	100	0,16	0,16
AQ200FUV25	25	20	18,0	3,5	100	0,25	0,25
AQ200FUV32	32	25	23,2	4,4	60	0,39	0,42
AQ200FUV40	40	32	29,0	5,5	40	0,59	0,66
AQ200FUV50	50	40	36,2	6,9	20	0,91	1,03
AQ200FUV63	63	50	45,8	8,6	20	1,45	1,63
AQ200FUV75	75	-	54,4	10,3	12	2,06	2,31
AQ200FUV90	90	65	65,4	12,3	8	2,94	3,32
AQ200FUV110	110	80	79,8	15,1	4	4,36	4,97
AQ200FUV125	125	90	90,8	17,1	4	5,61	6,47
AQ200FUV160	160	115	116,2	21,9	4	9,09	10,60

„K-Fiber UV Pipe PP-R” SDR 7,4 – S 3,2 (20° C/2,0 MPa – 60° C/1,0 MPa), length 4 meter


Material: PP-R GF
 Colour: outside layer black
 inside layer green
 Standards: DIN EN ISO 15874
 DIN 8077 / 8078

Code	D	DN	Di	S min.	Pack.	Weight kg/m	Water capacity l/m
AQ200PFUV20	20	15	14,4	2,8	100	0,16	0,16
AQ200PFUV25	25	20	18,0	3,5	100	0,25	0,25
AQ200PFUV32	32	25	23,2	4,4	60	0,39	0,42
AQ200PFUV40	40	32	29,0	5,5	40	0,59	0,66
AQ200PFUV50	50	40	36,2	6,9	20	0,91	1,03
AQ200PFUV63	63	50	45,8	8,6	20	1,45	1,63
AQ200PFUV75	75	-	54,4	10,3	12	2,06	2,31
AQ200PFUV90	90	65	65,4	12,3	8	2,94	3,32
AQ200PFUV110	110	80	79,8	15,1	4	4,36	4,97
AQ200PFUV125	125	90	90,8	17,1	4	5,61	6,47
AQ200PFUV160	160	115	116,2	21,9	4	9,09	10,60

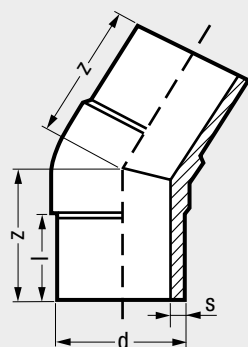
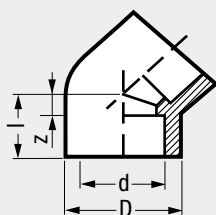


Fittings

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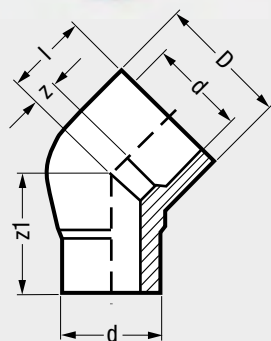
Elbow 45°



Code	d	D	l	z	s	kg	Pack.
AQ04520	20	29	21	6	-	0,02	300
AQ04525	25	35	24	8	-	0,02	200
AQ04532	32	44	29	10	-	0,05	100
AQ04540	40	52	32	11	-	0,06	70
AQ04550	50	65	37	13	-	0,10	48
AQ04563	63	84	44	16	-	0,21	24
AQ04575	75	99	50	20	-	0,32	14
AQ04590	90	120	58	25	-	0,58	9
AQ045110	110	148	69	32	-	0,98	4
AQ045125	125	165	78	37	-	1,53	2
SDR 11*							
AQ045160	160	-	143	190	14,6	2,6	1
AQ045200	200	-	153	207	18,2	4,5	1
AQ045250	250	-	134	220	22,7	7,9	1
AQ045315	315	-	155	250	28,6	13,7	1

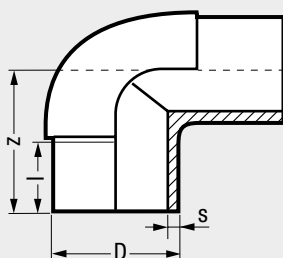
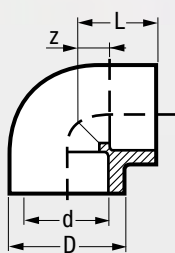
* SDR 11 jointing techniques: butt-fusion or electrofusion welding

Elbow 45° (Female/male)



Code	d	D	l	z	z1	kg	Pack.
AQ04120	20	29	20	5	28	0,02	300
AQ04125	25	34	22	6	34	0,02	200

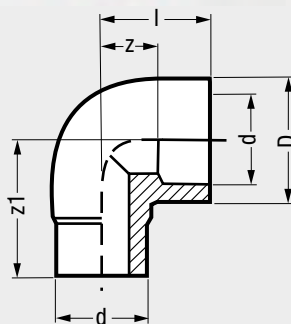
Elbow 90°



Code	d	D	L	l	z	s	kg	Pack.
AQ09020	20	29	27	-	14	-	0,02	300
AQ09025	25	35	31	-	16	-	0,02	180
AQ09032	32	44	37	-	20	-	0,05	100
AQ09040	40	52	44	-	23	-	0,07	60
AQ09050	50	65	52	-	28	-	0,14	36
AQ09063	63	84	62	-	34	-	0,27	22
AQ09075	75	101	71	-	41	-	0,44	10
AQ09090	90	120	83	-	50	-	0,79	6
AQ090110	110	148	99	-	62	-	1,30	4
AQ090125	125	165	125	-	84	-	2,17	2
SDR 11*								
AQ090160	160	-	-	143	238	14,6	3,2	1
AQ090200	200	-	-	153	262	18,2	5,7	1
AQ090250	250	-	-	133	292	22,7	10,4	1
AQ090315	315	-	-	154	370	28,6	19,6	1

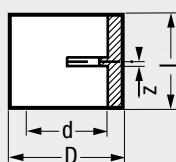
* SDR 11 jointing techniques: butt-fusion or electrofusion welding

Elbow 90° (Female/male)



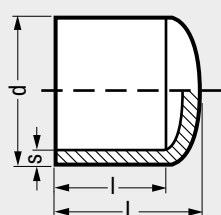
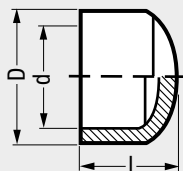
Code	d	D	l	z	z1	kg	Pack.
AQ09120	20	29	27	12	36	0,02	300
AQ09125	25	34	30	14	41	0,02	180

Socket



Code	d	D	l	z	kg	Pack.
AQ27020	20	29	34	5	0,01	500
AQ27025	25	35	37	5	0,02	300
AQ27032	32	44	41	5	0,03	160
AQ27040	40	52	46	5	0,05	80
AQ27050	50	65	52	5	0,07	60
AQ27063	63	84	60	5	0,13	45
AQ27075	75	99	65	5	0,20	28
AQ27090	90	120	76	10	0,35	15
AQ270110	110	148	80	6	0,57	10
AQ270125	125	160	102	7	0,90	5

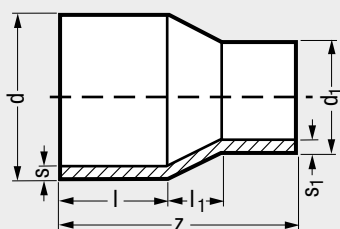
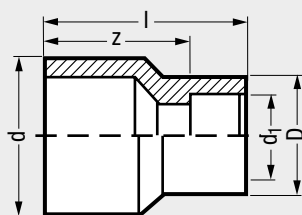
Cap



Code	d	D	l	L	s	kg	Pack.
AQ30120	20	29	25	-	-	0,01	600
AQ30125	25	34	28	-	-	0,01	400
AQ30132	32	43	32	-	-	0,02	255
AQ30140	40	52	36	-	-	0,03	160
AQ30150	50	65	41	-	-	0,06	100
AQ30163	63	79	48	-	-	0,09	60
AQ30175	75	99	54	-	-	0,18	30
AQ30190	90	120	66	-	-	0,35	18
AQ301110	110	148	79	-	-	0,59	10
AQ301125	125	162	87	-	-	0,85	5
SDR 11*							
AQ301160	160	-	-	162	14,6	1,1	3
AQ301200	200	-	-	180	18,2	2	1
AQ301250	250	-	-	217	22,7	5	1
AQ301315	315	-	-	256	28,6	7,6	1

* SDR 11 jointing techniques: butt-fusion or electrofusion welding

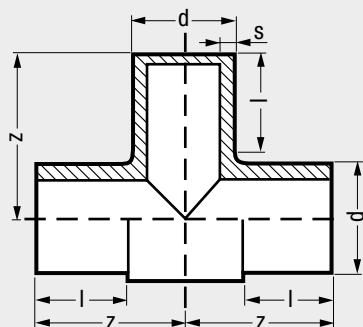
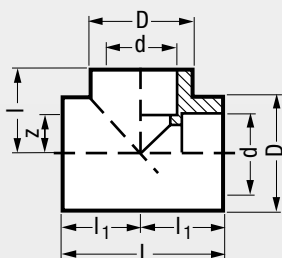
Reducing bush



Code	d	d ₁	D	l	z	l ₁	s	s ₁	kg	Pack.
AQ2432520	25	20	29	36	22	-	-	-	0,01	400
AQ2433220	32	20	29	37	23	-	-	-	0,02	325
AQ2433225	32	25	34	39	23	-	-	-	0,02	320
AQ2434020	40	20	34	43	28	-	-	-	0,02	210
AQ2434025	40	25	34	43	27	-	-	-	0,02	195
AQ2434032	40	32	43	45	27	-	-	-	0,03	180
AQ2435020	50	20	43	51	36	-	-	-	0,04	160
AQ2435025	50	25	43	51	35	-	-	-	0,04	120
AQ2435032	50	32	43	51	33	-	-	-	0,05	120
AQ2435040	50	40	52	53	33	-	-	-	0,05	80
AQ2436320	63	20	34	56	42	-	-	-	0,08	75
AQ2436325	63	25	34	56	40	-	-	-	0,08	60
AQ2436332	63	32	43	58	40	-	-	-	0,08	60
AQ2436340	63	40	52	60	40	-	-	-	0,08	60
AQ2436350	63	50	65	63	40	-	-	-	0,09	60
AQ2437550	75	50	65	67	44	-	-	-	0,12	36
AQ2437563	75	63	80	71	44	-	-	-	0,15	34
AQ2439063	90	63	80	78	51	-	-	-	0,17	24
AQ2439075	90	75	99	81	51	-	-	-	0,23	30
AQ24311063	110	63	80	88	60	-	-	-	0,32	17
AQ24311075	110	75	100	90	60	-	-	-	0,37	14
AQ24311090	110	90	110	93	61	-	-	-	0,45	12
AQ243125110	125	110	151	92	57	-	-	-	0,68	1
SDR 11*										
AQ24316090	160	90	-	109	254	-	14,6	8,2	1,3	1
AQ243160110	160	110	-	110	254	-	14,6	10	1,3	1
AQ243160125	160	125	-	110	254	-	14,6	11,4	1,4	1
AQ243200160	200	160	-	122	277	-	18,2	14,6	2,4	1
AQ243250160	250	160	-	149	309	-	22,7	14,6	4	1
AQ243250200	250	200	-	151	324	-	22,7	18,2	4,7	1
AQ243315250	315	250	-	150	365	-	28,6	22,7	8,1	1

* SDR 11 jointing techniques: butt-fusion or electrofusion welding

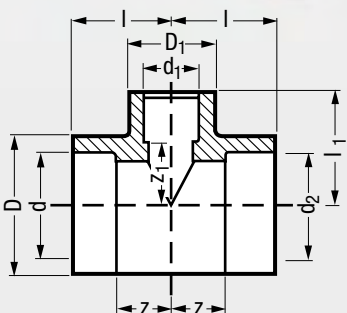
Tee



Code	d	D	l	L	l ₁	z	s	kg	Pack.
AQ13020	20	29	27	55	27	11	-	0,02	200
AQ13025	25	35	32	62	31	13	-	0,04	100
AQ13032	32	44	37	74	37	16	-	0,06	60
AQ13040	40	52	43	88	44	23	-	0,09	42
AQ13050	50	65	51	104	52	28	-	0,17	28
AQ13063	63	84	62	124	62	34	-	0,34	15
AQ13075	75	100	73	142	71	41	-	0,54	12
AQ13090	90	120	84	166	83	50	-	0,95	6
AQ130110	110	148	100	198	99	62	-	1,56	4
AQ130125	125	165	125	250	124	78	-	2,70	1
SDR 11*									
AQ130160	160	-	104	-	-	206	14,6	4,1	1
AQ130200	200	-	124	-	-	253	18,2	7,4	1
AQ130250	250	-	132	-	-	291	22,7	13,1	1
AQ130315	315	-	153	-	-	351	28,6	26,5	1

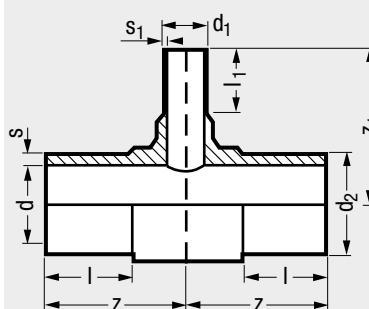
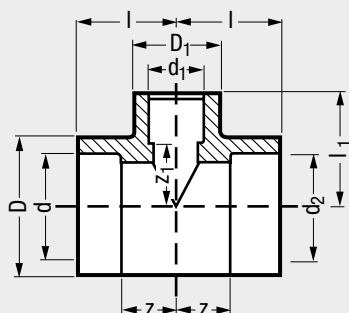
* SDR 11 jointing techniques: butt-fusion or electrofusion welding

Reducing tee



Code	d	d ₁	d ₂	D	l	z	l ₁	D ₁	z ₁	s	s ₁	kg	Pack.
AQ1302520	25	20	25	44	32	16	32	29	17	-	-	0,04	150
AQ1303220	32	20	32	43	38	20	36	34	21	-	-	0,06	80
AQ1303225	32	25	32	43	38	20	36	34	20	-	-	0,06	80
AQ1304020	40	20	40	52	44	24	39	43	24	-	-	0,08	48
AQ1304025	40	25	40	52	44	23	40	43	24	-	-	0,09	48
AQ1304032	40	32	40	52	44	23	40	43	22	-	-	0,09	48
AQ1305020	50	20	50	65	52	28	46	43	31	-	-	0,16	36
AQ1305025	50	25	50	65	52	28	46	43	30	-	-	0,16	30
AQ1305032	50	32	50	65	52	28	62	43	28	-	-	0,16	30
AQ1305040	50	40	50	65	62	39	62	65	35	-	-	0,16	30

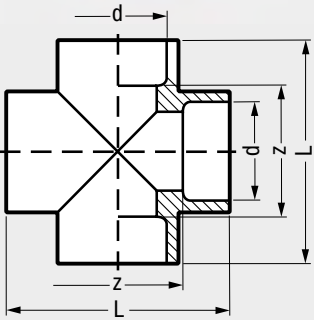
Reducing tee



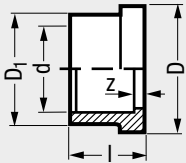
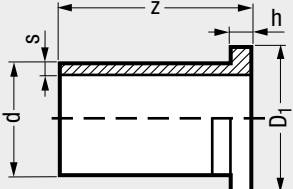
Code	d	d ₁	d ₂	D	l	z	l ₁	D ₁	z ₁	s	s ₁	kg	Pack.
AQ1306320	63	20	63	85	62	35	62	43	48	-	-	0,31	16
AQ1306325	63	25	63	85	62	35	62	43	46	-	-	0,31	18
AQ1306332	63	32	63	85	62	35	62	43	44	-	-	0,31	18
AQ1306340	63	40	63	85	62	39	62	85	42	-	-	0,37	18
AQ1306350	63	50	63	85	62	35	62	85	39	-	-	0,32	16
AQ1307520	75	20	75	85	71	41	71	43	57	-	-	0,51	12
AQ1307525	75	25	75	100	71	41	71	43	55	-	-	0,51	12
AQ1307532	75	32	75	100	71	41	71	43	53	-	-	0,51	12
AQ1307540	75	40	75	100	71	41	71	65	51	-	-	0,51	12
AQ1307550	75	50	75	100	71	41	71	65	48	-	-	0,51	12
AQ1307563	75	63	75	100	71	41	71	101	44	-	-	0,52	12
AQ1309063	90	63	90	120	83	50	83	120	55	-	-	0,91	5
AQ1309075	90	75	90	120	83	50	83	120	53	-	-	0,91	5
AQ13011063	110	63	110	148	99	62	99	85	71	-	-	1,53	4
AQ13011075	110	75	110	148	99	62	99	100	69	-	-	1,54	4
AQ13011090	110	90	110	148	99	62	99	120	66	-	-	1,57	4
AQ130125110	125	110	125	165	124	84	110	148	87	-	-	2,60	1
SDR 11*													
AQ13016090	160	90	160	-	104	206	83	-	190	14,6	8,2	3,6	1
AQ130160110	160	110	160	-	104	207	89	-	200	14,6	10	3,8	1
AQ13020090	200	90	200	-	124	250	82	-	216	18,2	8,2	6,9	1
AQ130200110	200	110	200	-	124	250	86	-	219	18,2	10	7,1	1
AQ130200160	200	160	200	-	120	250	101	-	253	18,2	14,6	7,6	1
AQ130250110	250	110	250	-	133	288	85	-	248	22,7	10	11,9	1
AQ130250160	250	160	250	-	134	292	102	-	266	22,7	14,6	12,4	1
AQ130315110	315	110	315	-	155	346	85	-	282	28,6	10	22,1	1
AQ130315160	315	160	315	-	155	352	101	-	301	28,6	14,6	22,8	1
AQ130315250	315	250	315	-	154	346	135	-	331	28,6	22,7	24,4	1

* SDR 11 jointing techniques: butt-fusion or electrofusion welding

Cross

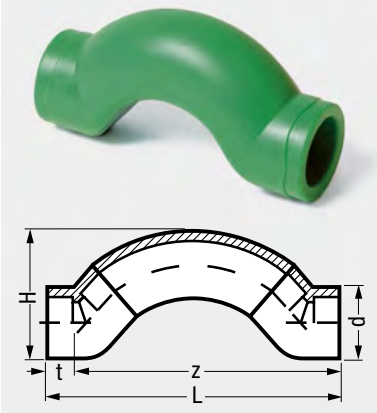
 	Code	d	L	z	kg	Pack.
	AQ18025	25	60	27	0,06	80
	AQ18032	32	75	34	0,08	50

Flange adaptor

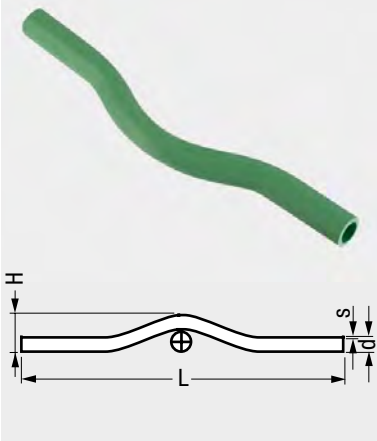
  	Code	d	D	l	z	h	D ₁	s	kg	Pack.
	AQ79040	40	60	29	7,4	7,8	50	-	0,03	80
	AQ79050	50	70	33	9,7	9,5	60	-	0,05	60
	AQ79063	63	89	40	12,9	15,5	76	-	0,08	48
	AQ79075	75	105	37	7,5	15	89	-	0,13	40
	AQ79090	90	125	46	9,5	19,5	109	-	0,25	26
	AQ790110	110	158	57	13	18	132	-	0,38	12
	AQ790125	125	162	62	13	21	146	-	0,48	10
	SDR 11*									
	AQ790160	160	-	-	207	25	212	14,6	1,8	1
	AQ790200	200	-	-	201	32	269	18,2	3	1
	AQ790250	250	-	-	220	35	320	22,7	4,9	1
	AQ790315	315	-	-	239	35	370	28,6	7,5	1

* SDR 11 jointing techniques: butt-fusion or electrofusion welding

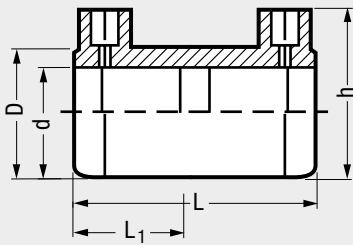
Cross over

	Code	d	L	z	H	t	kg	Pack.
	AQ28720	20	90	63	45	14	0,04	140
	AQ28725	25	104	80	55	16	0,06	90
	AQ28732	32	126	98	70	23	0,13	65

Cross over pipe

	Code	d	s	H	L	kg	Pack.
	AQ28520	20	3,4	53	365	0,06	100
	AQ28525	25	4,2	56	370	0,09	70
	AQ28532	32	5,4	68	370	0,15	50

Electrofusion socket



Code	d	D	L	h	L ₁	kg	Pack.
AQ27120	20	33	70	52	34	0,05	60
AQ27125	25	38	70	58	34	0,05	50
AQ27132	32	45	70	65	34	0,07	40
AQ27140	40	55	85	75	41	0,11	30
AQ27150	50	68	88	87	42	0,15	20
AQ27163	63	82	98	100	47	0,22	20
AQ27175	75	98	125	114	61	0,34	10
AQ27190	90	113	146	130	72	0,50	6
AQ271110	110	136	155	144	77	0,66	5
AQ271125	125	156	166	167	82	1,00	5
AQ271160	160	190	175	201	86	1,50	1
AQ271200	200	232	185	243	91	2,17	1
AQ271250	250	296	212	300	106	4,46	1
AQ271315	315	372,5	240	373	120	9,65	1

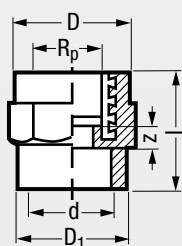


Transition Fittings

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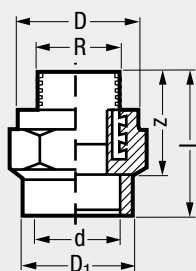
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Adaptor socket (Female thread)



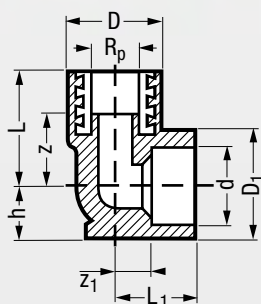
Code	d	Rp	D	D ₁	l	z	kg	Pack.
AQ270G2012	20	1/2"	35	29	41	11	0,07	200
AQ270G2034	20	3/4"	43	34	42	11	0,10	240
AQ270G2512	25	1/2"	35	34	41	11	0,07	200
AQ270G2534	25	3/4"	43	34	41	11	0,11	180
AQ270G3234	32	3/4"	43	43	44	11	0,11	160
AQ270G321	32	1"	50	43	48	12	0,15	130
AQ270G40114	40	1 1/4"	62	52	54	13	0,22	70
AQ270G50112	50	1 1/2"	69	64	57	14	0,24	48
AQ270G632	63	2"	84	79	68	19	0,49	30
AQ270G75212	75	2 1/2"	113	99	82	22	0,81	14
AQ270G903	90	3"	129	124	92	27	1,44	6
AQ270G1104	110	4"	160	151	165	27	2,15	4

Adaptor socket (Male thread)



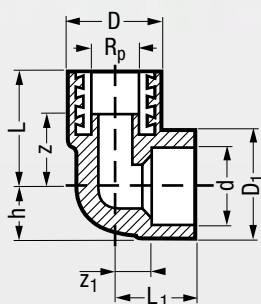
Code	d	R	D	D ₁	l	z	kg	Pack.
AQ243G2012	20	1/2"	35	29	53	40	0,08	200
AQ243G2034	20	3/4"	43	34	58	42	0,12	140
AQ243G2512	25	1/2"	35	34	53	40	0,08	160
AQ243G2534	25	3/4"	43	34	58	42	0,12	160
AQ243G3234	32	3/4"	43	43	58	42	0,13	150
AQ243G321	32	1"	50	43	66	48	0,19	75
AQ243G40114	40	1 1/4"	62	52	74	53	0,31	48
AQ243G50112	50	1 1/2"	69	64	77	54	0,35	36
AQ243G632	63	2"	84	79	92	65	0,65	24
AQ243G75212	75	2 1/2"	112	99	112	82	1,19	8
AQ243G903	90	3"	134	124	143	111	1,98	6
AQ243G1104	110	4"	169	151	161	124	2,80	3

Elbow/Wall bracket 90° (Female thread)



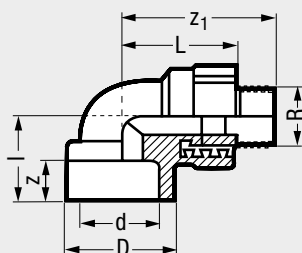
Code	d	Rp	D	L	z	h	D ₁	L ₁	z ₁	kg	Pack.
AQ472G2012	20	1/2"	35	35	21	15	29	27	11	0,08	150
AQ472G2512	25	1/2"	35	37	23	17	34	30	14	0,09	130
AQ472G2534	25	3/4"	43	43	28	22	34	35	19	0,14	90
AQ472G3234	32	3/4"	43	43	28	22	43	35	17	0,15	60
AQ472G321	32	1"	43	43	28	22	43	35	17	0,17	60

Elbow bracket 90° (Female thread)



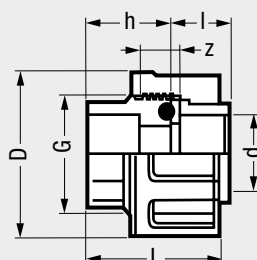
Code	d	Rp	D	L	z	h	D ₁	L ₁	z ₁	kg	Pack.
AQ090G2012	20	1/2"	35	35	21	13	29	27	11	0,08	200
AQ090G2512	25	1/2"	35	37	23	15	34	30	14	0,09	130
AQ090G2534	25	3/4"	43	39	28	20	34	32	19	0,14	120
AQ090G3212	32	1/2"	48	43	28	18	43	36,5	18	0,10	60
AQ090G321	32	1"	48	45	28	20	43	39,5	17	0,17	60

Elbow 90° (Male thread)



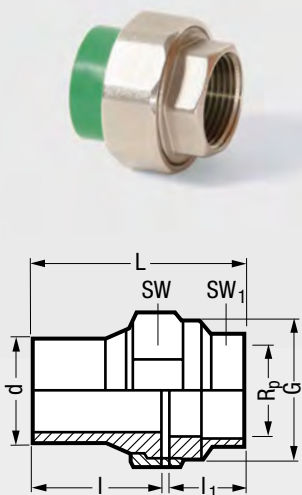
Code	d	R	D	L	z	L ₁	z ₁	kg	Pack.
AQ092G2012	20	1/2"	29	28	14	34	49	0,09	180
AQ092G2512	25	1/2"	34	32	16	38	53	0,10	150
AQ092G2534	25	3/4"	34	32	16	40	56	0,15	100
AQ092G321	32	1"	43	38	20	48	66	0,21	60

Union

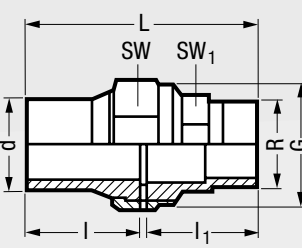


Code	d	G	D	L	L ₁	z	z ₁	kg	Pack.
AQ330A20	20	1"	46	18	44	15	26	0,04	120
AQ330A25	25	1 1/4"	56	19	48	15	28	0,06	100
AQ330A32	32	1 1/2"	66	21	52	15	30	0,09	40
AQ330A40	40	2"	79	23	58	17	34	0,14	30
AQ330A50	50	2 1/4"	87	26	66	19	39	0,16	30
AQ330A63	63	2 3/4"	107	30	79	23	47	0,27	10

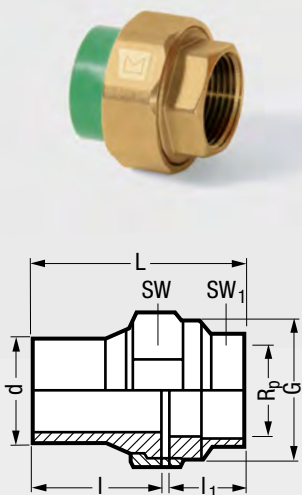
Metal union with PP-R nut (Female thread)

	Code	d	Rp	DN	G	L	I	I ₁	SW	SW ₁	Pack.
	AQ54220	20	1/2"	15	1	38	19	16	38	26	100
	AQ54225	25	3/4"	20	1 3/4	43	22	18	48	32	100
	AQ54232	32	1"	25	1 1/2	48	23	23	54	37	100
	AQ54240	40	1 1/4"	32	2	55	26	26	73	47	25
	AQ54250	50	1 1/2"	40	2 1/4	58	29	26	85	55	25
	AQ54263	63	2"	50	2 3/4	63	32	28	107	64	18

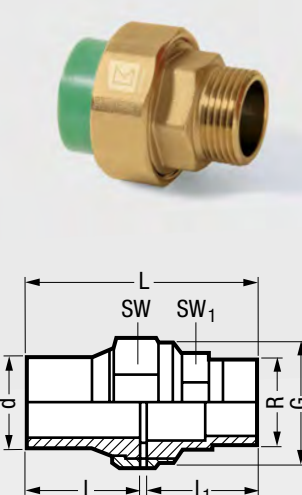
Metal union with PP-R nut (Male thread)

	Code	d	R	DN	G	L	I	I ₁	SW	SW ₁	Pack.
	AQ54720	20	1/2"	15	1	51	19	29	38	26	100
	AQ54725	25	3/4"	20	1 3/4	57	22	31	48	32	100
	AQ54732	32	1"	25	1 1/2	65	23	38	54	37	100
	AQ54740	40	1 1/4"	32	2	73	26	44	73	47	25
	AQ54750	50	1 1/2"	40	2 1/4	76	29	44	85	55	25
	AQ54763	63	2"	50	2 3/4	85	32	50	107	64	18

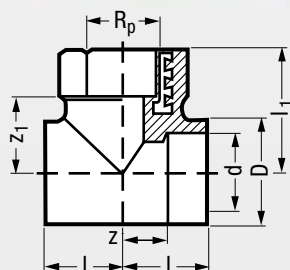
Metal union with PP-R nut in yellow brass (CW617N) - (Female thread)

	Code	d	Rp	DN	G	L	I	I ₁	SW	SW ₁	Pack.
	AQ53220	20	1/2"	15	1	38	19	16	38	26	100
	AQ53225	25	3/4"	20	1 3/4	43	22	18	48	32	100
	AQ53232	32	1"	25	1 1/2	48	23	23	54	37	100
	AQ53240	40	1 1/4"	32	2	55	26	26	73	47	25
	AQ53250	50	1 1/2"	40	2 1/4	58	29	26	85	55	25
	AQ53263	63	2"	50	2 3/4	63	32	28	107	64	18

Metal union with PP-R nut in yellow brass (CW617N) - (Male thread)

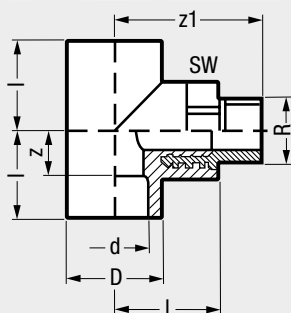
	Code	d	R	DN	G	L	I	I ₁	SW	SW ₁	Pack.
	AQ53720	20	1/2"	15	1	51	19	29	38	26	100
	AQ53725	25	3/4"	20	1 3/4	57	22	31	48	32	100
	AQ53732	32	1"	25	1 1/2	65	23	38	54	37	100
	AQ53740	40	1 1/4"	32	2	73	26	44	73	47	25
	AQ53750	50	1 1/2"	40	2 1/4	76	29	44	85	55	25
	AQ53763	63	2"	50	2 3/4	85	32	50	107	64	18

Tee 90° (Female thread)



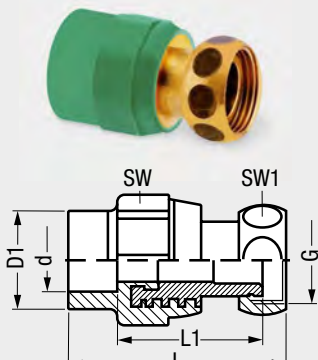
Code	d	Rp	D	l	z	l ₁	z ₁	kg	Pack.
AQ130G2012	20	1/2"	29	28	14	33	20	0,08	180
AQ130G2512	25	1/2"	35	31	16	37	24	0,09	100
AQ130G2534	25	3/4"	34	32	16	40	25	0,13	80
AQ130G3234	32	3/4"	43	38	20	45	30	0,15	60
AQ130G321	32	1"	43	38	20	48	30	0,20	60

Tee 90° (Male thread)



Code	d	R	D	l	z	l ₁	z ₁	kg	Pack.
AQ133G2012	20	1/2"	29	28	14	34	49	0,11	100
AQ133G2034	20	3/4"	29	28	14	35	50	0,15	100
AQ133G2512	25	1/2"	34	32	16	38	53	0,11	80
AQ133G2534	25	3/4"	34	32	16	40	55	0,15	80

Union for watermeters

	Code	d	Nut thread	D1	L	L1	SW	SW1	kg	Pack.
	AQ3322012	20	G 1/2	29	67	44	36	23	0,11	180
	AQ3322034	20	G 3/4	29	67	44	36	31	0,13	160
	AQ3322534	25	G 3/4	34	67	44	36	31	0,13	150

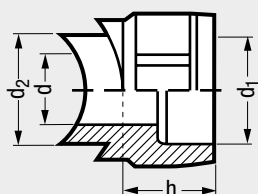


Weld-in saddles

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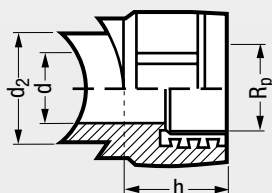
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Weld-in saddle



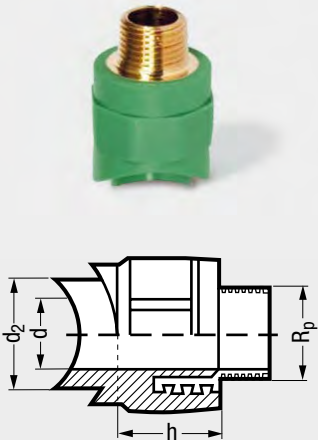
Code	d	d ₁	d ₂	h	Pack.
AQ130S406320	40 - 63	20	25	29	150
AQ130S406325	40 - 63	25	25	29	90
AQ130S406332	40 - 63	32	25	29	90
AQ130S7512525	75 - 125	25	25	29	60
AQ130S7512532	75 - 125	32	32	35	60
AQ130S7512540	75 - 125	40	40	38	60
AQ130S16025025	160 - 250	25	25	29	40
AQ130S16025032	160 - 250	32	32	35	40
AQ130S16025040	160 - 250	40	40	38	40
AQ130S16025050	160 - 250	50	50	39	30
AQ130S16025063	160 - 250	63	63	45	20

Weld-in saddle (Female thread)



Code	d	R _p	-d ₂	h	Pack.
AQ270S406312	40 - 63	1/2"	25	43	150
AQ270S406334	40 - 63	3/4"	32	50	90
AQ270S7512512	75 - 125	1/2"	25	43	90
AQ270S7512534	75 - 125	3/4"	32	50	60

Weld-in saddle (Male thread)

	Code	d	Rp	-d ₂	h	Pack.
	AQ243S406312	40 - 63	1/2"	25	43	150
	AQ243S406334	40 - 63	3/4"	32	50	90
	AQ243S7512512	75 - 125	1/2"	25	43	90
	AQ243S7512534	75 - 125	3/4"	32	50	60



Valves

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Concealed valve (only upper part) chrome heavy part

	Code	Dimension G	kg	Pack.
	AQ5993	$\frac{3}{4}$ "	0,31	1

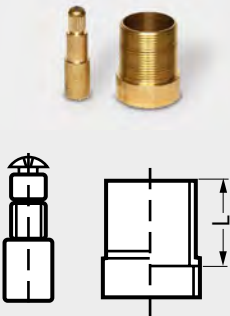
Concealed valve (only upper part) chrome light part

	Code	Dimension G	kg	Pack.
	AQ5992	$\frac{3}{4}$ "	0,21	1

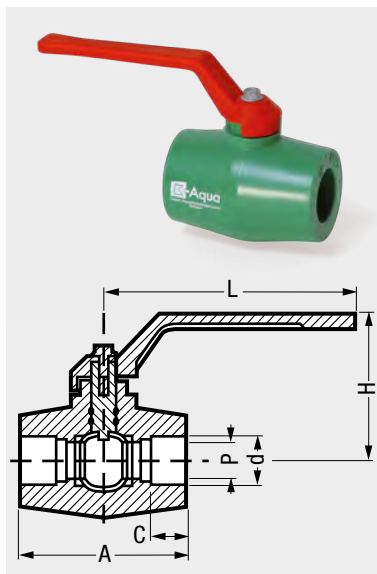
Straight seat valve (only upper part) green handle

	Code	Dimension G	kg	Pack.
	AQ5991	$\frac{3}{4}$ "	0,12	1

Elongation pieces

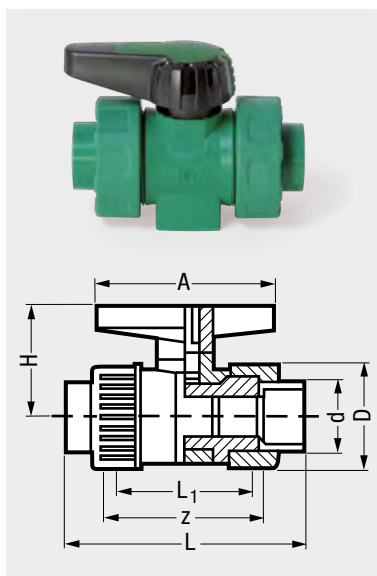
	Code	L	kg	Pack.
	AQ599E	30	0,05	1

PP-R Ball valve (Ball in brass, chromium plated)



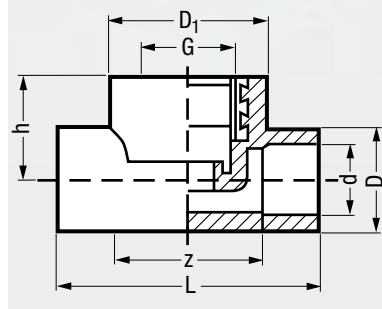
Code	d	A	C	H	L	P
AQ85020	20	67,5	14,5	60	102	15
AQ85025	25	70,5	16	60	102	15
AQ85032	32	79,5	18	63	102	20
AQ85040	40	94	20,5	78	119,5	25
AQ85050	50	109	23,5	83	119,5	32
AQ85063	63	130	27,5	103	146	40
AQ85075	75	151	31	110	146	50
AQ85090	90	173	35,5	133	205	65

PP-R Ball valve (Ball in PP)



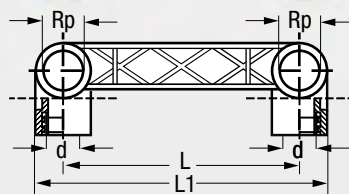
Code	d	D	L	z	H	A	L ₁	kg	Pack.
AQ85220	20	46	98	70	51	68	63	0,11	1
AQ85225	25	56	113	82	61	78	75	0,19	1
AQ85232	32	66	121	87	70	88	79	0,28	1
AQ85240	40	79	138	98	81	98	91	0,44	1
AQ85250	50	87	148	101	90	108	95	0,54	1
AQ85263	63	107	175	121	110	118	115	0,93	1

Tee 90° (Female thread) for internal valve



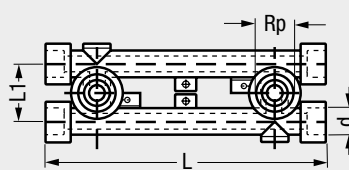
Code	d	G	D	L	D ₁	z	h	kg	Pack.
AQ599A2034	20	3/4"	33	80	45	46	33	0,11	100
AQ599A2534	25	3/4"	36	80	45	43	33	0,12	80
AQ599A3234	32	3/4"	43	75	45	39	33	0,14	80
AQ599A321	32	1"	44	94	45	58	33	0,19	80

Battery (Female thread)



Code	d	Rp	L	L1	kg	Pack.
AQ490G2012	20	1/2"	150	185	0,18	1
AQ490G2512	25	1/2"	150	185	0,19	1

Adjustable battery (Female thread)



Code	d	Rp	L adjustable	L1	kg	Pack.
AQ492G2012	20	1/2"	100-135-150	230	0,21	1
AQ492G2512	25	1/2"	100-135-150	230	0,23	1

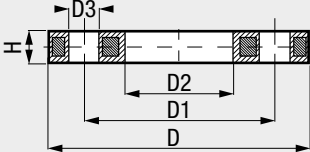


Accessories

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Flat gasket	110 - 111
Pipe clamps	111
Plug	112

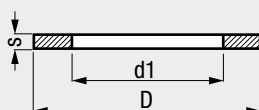
Backing flange PP-Steel for socket fusion system (SF) or butt fusion system (BF)

 	Code	d	D	D ₁	D ₂	D ₃	H	System	kg	Pack.
	AQ75040	40	140	100	51	18	16	SF/BF	0,62	20
	AQ75050	50	150	110	62	18	18	SF/BF	0,82	17
	AQ75063	63	165	125	78	18	18	SF/BF	0,94	15
	AQ75075	75	185	145	92	18	18	SF/BF	1,35	11
	AQ75090	90	200	160	110	18	20	SF	1,39	13
	AQ750110	110	220	180	133	18	20	SF	1,41	13
	AQ750125	125	250	210	150	18	20	SF	1,41	12
	AQ750160	160	285	240	178	22	24	BF	3,60	1
	AQ750200	200	340	295	235	22	27	BF	5,20	1
	AQ750250	250	395	350	288	22	30	BF	6,63	1
	AQ750315	315	445	400	338	22	34	BF	8,40	1

Flat gasket

	Code	d	Pack.
	AQ71440	40	1
	AQ71450	50	1
	AQ71463	63	1
	AQ71475	75	1
	AQ71490	90	1
	AQ714110	110	1
	AQ714125	125	1
	AQ714160	160	1
	AQ714200	200	1
	AQ714250	250	1
	AQ714315	315	1

Flat gasket for unions PP-R



Code	d	R/Rp	Nut thread	D	d ₁	s	Pack.
AQ490F20	20	1/2"	G 3/4"	27	20	3	1
AQ490F25	25	3/4"	G 1"	35	25	3	1
AQ490F32	32	1"	G 1 1/4"	38	28	3	1

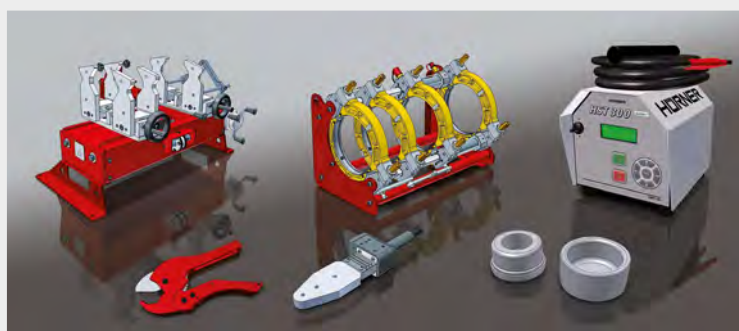
Pipe clamps



Code	d	kg	Pack.
AQ50020	20	0,06	100
AQ50025	25	0,06	100
AQ50032	32	0,07	75
AQ50040	40	0,08	50
AQ50050	50	0,08	50
AQ50063	63	0,13	50
AQ50075	75	0,20	25
AQ50090	90	0,21	25
AQ500110	110	0,24	25
AQ500125	125	0,30	25
AQ500160	160	0,35	25
AQ500200	200	0,37	10
AQ500250	250	0,52	10

Plug

	Code	G	kg	Pack.
	AQ90912	1/2"	0,02	1



Tools

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Welding tool for repairing plug	116
Repairing plug	116
Welding machine 50-125	116
Butt welding machine 90-250	117
Electrofusion machine	117

Pipe cutter 20-40



Code	Pack.
AQ97040	1

Pipe cutter 50-125



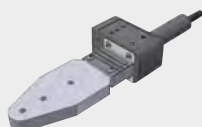
Code	Pack.
AQ975125	1

Pipe scraper



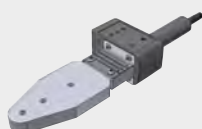
Code	Pack.
AQ974	1

Hand welding machine 20-63 (complete set)



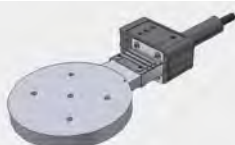
Code	Pack.
AQ98063	1

Hand welding machine 20-32 (complete set)



Code	Pack.
AQ98032	1

Hand welding machine (only mirror) 50-125



Code	Pack.
AQ991125	1

Welding tool



Code	d	Pack.
AQ98220	20	1
AQ98225	25	1
AQ98232	32	1
AQ98240	40	1
AQ98250	50	1
AQ98263	63	1
AQ98275	75	1
AQ98290	90	1
AQ982110	110	1
AQ982125	125	1

Welding tool for weld in saddles



Code	d	Pack.
AQ98504006325	40-63x25	1
AQ98507512525	75-125x25	1
AQ98507512532	75-125x32	1
AQ98507512540	75-125x40	1
AQ98516025025	160-250x25	1
AQ98516025032	160-250x32	1
AQ98516025040	160-250x40	1
AQ98516025050	160-250x50	1
AQ98516025063	160-250x63	1

Drilling tool for weld in saddle



Code	Pack.
AQ98625	1
AQ98632	1
AQ98640	1
AQ98650	1
AQ98663	1

Welding tool for repairing plug



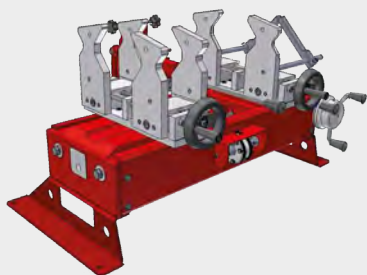
Code	d	Pack.
AQ9837	7	1
AQ98311	11	1

Repairing plug



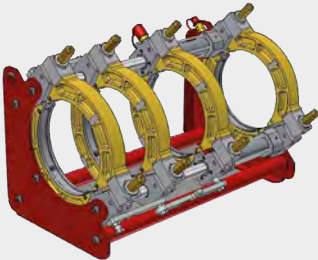
Code	d	Pack.
AQ5937	7	1
AQ59311	11	1

Welding machine 50-125 (complete set)



Code	Pack.
AQ988125	1

Butt welding machine 90-250

	Code	Pack.
	AQ989250	1

Electrofusion machine

	Code	Pack.
	AQ990	1

1. General

The supply of our goods and services are subject to the following terms and conditions provided that the Customer is a person acting in the course of his business or is a public authority. Any terms and conditions of the Customer which conflict with or deviate from the following shall not apply except where we have expressly agreed to the same in writing. Our unconditional supply of goods and services or receipt of payment by us shall not be construed as acceptance of conflicting terms and conditions.

These terms and conditions shall govern all future purchase orders placed by the Customer.

The goods and services supplied by us are in conformity with the laws and standards in force in Germany. We do not warrant conformity with the laws and standards of other countries (see clause 9). Insofar as it is necessary for the handling of our business we are entitled to store and process the Customer's data in digital form to the extent permitted by data protection laws (in particular §28 of the German Data Protection Act).

2. Contractual Representations

Our quotations are without obligation in terms of price, quantity, delivery term and availability. Our order acknowledgement in text form is definitive in terms of confirming acceptance of your order, the scope of delivery and delivery date.

3. Prices, Payment Terms, Set Off

Except where otherwise agreed our prices are subject to the price list valid at the date of the contract. They are ex works (EXW Incoterms 2010) and do not include packing, freight, customs duties, ancillary import charges, insurance and VAT. VAT shall be charged at the rate current at the date of performance. All items are subject to a packing charge of 3% of the list price. Assembly, installation or any other services will be charged on a time and material basis.

Where the delivery or performance date is more than 3 months after the contract date we are entitled upon timely notification to the Customer and prior to delivery or performance to adjust the price in such a manner as is necessitated by any general price development beyond our control (e.g. exchange rate fluctuations, currency regulations, customs duties changes, increases in material and production costs) or by changes of suppliers. For supplies of goods or services within three months from the contract date the contract price shall apply in any event. In relation to framework agreements with a price clause the three month period shall begin to run upon the effective date of the agreement. In relation to framework agreements with a price clause the three month period shall begin to run upon the effective date of the agreement.

Except where otherwise agreed the Customer shall remit payment of the agreed price without deduction within 30 days of the date of invoice. Upon expiry of the payment term the customer shall be deemed in default of payment. Payment remitted within 10 days of the invoice date are subject to a 3% discount. Cheques and bills of exchange will not be accepted as a means of payment.

The Customer may only set off undisputed counterclaims or counterclaims against which we have no further recourse to appeal; his rights under warranty will not be limited by the aforesaid. The Customer shall only be entitled to rights of retention in so far as these are based on the same legal transaction.

4. Delivery Date

The commencement of the delivery term quoted shall be subject to clarification of all technical matters and proper and timely performance by the Customer of his obligations.

Where we fail to deliver upon an agreed delivery date and such failure is caused by an act or omission on our part the Customer shall grant us an extension in writing of not less than 2 weeks. Where upon the expiry of the grace period, delivery is still not forthcoming and the Customer desires to rescind the contract or demand damages in lieu of performance, the Customer shall prior thereto set a final and reasonable deadline in writing expressly indicating his intention. The Customer is obliged at our request to declare within a reasonable period whether he shall rescind the contract due to the delay in delivery and/or demand damages in lieu of performance or insist upon performance.

Where assembly, installation or commissioning is delayed due to reasons for which we are not responsible the Customer shall bear the reasonable cost of any waiting time and additional travel expenses incurred by our service personnel.

5. Shipment, Transfer of Risk

We supply our goods and services „ex works“ (EXW Incoterms 2010). The risk of loss, destruction or damage passes to the Customer upon loading at our warehouse or when the goods cannot or are not to be shipped upon notification of their readiness for shipment. This shall also apply where we deliver by installment or where we have assumed other duties to include without limitation transportation or to supply to the delivery location or assembly of the goods.

Where we instructed to supply to the delivery location, we reserve the right to select the transport route and the mode of transport. We shall select the packaging, protection aids and / or means of transportation on the basis of our experience at the expense of the Customer. Extra costs incurred by special shipping requirements of the Customer shall be for the account of the Customer. The same applies to increases in freight charges that occur after the contract is concluded, any additional costs for re-routing, warehouse charges etc. except where delivery carriage paid is agreed.

Supply of goods or services by installment is permitted except where unreasonable for the Customer.

In relation to make and hold orders approximately the same quantities shall be called off each month except where otherwise agreed in writing. The entire order quantity shall be deemed called off one month after the expiration of the call off deadline, or in the absence thereof twelve (12) months after the contract date. Where the Customer fails to assign goods ordered to a certain delivery within one month after the expiry of the deadline for such assignment, or in the absence of an agreed deadline, within one month of our request for such assignment, we reserve the right to assign the goods at our discretion and deliver the same.

Excess or short deliveries are permitted insofar as they are within the limits which are customary in the trade.

6. Sub-Contractors

We are entitled to use sub-contractors for performance of our contractual obligations.

7. Act of God

In the event of act of god we are entitled to suspend performance of our obligation to deliver. Where there is a considerable change in the circumstances prevalent at the formation of the contract, we reserve the right to rescind the contract. The same shall apply in relation to shortages of energy or raw materials, strikes, governmental decrees, disruption of operations or infrastructure or where, for the above reasons, a sub-supplier fails to supply us properly, on time or at all.

8. Retention of Title

Goods sold shall remain our property until payment of all claims under the business relationship has been received from the Customer ("Reserved Goods").

If the Customer works or processes the Reserved Goods our reservation of title shall be extended to cover the whole of the new article. In the case of processing, combining or mixing with external goods by the Customer we shall acquire title in the fraction that corresponds to the invoice value of our goods to that of the other objects used by the Customer at the time the processing, combining or mixing took place.

If the Reserved Goods are combined or mixed with a principle good belonging to the Customer, the latter hereby assigns in addition his rights in the new good to us. If the Customer combines or mixes the Reserved Goods with a principle good belonging to a third party against payment, he hereby assigns his claims for payment against the third party to us.

The Customer may resell Reserved Goods in the framework of due business activities. If the Customer sells such goods without receiving the full purchase price in advance or contemporaneously with the surrender of the item purchased, he shall agree a reservation of title with his customers in accordance with these conditions. The Customer hereby assigns to us his claims under this resale and the rights under the reservation of title agreed. He agrees at our request to notify the assignment to Customers and to provide us with the information required to claim our rights against the Customers and to hand over documents. Irrespective of the assignment the Customer shall only be authorised to collect payments under the resale as long as he complies duly with his obligations to us.

If the value of the securities provided to us exceeds our claims by more than 10 per cent, we undertake to release securities of our choice on demand by the Customer.

In the event that the above retention of title clauses are void or unenforceable according to the law of the state/country in which the goods are situated, the collateral security which corresponds to the retention of title in that state/country is deemed to be agreed.

9. Customer's Rights in the case of Defects

The goods supplied by us correspond to the German regulations and standards currently in force. We give no guarantee that the goods comply with other national regulations. Where the goods are to be put into operation overseas it is the responsibility of the Customer to ensure that the goods are in conformity with the relevant legal requirements and standards and where required to make appropriate adaptations. The Customer shall not be entitled to make a claim based on defective delivery or performance where the reduced value or merchantability of the goods delivered or services supplied is nominal.

Where the goods delivered by us are defective and the Customer has notified us of the same in writing not later than 28 days after the delivery date we shall at our option deliver a replacement or remedy the defect. The Customer shall grant us a reasonable period of not less than 15 working days to carry out the same.

The Customer is entitled to demand reimbursement of the costs incurred by reason of the replacement delivery or remedying of the defect insofar as such costs are not increased due to the subsequent transportation of the goods delivered to a location other than the original shipment location unless the purpose for which the goods are intended requires the same.

In the event that we are not in a position to remedy the defect or deliver a replacement the Customer is entitled to rescind the contract or to demand a reasonable reduction in the purchase price. Rescission of the contract is only permissible where the Customer prior thereto sets a final and reasonable deadline in writing expressly indicating his intention.

The Customer shall retain a statutory right or recourse against us insofar as the Customer has not agreed terms with its customer which exceed the statutory liability for defects.

10. Liability

Except where otherwise provided below any claim of the Customer for damages other than those claims set out in § 9 aforesaid are hereby excluded irrespective of the legal grounds upon which it is based. Accordingly we do not accept liability for any damage not incurred by the goods themselves nor do we accept any liability for loss of profit or any other pecuniary loss suffered by the Customer. To the extent that our contractual liability is excluded or limited, such exclusion or limitation shall apply in relation to the personal liability of employees, representatives and vicarious agents.

The aforesaid limitation of liability shall not apply where the damage incurred has been caused by willful default, by gross negligence or breach of guarantee, where personal injury has been suffered or where damages are claimed under the product liability act.

Where we are in negligent breach of a material term of the contract our liability to compensate damage to property shall be limited to such loss as was typically foreseeable at the time the contract was made. A material term of the contract shall be any term which places the Customer in the legal position provided for under the contract in terms of its content and purpose and any term which must be complied with in order to ensure proper performance of the contract and upon the performance of which the Customer relied or could be reasonably expected to rely.

Any other liability in damages is hereby excluded.

Any assignment of the Customer's claims provided for in § 9 and 10 above is not permitted, insofar as this prohibition is permissible at law.

11. Limitation

The limitation period for claims based on the supply of defective goods and services (§9) as well as for claims for damages (§10) is one year from the date of delivery. The limitation period aforesaid shall not apply in relation to claims based on willful default, gross recklessness, breach of guarantee or to personal injury claims and to claims under product liability laws (§9 sub-para. 2). The limitation period aforesaid shall further not apply where longer limitation periods are prescribed by law including without limitation claims relating to buildings, defects in construction and rights of recourse.

12. Product Description

A description of all of our goods is set out in our „General Technical Data Sheets“, which will be sent to the Customer on request. It is the responsibility of the Customer to test our goods and services as to their suitability for the purpose intended

13. Jurisdiction, Choice of Law

For Customers whose principle place of business is situated in the European Union all disputes arising out of these terms and conditions shall be resolved before the courts with jurisdiction located at our principle place of business. We reserve the right to sue the Customer at its principle place of business.

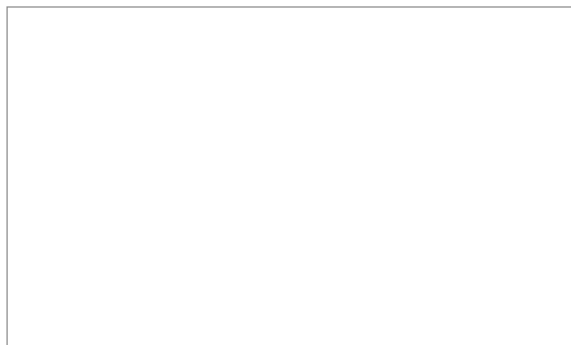
For Customers whose principle place of business is situated outside the European Union all disputes arising out of these terms and conditions shall be finally settled in accordance with the Rules of the German Institution of Arbitration e.V. (DIS) without recourse to the ordinary courts of law. The place of arbitration shall be Frankfurt am Main, Germany. The laws of the Federal Republic of Germany shall govern all the legal relationships between us and the Customer.



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Made in Germany